# Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography

B. Müller Alexander Osika Kai Xue Ben Cassie Brieuc de La Fournière Daniel Ari Friedman

October 31, 2026

**Paper release:** `r2056` **Released:** October 31, 2026

**Author affiliations:** Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

**Abstract**

This paper develops the carrier layer of Observer-Patch Holography (OPH): the hypothesis that federations of twelve-port icosahedral carriers, repaired by the native seam law, are the substrate on which observers, clocks and geometry execute. On one carrier, oriented incidence and an integer atom-counting grammar with normalized readback cost give twelve unit port lines with exact gap two, the inverse-port pairing, the proper action of $A_5$ and an exact rank-three Gram frame. The classical seam rule follows from four stated requirements, its quantum record laws form one convex family classified by five coherence multipliers, and the slowest band of the uniform native repair mean is the normalized rank-three Gram band. Any finite-range cosine kinetics invariant under the sixty proper icosahedral rotations is isotropic through angular rank five at every order in the cell size; its first directional term is a multiple of the unique rank-six invariant, which has an exact sixty-two-point Morse census and enters at order $a^4k^6$ whenever the weighted sixth moment is nonzero. On the thirty seams we prove the exact integer current image, finite continuity, a rational Coulomb Green function, local face-curvature and step-scaled Maxwell-shaped actions with sharp stability threshold $h^2(3+\sqrt{5}) < 4$, and a continuum limit of the complete seam symbol on a supplied Euclidean domain. Stationary native repair has an exact covariance and record-power law. Central records carry the Born–Lüders identities for a declared algebra-state representation, and checkpoint data fix an observer's future law. Federations of 20, 80 and 320 carriers serve larger domains and refinement; their cross-carrier routing is a declared convention, and the existing port-ancestry map does not preserve seam adjacency. The carrier's oriented face complex has vanishing first cohomology. The distinct two-load configuration complex nevertheless has $\mathbb{Z}/2$ exchange topology, and a specified coherent ordering memory protects its two quantum characters. This finite transport sector does not yet supply the localized overlap defect required for physical matter.

**Interpretation and state variables.** OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements.

A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

**Basis, premise labels and paper set.** OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

**Time conventions.** Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

**One theory and its executable hypothesis.** OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/BASIS.md

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

**The tower.** The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

**0. Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

**1. Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

**2. Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

**3. Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

**4. Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

**5. Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

**6. Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

**7. Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology pa-

pers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

**Completion and stopping.** The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

**Fixed physical boundaries.** Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

**From repair topology to a field experiment.** A specified repair process admits protected quantum history sectors, fermionic exchange and an executable coupling to field and clock observables. Its ingredients are specified observer, scheduler, repair and memory laws; they need not be uniquely selected to define a mathematical model. What must be established is that the resulting operations share a carrier, retain complete quantum outputs and reproduce the intended observables. The constructions below establish several such joins, including a joint source experiment. They do not identify a complete physical theory.

**Protected transport sectors.** On the native 66-state binary two-load configuration complex, an exact noncommutative reduction of the face and disjoint-move relations gives $\pi_1 = \langle g \mid g^2 \rangle = \mathbb{Z}/2$. The surviving loop has odd ordered-cover parity. This is topology of the declared transport and deformation domain, not topology of a localized field defect. The complete active branch $*$-algebra on either the unextended 78-state source or its binary shell is the full matrix algebra. A Hermitian source charge commuting with every branch is therefore scalar. This excludes that charge class, not every invariant subspace, expectation-conserved observable or controlled lifetime.

A coherent ordering bit changes the algebra. If a native branch maps $x$ to $f_h(x)$, let $p_h(x)$ be the parity of the transported order and define

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I_{66} \otimes X.$$

Every branch commutes with $Q$, so its two sharp sectors survive arbitrary finite record-adaptive schedules. An actual odd exchange loop returns the source and applies $X$ to the bit: the memory participates in transport. Sector preservation does not assert preservation of every coherence between the sectors. For the all-seam C or E family, effect separation, a connected movement tree

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/docs/COMPLETION_PROGRAM.md

and the odd loop generate exactly $M_{66}(\mathbb{C}) \oplus M_{66}(\mathbb{C})$, with commutant $\mathrm{span}\{I, Q\}$, although the classical ordered cover is connected. The states $I_{66}/66 \otimes |\pm\rangle\langle\pm|$ have identical source and complete classical-record marginals; their distinction is quantum memory. Computational-basis reset or dephasing destroys it. Protection thus depends on the specified memory law preserving these sectors.

Literal source words implement this lift. Complete execution covers all 132 input columns on seam $(0, 5)$, and exact word audits cover all 1024 addresses of the 90 seam/current feedback words. Native outcomes, padding and source-service failures remain in the accounting. The result is a protected sector of the enlarged process; a spatially localized source realization of its charge remains an additional construction.

This same carrier also separates operator action from scalar repair cost. Writing $P_- = (I - X)/2$ and $M_h = \sum_{x \in D_h} |f_h(x)\rangle\langle x|$,

$$A_h = \pi \sum_{x \in D_h} p_h(x) |f_h(x)\rangle\langle f_h(x)| \otimes P_-, \qquad K_h = e^{-iA_h}(M_h \otimes I)/\sqrt{2}.$$

Two closed histories with identical multisets of complete branch labels and identical compiled resource totals induce respectively $I$ and $X$. An additive scalar cost of those labels cannot determine this response; the displayed operator action does. Its phase is not physical energy per unit time. Nor is the reduced memory autonomous: its one-step flip probability is $26/495$, while the two-step $Z$ multiplier differs from the squared one-step multiplier by $398/245025$. The continuing source must be retained in any proposed Hamiltonian or memory description.

**Internal states and fermionic exchange.** Give each occupied port an independent internal space $V$. The map

$$|\{x < y\}; a, b\rangle \longmapsto c^\dagger_{x,a} c^\dagger_{y,b} |0\rangle$$

fixes the CAR convention. Passing the occupied spectator reverses the internal slots and crosses one creation operator, hence

$$K_h^V = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes (-\mathrm{SWAP})^{p_h(x)}.$$

The complete effects remain the original domain projectors divided by two. Independent CAR calculations check every branch for $\dim V = 2, 3$; the crossing argument applies to every finite $V$. The odd loop acts as $-\mathrm{SWAP}$, giving $-1$ on two equal internal states. Global Fock parity on this two-particle sector is $+1$ and cannot replace that exchange.

The ordering memory implements the same operation: for $J = -\mathrm{SWAP}$,

$$E_V \psi = (|0\rangle\psi + |1\rangle J\psi)/\sqrt{2}, \qquad (X \otimes I) E_V = E_V J.$$

Every native branch, and every common $U \otimes U$, intertwines through $E_V$. The negative exchange character acts on $\mathrm{Sym}^2 V$; the positive one acts on $\Lambda^2 V$. Both are compatible with identical fermions carrying independent internal states.

The finite algebra constrains but does not fix internal encodings. An untouched fiber of dimension $m$ enlarges the repair commutant to $M_m \oplus M_m$. A separate collective-label encoding that puts all fifteen states of $\Lambda^2\mathbb{C}^5 \oplus \overline{\mathbb{C}^5}$ in the negative sector requires $m \geq 15$ and has an explicit inclusion at equality. This is not a universal memory bound for the two-particle construction. No element of the supplied $S(U(3) \times U(2))$ gauge group acts as $-I$ on the whole faithful fifteen-state module:

its nonidentity involutions have signature $(7, 8)$. The smaller weak-parity encoding is consequently an internal-grading control, not an all-fermion exchange assignment.

An exact basis permutation joins that exterior module and all twelve currents to the retained chiral action. Specified two-component and three-family factors give $\dim V = 90$; neither family count nor physical Lorentz spin is derived by this basis change. On twelve ports the hard-core two-particle dimension is 534,600, whereas the complete 1080-mode two-particle space has dimension 582,660. The remaining 48,060 states are same-port collisions and must be included when propagation or interactions can reach them.

**One calibrated action and its observables.** The Weyl–Higgs action uses explicit charged Yukawa, Higgs-background and CKM inputs with recorded matching conventions. Recovering them from its response verifies implementation; it is not an independent mass prediction. The creation-pairing matrix $\Delta = \epsilon_{\text{spin}} \otimes \overline{K}$ is antisymmetric on 90 modes. At rest,

$$B_0 = \begin{pmatrix} 0 & \Delta \\ \Delta^\dagger & 0 \end{pmatrix}, \qquad H = E_{\text{vac}} + \sum_{a=1}^{90} E_a \gamma_a^\dagger \gamma_a.$$

The canonical chart retains 84 positive charged directions, six zero-energy neutrino annihilators and every particle–hole partner. Choosing the annihilator half of the zero block specifies the static reference; it does not supply a positive-energy continuation at nonzero momentum. This minimal action has massless neutrinos; a separate neutrino mass extension is not silently included.

Copying this internal system to twelve ports gives 1080 CAR modes. The static hard-core two-quasiparticle sector supports the colored construction, whereas a bare two-particle sector need not survive pairing. The compiled static convention subtracts $11E_{\text{vac}}$ from the replicated vacuum scalar $12E_{\text{vac}}$; the remaining common $E_{\text{vac}}$ does not affect fixed-sector Ramsey gaps. Every extension across particle-number sectors must specify the same reference convention explicitly.

Complete source-controlled Ramsey observations recover an energy difference between equal-charge modes, with an opposite-charge partner. A separate neutral-ancilla phase probe reads the same single-particle charge, retaining all seven charge values and all pointer outcomes. Neither measures an absolute rest energy. The common electromagnetic phase cancels on the neutral pair and is not an electromagnetic-response signal. These executions retain all 128 padded internal addresses, native records and quantum output factors.

Currents are transformed with the Nambu chart, including their anomalous blocks. The neutral Higgs background has a nine-dimensional unbroken stabilizer, color and electromagnetism; three complementary gauge directions are broken. General weak operations cannot therefore be represented solely by $U \otimes U$ within fixed quasiparticle number. A time-dependent Higgs background requires the complete CAR domain.

**Spatial dynamics, clock and particle creation.** A Hamiltonian cycle of the actual twelve-port seam graph supplies a finite spatial ring. With specified spacing and $c = 1$ units, its bare action is

$$h = -D \otimes \sigma_z \otimes I_{45}, \qquad D_{r,r+1} = -i/(2a), \quad D_{r+1,r} = i/(2a),$$

with the same local pairing matrix. The static symbol satisfies $E(k)^2 = \sin^2(k)/a^2 + E(0)^2$. All twelve momenta, all 2160 Nambu directions and both kinetic zeros remain. This is a one-dimensional witness, not a chiral $3 + 1$-dimensional limit. Its 1080 CAR modes retain every occupation sector, including collisions. The pairing graph splits into closed internal components of sizes one, two and six, so the largest spatial component has 72 modes. Complete four-state Fock

source gates and fermionic routing determine their full Gaussian unitary maps without allocating arbitrary $2^{1080}$-component vectors. A source-prepared localized excess and its bond current have complete occupation readouts, with purifiers retained. The executed finite symmetric product is specified as such; its error relative to the summed Hamiltonian is separate from its source-gate error.

Two source-prepared clock hands, each with faithful eigenvalues $1/6, 5/6$ and a retained purifier, share the reference energy $\Omega_{\mathrm{ref}} = 1/4\,\mathrm{GeV}$. Both hands and every field factor use $\lambda_j = 1 + j/4$ for the actual native current. Complementary readouts infer $\tau_{\mathrm{eff}} = \sum_h \lambda_{j_h} \Delta\tau$ on a declared unaliased interval. The inverse receives complete probabilities and error bounds, not the programmed duration or a species mass. Rational enclosures and finite-shot bounds distinguish ensemble inference from a single clock reading. The energy-unit calibration and $\hbar = h/(2\pi)$ set seconds; repair counts and host runtime do not set this clock.

The particle-changing drive follows the actual common Higgs field $H = (0, ve^{i\phi})$ with fixed gauge background. Its creation phases have opposite signs in the up and down/charged-lepton channels, according to their conjugate-Higgs tensors. In a compatible static chart it comprises 42 neutral two-mode pairs and six massless spectators per site, with

$$H_{E,\phi} = E\cos\phi\,\mathrm{diag}(-1,0,0,1) + E\sin\phi\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Literal source gates retain each complete four-state space and all nonadjacent CAR signs. Across twelve sites this gives 504 pairs and 72 spectator modes. An externally driven quadratic field is not a quantized interacting gauge–Higgs theory.

Repair itself extends to every occupation sector. Binary endpoint occupancies follow the original C/E rule; nonbinary endpoints are held. A coherent extension includes that identity block in the existing zero-current Kraus operator. The binary and nonbinary output ranges are orthogonal, so the complete partial-injection effects sum to identity. On states with even occupancy at both endpoints the unread channel is exactly the identity, including coherences and references. The extension is unital before arbitrary record-conditioned controls; its faithful tracial reference has trivial modular flow. General completeness is algebraic. Literal source dilations verify the bounded collision and spectator-crossing cases, not a full 1080-mode dilation.

A separately flagged hold is retained as a control. It measures a vacuum/pair distinction that the coherent hold preserves. If $A$ is the probability of no created pair on the repaired sites, a drive–repair–inverse echo returns $A^2 + (1-A)^2$ for the flagged policy and one for the coherent policy on this even-endpoint domain. For the stated drive at $\tau = 0.002\,\mathrm{GeV}^{-1}$, $j = 0$ and $\phi = \pi/2$, the flagged return is 0.63176674. Coherent collision holding does not remove backaction from the original current or E-parity records.

**A joint source experiment.** Selected source factors test repair, propagation, particle creation and clock readout together. Four active CAR modes on two sites carry the same charged component. A spatial superposition is source-prepared with a path reference and an opposite-charge spectator at a third site. The quasiparticle-to-bare chart is fixed by both static and driven terms; diagonalizing the static mass alone does not suffice. Each of two events applies complete native repair, routed bare hopping in that chart, and the local Higgs factors. The same current advances the clock without reset. All 16 active states, spectator states, references and clock purifiers remain, including states with three active particles.

For both C/E laws and both collision policies, all $32^2 \times 4$ outcomes are retained: 16,384 in total. Complete effects are checked on the 64-dimensional field/clock input algebra, together with the untouched reference marginal. The coherent C process has collision population 0.23404814

before the second repair and final three-particle probability 0.53649765, versus 0.36223079 for the flagged control. Their unread final-state trace distance is 0.20709787. E distinguishes the relevant odd binary occupancies, so the two policies agree on this witness within numerical admission. The composed all-outcome total-variation bound is $4.200002 \times 10^{-7}$, conditional on the full-column numerical source checks. It is not an interval-certified hardware bound. Boundary hops out of the selected sites are omitted: this experiment joins actual source factors but is not an invariant two-site restriction of the full spatial field.

**Energy and modular observations on the same field.** The selected four-mode factors admit energy and reference-state calculations on their complete 16-state algebra. Let $H$ be the sum of the static hopping and Higgs terms, with its vacuum scalar retained. The executed ordered product is not silently replaced by $e^{-itH}$. An explicit piecewise control schedule reproduces that product, separates switching work from energy transfer during complete repair, and uses the same clock duration. The latter transfer is not assigned to heat or controller work without an apparatus model. On the spatial action, onsite terms and half of each incident bond define

$$H = \sum_x h_x, \qquad J_{xy} = i[h_x, h_y] = -J_{yx}, \qquad \frac{dh_x}{dt} = -\sum_y J_{xy} + \partial_t h_x.$$

These identities hold on all occupation sectors. Equality of full unitaries after a scalar energy-reference shift requires the corresponding global source phase; density evolution, currents and gaps do not determine that scalar. Complete, charge-preserving source instruments on all 64 apparatus input columns reconstruct energy, drive-power and ring-current expectations from separate quadratic-group settings. This is expectation readout, not a projective measurement of the summed $H$.

For the declared $\beta = 1\,\mathrm{GeV}^{-1}$, the same static $H$ gives a faithful $\rho = e^{-\beta H}/Z$. A complete eight-qubit source word prepares its field–purifier state. Signed CAR restrictions and compatible Petz observations compose on every matrix unit, and a native recovery isometry retains all four regional inputs, sixteen field states and four environment states. This is a new Gibbs-input experiment with the same field instrument, not a replacement of the earlier localized input. The modular convention is

$$\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \tau_{-\beta s}(A);$$

the dimensionless parameter $s$ does not supply a second physical clock.

The observation is not a regional retraction. The cross-boundary hopping has vanishing partial traces and squared Hilbert–Schmidt norm exactly 4 in the specified units. Its nonadditivity makes the Gibbs density nonproduct for $\beta > 0$, obstructing a reference-preserving UCP retraction onto the full site factor. Compatible non-idempotent Petz observations remain valid. Nor is this Gibbs density stationary under the driven, repaired history. The complete quantum entropy balance is

$$S(\sigma) - S(\rho) = \beta \operatorname{Tr}[H(\sigma - \rho)] - D(\sigma\|\rho)$$

for normalized states; retaining classical records gives an identity involving conditional relative entropies. Numerical trace drift contributes $(\operatorname{Tr}\sigma - 1)\log Z$, and the quadratic Bregman remainder subtracts $\epsilon \operatorname{Tr}\delta$ from $D(\rho + \epsilon\delta\|\rho)$. These reported entropies concern the unnormalized continuing component. First-export abort weights and output families are retained separately; their entropy is not included. A geometric cap kernel, local stress dictionary and generalized-entropy stationarity do not follow from these finite identities.

**The Weyl/contact branch and its finite premise closures.** The following constructions use the supplied Weyl/contact action with $c = 1/8$ and $U = 1$, separately from the calibrated Weyl–Higgs field above. Its dimer and plaquette restrictions retain their actual reference states. The charged-block service is first defined on decoupled dimers. The three-dimensional cube below instead uses bipartite eta symmetry to join a larger protected interacting sector to the complete field. The common local coefficients do not identify distinct global domains.

**A complete charged internal sector.** The same lower charged dimer doublets admit a sharper finite native-record attachment than the fixed-column dictionary. Retain both internal columns at each of two occupied blocks among twelve identical dimers, with the same even half-filled ground at every other block. Their carrier has dimension $66 \cdot 4 = 264$. For the declared graded whole-block repair, a movement that crosses the spectator acts internally as minus the swap; a hold acts as the identity. In the singlet–triplet basis every complete native branch is therefore

$$K_{\text{spinful}} = K_+ \otimes I_1 \ \oplus \ K_- \otimes I_3.$$

An integer certificate checks all 720 raw C/E slots and 31,680 internal input columns, including zero slots. The triplet qutrit and its untouched reference correlations survive every admitted complete history without intermediate recovery. For initially correlated spatial/internal inputs, this assertion concerns the complete internal marginal; conditioning on a record can still steer it.

The raw effects separate all 66 positions. A connected transition tree and an odd exchange loop then classify the entire passive source commutant as $I_{66} \otimes (M_1 \oplus M_3)$, of complex dimension ten. Thus the three-dimensional factor is maximal under the full branch grammar; singlet–triplet coherences do not give a protected four-state factor. Positive all-seam coverage and the checked invariant maximally mixed state supply the corresponding fixed-algebra premises. This enlarged field carrier is distinct from the bare H66 source with scalar fixed algebra.

**Complete extension and an exact spectral exclusion.** Let $P$ be the sum of the orthogonal two-doublet/remaining-ground block projectors. The native outcomes have total effect $P$; adjoining the outside outcome $I - P$ gives a trace-preserving instrument on the full 48-mode Fock space. This is a formal operator construction, with bounded checks on all 4,096 states of three dimers, not a 48-mode source execution. Every definite occupied pair has the unchanged ground on its entire exterior. With the original supplied $K_j = H_j - N_j/2$, $c = 1/8$ and $U = 1$, the same defect has

$$\Delta = \frac{\sqrt{17}}{8} - \frac{1}{16}, \qquad (K_{\text{tot}} - 12E_g I)|_P = 2\Delta I_P.$$

Its excess CAR number is $q = -1$ for the hole and $q = +1$ for the particle; native currents transport $q$ times the defect count. All three protected triplet states have the same gap and total charge $2q$. A common scale or chemical potential cannot split them into three different masses. The triplet is an internal spin-one representation, not three generations.

The same sector is not invariant under the joined uniform four-site Weyl/contact field. For its four charged-pair columns $V$, exact dyadic arithmetic gives

$$\|(I - VV^*)H_{\text{uniform}}V\|_F^2 = \frac{3}{256} > 0$$

for both particle and hole pairs, whereas the decoupled sum gives zero. Indeed, with $L = (I - VV^*)H_{\text{uniform}}V$, the stronger identity $L^*L = (3P_{\text{singlet}} + P_{\text{triplet}})/512$ excludes every nonzero nonleaking input, including every triplet state. This generator leakage preserves the distinction

between the supplied protecting block service and the interacting field law. No binding, topological boundary charge, physical clock, Lorentz spin, observed species or absolute mass follows; the physical particle and common-field gates remain open.

**A protected interacting sector on the original three-dimensional cube.** The source-bound `cube_eta_protection` packet retains the original open $2\times2\times2$ Weyl/contact action, $c = 1/8$, $U = 1$, on all $2^{16} = 65\,536$ occupation states. In the physical onsite basis define

$$\eta^+ = \sum_x (-1)^{x_1+x_2+x_3} c^\dagger_{x\uparrow} c^\dagger_{x\downarrow}, \qquad \eta^- = (\eta^+)^\dagger, \qquad \eta_z = (N-8)/2.$$

Onsite singlets are invariant under each supplied $i\sigma_i$ link; the bipartite signs cancel the hopping commutator. Thus $[H, \eta^+] = U\eta^+$, checked exactly on the complete field action. For $X = \eta^+|_{N=4} : \mathcal{H}_4 \to \mathcal{H}_6$,

$$X^\dagger X = (\eta^-|_{N=4})^\dagger (\eta^-|_{N=4}) + 4I, \qquad H_6 X = X(H_4 + UI).$$

Consequently $X$ is injective and its polar isometry intertwines the full interacting four-particle parent with a protected candidate of dimension $\binom{16}{4} = 1820$. Its range is precisely the $\eta \geq 2$ subspace at $N = 6$.

The executed repair domain is a finite four-block service: the four $x_1$-directed dimers, native ports $(0, 1, 5, 7)$, and all five inherited seams $(0, 1), (0, 5), (0, 7), (1, 5), (1, 7)$. Two blocks carry the unchanged lower-hole doublet and two the unchanged dimer ground. This gives a 24-dimensional $N = 6$ support $P$. Its physical columns are expanded exactly as $A + \sqrt{17}B$, with integer Gaussian matrices $A, B$; $\eta^- A = \eta^- B = 0$, so the support has $\eta = 1$. The columns are also bound to the consumed normalized analytic dictionary: its exact normalization is $n = 136 - 32\sqrt{17}$, with $(A + \sqrt{17}B)^\dagger (A + \sqrt{17}B) = n^2 I$. Every native C/E coin and raw slot is checked against its actual graded whole-block permutation. With the complete outside outcome $Q = I - P$, all 130 slots are retained: 120 native and ten outside over the ten seam/law choices. Orthogonality gives

$$PX = 0, \qquad L_r X = 0 \quad (r \text{ native}), \qquad QX = X.$$

Every admitted complete repair is therefore exactly the identity on this sector, including arbitrary reference cross terms; interleaved field evolution remains unitary and interacting. This mechanism requires no alignment of the transverse dimer spin frames. It does not assert a twelve-block execution, arbitrary repair invariance, or maximality of the protected subspace.

**Exact interacting spectral census and a local fixed-number probe.** Let $v = \eta^+|0\rangle$, $v^\dagger v = 8$, and $\Pi_3 = I - vv^\dagger/8$ on the 120-dimensional $N = 2$ space. For $T = (\eta^+)^2|_{N=2}$,

$$T^\dagger T = 60I + 24vv^\dagger, \qquad H_6 T = T(H_2 + 2UI).$$

Hence $T/\sqrt{60}$ is isometric on the rank-119 $\eta = 3$ parent. The hidden-spin census is 35 singlets and 84 triplet states; this is neither Lorentz-spin identification nor a family count. Exact integer characteristic polynomials in the actual onsite Hubbard frame give, with $z = 16(E_{N=6} - 2)$,

$$p_{119}(z) = z^{57}(z^2 - 12)^{21} q(z) c_{64}(z)^3 c_{128}(z)^3,$$
$$q(z) = z^2 - 16z - 12, \qquad c_a(z) = z^3 - 16z^2 - 12z + a.$$

These are eleven distinct energies: multiplicity 57 at zero, 21 at each $\pm\sqrt{12}$, one at each quadratic root and three at each of the six cubic roots. The last twenty states have contact-shifted energies.

Squarefreeness, coprimality and rational root isolation are exact; approximate displayed response weights are separate diagnostics.

The $N = 6$ reference and one-site density probe are

$$|R\rangle = \frac{(\eta^+)^3|0\rangle}{\sqrt{2016}}, \qquad H|R\rangle = 3|R\rangle, \qquad |s\rangle = \frac{4n_0 - 3}{\sqrt{15}}|R\rangle.$$

Here $\langle n_0\rangle_R = 3/4$, $\mathrm{Var}_R(n_0) = 15/16$; $|s\rangle$ is an $\eta = 3$ singlet with mean energy 3 and variance 3/112. Its normalized two-particle parent is $(8|\uparrow\downarrow\rangle_0 - v)/\sqrt{56}$. The exact parent Krylov Gram determinant is $14883950729121848391888005329065541569085 44 > 0$. Together with the annihilator $qc_{64}c_{128}$ this proves positive response weight at all eight interacting energies and zero weight at the three free-energy values.

The local number-preserving phase $F_0 = e^{i\pi n_0/2}$ satisfies

$$F_0|R\rangle = \frac{|R\rangle - \sqrt{15}|s\rangle}{4}, \qquad \mathrm{Tr}_0(F_0|R\rangle\langle R|F_0^\dagger) = \mathrm{Tr}_0\,|R\rangle\langle R|.$$

Thus the entire exterior reduced state is unchanged initially, while the global pure-state trace distance is $\sqrt{15}/4$ and the disturbed state has energy variance 45/1792. This is a finite local unitary disturbance of an entangled reference, not yet a boundary charge or an asymptotic localized particle. Both components remain protected. A fixed-$N = 6$ pointer Ramsey construction using controlled $F_0$, common field evolution and controlled $F_0^\dagger$ has

$$C(t) = \langle R|F_0^\dagger e^{-it(H-3)}F_0|R\rangle = \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it(z_j/16-1)}, \qquad w_j > 0, \quad \sum_j w_j = 1.$$

No vacuum/charged-sector phase reference is needed in this protocol. The specified $H$ and reference energy fix the frequency origin; these finite spectral gaps are not physical masses.

**Native preparation and the remaining detector premise.** The actual source prepares the 56 signed three-doublon amplitudes using an eight-bit, 5379-pulse amplitude tree on the even modes, with the odd modes blank. Eight native CNOT words copy into the odd modes in 24 pulses, followed by the three-pulse one-site phase including its scalar phase. The compact tree evaluation is an exact spectator blank isometry; all 65 536 rows are retained for copying and the phase. The independent exact trigonometric and literal-port certificate bounds the complete source-state error by $8.48 \times 10^{-13}$, including every source export. All 130 following repair records are retained without success normalization. The native seventeen-bit time-dependent detector and native field continuation are not executed here: the fixed-number response above is analytic, while the reference preparation and local kick are executed. Hardware precision, growing-volume localization, deformed excitation transport, a continuum limit and empirical species identification remain open. The earlier small-carrier leakage and transverse hidden-spin obstruction remain valid; the larger eta sector supplies a different common protection mechanism.

The preceding preparation packet stops before the time-dependent detector. The following extension executes that detector and field continuation on the same cube, closing this finite readout premise.

**An internal observer reads the protected response.** The `protected_particle_readout` construction attaches the preceding cube spectrum to complete native pointer records. Its identified object is the eight-dimensional cyclic response generated by the centered local density probe, inside

the 119-state $\eta = 3$ sector and the larger 1820-dimensional protected space. It is a neutral, fixed-$N = 6$ response; these three dimensions are not interchangeable particle counts. The construction uses Ramsey interference to read a correlation function, a standard measurement principle, and supplies its actual source implementation for this field.[3]

The apparatus retains sixteen field bits and one pointer, hence all $2^{17} = 131\,072$ field/pointer occupation rows. The unchanged source prepares $|R\rangle$ and a pointer $|+\rangle$, then applies controlled $F_0 = \exp(i\pi n_0/2)$. There are two field segments of duration $1/2$: the first is followed by complete C repair on all five admitted seams, and the second by complete E repair on those seams. Controlled $F_0^\dagger$, a pointer phase $\phi \in \{0, \pi/2\}$, and the inverse preparation rotation precede the complete six-outcome pointer export. Every source first-failure record and all 130 repair records remain present. An outside repair outcome continues; a native repair outcome or source failure terminates in its own retained record. No probability is conditioned on survival.

With exact field evolution, each repair acts identically on both ideal arms. Immediately before the final pointer rotation their field vectors are

$$b_0 = e^{-itH}|R\rangle, \qquad b_1 = e^{i\phi}F_0^\dagger e^{-itH}F_0|R\rangle.$$

Consequently the unconditional ideal pointer-zero probability is

$$\begin{aligned} p_0(t,\phi) &= \tfrac{1}{2}\{1 + \mathrm{Re}[e^{i\phi}C(t)]\}, \\ C(t) &= \frac{1}{16} + \frac{15}{16}\sum_{j=1}^{8} w_j e^{-it\omega_j}, \qquad \omega_j = z_j/16 - 1. \end{aligned}$$

Thus the two settings read $\mathrm{Re}\, C = 2p_0(t,0) - 1$ and $\mathrm{Im}\, C = 1 - 2p_0(t,\pi/2)$. The phase $3t$ follows from the reference eigenenergy; it is not an adjustable frequency fit. A common scalar added to $H$, or a common number chemical potential on this fixed number domain, cancels between the arms. Absolute energy zero is therefore not determined by this experiment.

**Complete source and field error, without a sector projection.** The executor composes success and every first-export failure map of each local word on at most three work bits, then applies these maps to the full field/pointer array. This is finite block contraction of the same native word, not replacement by an ideal gate. Independent tests compare the contracted maps with direct source execution, including order-one failure controls. The exact $N = 6$ restriction is used only by the independent original-$H$ comparator, not by the native execution.

Each half-time field segment uses 32 symmetric product steps. The actual ordered support list has 24 hopping terms of norm $1/16$ and eight contact terms of norm one. Counting only possibly nonzero nested commutators gives the all-Fock coefficient $K = 629/256$. The two segments therefore contribute

$$\varepsilon_{\mathrm{split}} = 2\,\frac{629}{256}\,\frac{(1/2)^3}{32^2} = \frac{629}{1048576}.$$

Independent rational trigonometric and literal-port bounds add less than $8.56 \times 10^{-11}$ for the preparation, controlled pointer operations, and both complete source field words. The complete output-vector error is consequently $\varepsilon < 6.00 \times 10^{-4}$. Every halted record is part of this norm. Complete repair and export preserve the comparison by isometric extension or contraction; the all-input field bounds also hold with untouched references. The initial preparation bound consumes the specified initialized blank.

[3] For many-body Ramsey correlation measurements, see Knap et al., *Phys. Rev. Lett.* 111, 147205 (2013), https://arxiv.org/abs/1307.0006.

For normalized complete outputs, half trace distance is at most their vector distance. Each unconditional pointer probability therefore differs from its ideal value by at most $\varepsilon$, and each reconstructed quadrature by at most $2\varepsilon$. Rational root, residue and trigonometric enclosures at $t = 1$ imply, for the actual specified source,

$$0.993649 < p_0(1, 0) < 0.994850, \qquad 0.501379 < p_0(1, \pi/2) < 0.502580.$$

Both intervals exclude the corresponding no-response values, one and one half. Numerical source residuals have their separate declared admission; they are not substituted for these proved source bounds. The executed construction uses 681 283 native pulses per setting. It gives pointer-zero probabilities approximately 0.9942497011 and 0.5019798894, versus the independent original-$H$ values 0.9942494663 and 0.5019797169. The complete output-vector discrepancy is approximately $2.27 \times 10^{-6}$, below the rigorous bound.

The imaginary fringe also distinguishes the contact interaction from the same free cube. Set $U = 0$ while retaining the hopping, reference, local probe and ideal repair schedule. Bipartite occupation parity $\Gamma_{\mathrm{sub}} = (-1)^{\sum_{x\,\mathrm{odd}} n_x}$ satisfies

$$\Gamma_{\mathrm{sub}} H_0 \Gamma_{\mathrm{sub}} = -H_0, \qquad H_0|R\rangle = 0, \qquad \Gamma_{\mathrm{sub}} F_0|R\rangle = F_0|R\rangle.$$

The free characteristic function is consequently real at every time, so $p_{0,\mathrm{free}}(t, \pi/2) = 1/2$ exactly. Both free arms remain in the protected sector. The certified interacting interval therefore excludes this specified contact-free response. This is discrimination between two internal dynamics, not a fit of a physical particle mass.

For independent reset repetitions, Hoeffding's inequality and a union bound give simultaneous sampling error at most $1/2048$ in these two probabilities using $2^{24}$ trials per setting, except with probability less than $4(3/8)^8$.[4] The sampling allowance still separates both fringes from no response. Independent resets are the sampling-service premise; a seeded numerical sample is an illustration of that model, not laboratory data. The recorded PCG64 illustration, with seed 20261101, gives respectively 16 680 454 and 8 423 380 pointer-zero outcomes out of $2^{24}$ trials; the complementary count includes every other complete outcome.

**What eight-line resolution would additionally require.** The executed two settings attach the known protected spectral measure to an internally accessible record. They do not reconstruct eight unknown frequencies or individually resolve the eight lines. There is, however, a certified prospective experiment on the source-specified support. At times $t_n = 39n$, $n = 0, \ldots, 7$, use both pointer quadratures and set

$$y_n = \frac{16C(39n) - 1}{15}, \qquad V_{nj} = e^{-i39n\omega_j}, \qquad y = Vw.$$

Exact interval sine bounds in the Vandermonde product show

$$|\det V| = \prod_{j<k} 2\left|\sin\frac{39(\omega_k - \omega_j)}{2}\right| > 1400.$$

Hadamard's inequality bounds every $7 \times 7$ cofactor by $7^{7/2} < 343(8/3)$, so

$$\|V^{-1}\|_\infty < \frac{2744}{525}, \qquad \|\widehat{w} - w\|_\infty < \frac{43904}{2625}\, e,$$

[4] The bounded-independent-sum inequality is imported from Hoeffding, *J. Amer. Statist. Assoc.* 58 (1963), 13–30, https://doi.org/10.1080/01621459.1963.10500830.

where $e$ bounds the total error of each pointer probability. The latter includes both quadratures and removal of the known elastic atom. Finite precision evaluation of an implemented inverse would need its own error allowance.

A sufficient prospective budget is systematic probability error 1/65536 and sampling error 1/32768 at each of the sixteen settings. With $9 \cdot 2^{29}$ independent trials per setting, the simultaneous failure probability is less than $32(3/8)^9 < 1/100$ and the weight error is less than 343/448000. Twice that error is smaller than the least certified weight, so every line is separated from zero. The total is 77 309 411 328 trials and the largest model time is 273. This long-time accuracy and sampling schedule is not executed here. It identifies weights on the supplied support, not an arbitrary unknown spectrum from finitely many samples.

The finite result is an internally observable protected excitation response of the admitted simulator, with complete-history and finite-time errors. It does not require an external particle name or an SI calibration. It also does not establish that these neutral modes are single odd fermions, that the reference is a vacuum ground state, or that finite spectral lines are relativistic rest masses. Transported excitations, many-body continuum dynamics and comparison with nature remain separate constructions.

The one-time readout above leaves its long-time reconstruction prospective. The following certified accelerated calculation closes that known-support weight-resolution premise while retaining a bound to the full native word.

**Full weight resolution by certified accelerated emulation.** The preceding sixteen-setting proposal can now be executed as a certified emulation of the same source word. The result resolves all eight weights on the source-specified support; it does not infer arbitrary unknown frequencies or identify physical species. The observable remains the eight-dimensional neutral cyclic response of the excited reference, with elastic weight 1/16.

The actual chronological field word places the two disjoint spin hoppings of each spatial edge consecutively. They commute, and their sum is the complete Weyl edge commuting with physical $\eta^+$. The complete contact layer satisfies $D_6T = TD_2 + 2T$, where $T = (\eta^+)^2$. Thus every ideal Strang step has the exact identity

$$S_6(h)T = e^{-2ih}TS_2(h), \qquad S_6(h)|R\rangle = e^{-3ih}|R\rangle.$$

The complete protected repairs act identically on these ideal states. This is a property of the grouped ideal word, not an exact projection identity for its rounded native implementation.

The actual Hubbard-frame singlet chart has dimension36, retaining all35 $\eta = 3$ singlets and the $\eta = 4$ parent. If $U$ is the spatial forward half-hop matrix and $W = \mathrm{Sym}^2 U$ on normalized unordered pairs, the step is $W^T \exp(-ihD)W$. Transpose reverses the hopping order because every individual Hubbard hopping matrix is symmetric. Integer generator checks bind this chart to every original edge and the contact layer. Outward interval square roots supply the distinct-site normalization.

Use the common step $h = 39/2^{19}$. Time $39j$ takes $j2^{19}$ steps, with half before C repair and half before E repair. At time273 this is 3 670 016 steps. Their 38 755 368 960 field pulses are represented by the repeated source word; they are not individually executed. Instead, outward160-bit step entries are rounded to128-bit dyadics and powered by exact Gaussian-integer matrix multiplication. For approximate unitaries with errors $a, b$, one rounded product has error at most $a+b+ab+36\,2^{-128}$. The longest power needs24 products and has operator error $249561087/2^{128} < 7.34 \times 10^{-31}$.

Independent literal-source templates bound the complete 10 560-pulse step error by $1.323294 \times 10^{-12}$. Telescoping includes all native first-export failures, and the full-Fock splitting error is $(629/256)t^3/n^2$. With preparation, pointer operations, complete repairs, certified arithmetic and

exact binary64 probability-conversion error, every setting has systematic probability error below $8.568121 \times 10^{-6} < 1/65536$. Individual long-word failure weights are not materialized; their record addresses and complete-output norm bound are retained. No survival normalization is used.

The emulator draws counts from a binary64 proxy to the certified powered ideal-word probabilities, with PCG64 seed20261102. These are surrogate counts, not exact samples of the unmaterialized native failure instrument. There are $9\,2^{29}$ independent reset trials at each of sixteen settings, or77 309 411 328 overall. Zero is contrasted with all other complete outcomes. The stated independent binomial service gives simultaneous sampling error 1/32768 except with probability below $32(3/8)^9 < .01$. The same conditional recovery theorem applies to genuine native counts under that sampling event and the proved native systematic bounds.

Only the source frequency intervals enter the inverse. Rational Lagrange polynomials give a96-bit matrix $B$ for $V_{nj} = e^{-i39n\omega_j}$; outward arithmetic certifies $\|B\|_\infty < 2$ and $\|BV - I\|_\infty < 10^{-18}$. Counts determine $\widehat{y}$ and $\widehat{w} = \mathrm{Re}(B\widehat{y})$. Since the true source weights form a probability vector, the conditional error is bounded by

$$\|\widehat{w} - w\|_\infty \le \|B\|_\infty \frac{16}{5}(\varepsilon_{\text{systematic}} + \varepsilon_{\text{sampling}}) + \|BV - I\|_\infty < 0.000258393828.$$

In ascending frequency order the recovered weights are

$$(.002457894673,\ .005641397395,\ .005890881572,\ .007131698528,$$
$$.004528321839,\ .421578515341,\ .415796456292,\ .136974834359).$$

Every resulting interval is strictly positive and contains the independent exact residue, checked after inference. This closes weight resolution on the given support in the certified emulator. It does not turn $E_H - 3$ into an absolute rest mass, identify an unknown support, or establish physical particle species or native hardware execution.

**Comparing the actual interacting regional clocks.** The complete four-mode Weyl/contact Gibbs source supplies a useful finite alternative to assuming regional modular covariance. Let $W_S(s) = \exp(isH)$ be its state modular unitary at $\beta = 1$, and let $W_A(s) = \rho_A^{-is}$ be the embedded unitary of an actual CAR-reduced reference, with fixed scalar gauges understood. The relative operations

$$C_{AB}(s) = W_A(s)W_B(s)^*, \qquad C_{AB}(s)C_{BD}(s) = C_{AD}(s)$$

compare these clocks on every input and untouched reference. They obey the time-cocycle identity $C_{AS}(s+t) = C_{AS}(s)W_S(s)C_{AS}(t)W_S(s)^*$. The existing source words implement each global-to-regional correction at $s = 1/2$ with 2,826 pulses and ideal operator splitting bound 11/32768; the regional-to-regional comparison needs twenty pulses. All sixteen Fock columns and first export failures remain present. Such corrections are active: the first changes the common Gibbs reference by half trace distance about 0.00444. They do not assert that both regional flows preserve the same global state.

**The stronger identification fails on this reference.** For a linear modular intertwiner, an eigenoperator coefficient satisfies $(\nu_j - \omega_i)L_{ji} = 0$. The actual dimer at speed 1/8, contact $U = 1$, $\beta = 1$ and $\mu = 0$ has three positive regional frequencies, approximately 0.0001258906, 0.9988218345 and 0.9989477251. An exact source characteristic polynomial and nine partial-trace moment identities determine its reduced spectral weights. Rational bounds on the square root, exponential and logarithm then separate all three frequencies from every global energy difference by more than $10^{-4}$, in all 243 comparisons. Consequently every same-parameter linear intertwiner has range in

the six-dimensional regional modular centralizer and cannot transmit a moving local clock hand. The group and coefficient implications are Lean checked; the concrete spectral and interval attachment is checked separately by exact arithmetic. This excludes a strong identification on one finite reference, while the explicit relative operations provide a functioning comparison. The supplied action, coherent services, reference preparation and dimensionless parameter remain visible inputs. No physical duration, common energy calibration, geometric modular flow or Einstein premise is discharged.

**Controlled regional modular energy.** The same complete-Fock Weyl/contact action now supplies a quantitative regional modular attachment on its four-site open plaquette. All 256 field states and all sixteen states of the two-site region are retained. For the normalized signed partial trace $E$, put $H_c = H - I$ and $X(\beta) = E(e^{-\beta H_c})$. Exact Gaussian-integer and rational arithmetic constructs the first eight coefficients of $-\log X$. In particular, with $M_j = E(H_c^j)$,

$$C_1 = M_1, \qquad C_2 = \tfrac{1}{2}(M_1^2 - M_2), \qquad C_3 = \tfrac{1}{6}M_3 - \tfrac{1}{4}(M_1M_2 + M_2M_1) + \tfrac{1}{3}M_1^3.$$

For $0 \le x = \beta B < \log 2$, where $B \ge \|H_c\|$, the positive coefficient majorant $f(x) = -\log(2 - e^x) = \sum a_j x^j$ gives

$$\left\| -\log X - \sum_{j=1}^{p} \beta^j C_j \right\| \le f(x) - \sum_{j=1}^{p} a_j x^j.$$

This follows from the norm-convergent noncommutative logarithm series, without replacing the reduced reference by a local Gibbs ansatz. At the declared $\beta = 1/32$ and exact bound $B = 3$, an outward rational evaluation gives eighth-order error below $1.917 \times 10^{-9}$. Hence every Hermitian trace-zero regional variation obeys the entropy-tangent error bound $1.917 \times 10^{-9}\|\delta\|_1$. The normalization scalar in $K_A = -\log X + \log \operatorname{Tr}_A X$ cancels only on that tangent space.

The first nonscalar boundary correction is cubic and does not commute with the bare regional energy. Removing the contact interaction removes its interaction-dependent terms; removing the crossing bonds removes all nonscalar corrections. A full-versus-empty regional matrix-element witness and the same rational tail prove distance greater than $2.7 \times 10^{-8}$ from $\beta C_1 + aI$ for every scalar $a$ on this instance. This supplies both a corrected finite modular generator and a certified boundary to its bare-energy identification. The source preparation below supplies this plaquette reference to a separate controlled error. The action, temperature and lattice remain declared; the norm bound can grow with volume. Geometric cap weights, null stress, fixed-volume entropy–area stationarity, a physical scaling limit and positive Newton normalization remain additional obligations.

**A separately prepared interacting Gibbs reference.** On the complete 256-state plaquette at $\beta = 1/32$, the degree-eight Gibbs ensemble retains all 256 preparation records and every export abort. Independent outward arithmetic certifies its literal-source preparation error below $1.700 \times 10^{-12}$ on the initialized input. The actual regional success logarithm, including the exact normalization scalar, is within $2 \times 10^{-9}$ of the preceding $P_8$ comparison. The all-input continuation has a separate source and splitting bound. The proof in `model/certified_reference/README.md` supplies finite Gibbs preparation; it does not assert invariance under the complete repair. The following stationary reference has a different modular generator.

**The actual stationary source weight.** One identical literal native $Y(-\pi/3)$ request and complete dephasing export per occupation mode prepare $\sigma_M = \operatorname{diag}(a, b)^{\otimes M} = a^M e^{\gamma N}$, where

$a = |U_{00}|^2$, $b = |U_{10}|^2$ for the actual represented source unitary and $\gamma = \log(b/a)$. The exact one-mode error is below $1.894 \times 10^{-16}$, proving $a, b > 0$ and $b > a$; abort blocks remain separate. Number conservation and complete unital repair preserve this weight exactly. Tracing complete outside outputs before regional compression gives a $k$-mode logarithm error at most $k\delta/(1/4 - \delta)$, with $\delta = (2 + \epsilon)\epsilon$. This charge clock is trivial within fixed number. On the earlier six-site $N = 4, S = 1$ active sector, every common stationary modular generator stays at distance at least $|\beta|\sqrt{13/48}$ from $\beta K$, even allowing scalar and protected-spin terms. Thus the stationary reference and interacting Gibbs energy remain distinct. The exact source and reference proofs are in `model/continuing_reference/README.md`; the following construction closes its previously restricted scheduler premise.

**A stationary finite controller for native feedback.** The reference can now continue under an explicit record-adaptive scheduler. Use two classical working bits $(c, d)$; $c = 0$ selects the complete native C repair on seam $(0, 1)$, and $c = 1$ the complete E repair on $(0, 5)$. Append every raw record $r$, including zero slots and the outside outcome $Q = I - P$, and update

$$(c, d) \longmapsto (d \,\mathrm{XOR}\, f(r), c), \qquad f(r) = \text{coin XOR (raw mod 2)}, \quad f(Q) = 0.$$

At output $(u, v)$ every incoming term uses the same instrument $v$. Completeness and unitality of the unchanged repair therefore give

$$\sum_r L_r^{(v)} \frac{\sigma_M}{4} L_r^{(v)*} = \frac{\sigma_M}{4}, \qquad \Omega = \sigma_M \otimes I_4/4, \quad \sigma_M = \mathrm{diag}(a, b)^{\otimes M}.$$

Here $a, b$ are the actual literal-source weights, so the successful block still has trace $(a + b)^M$. All initial export-failure sectors are retained and frozen, emitting a bypass record. Separate unitary layers selected by the current working memory and generated by the original number-conserving field Hamiltonian also preserve this marginal. The controller is a declared reversible classical law on $\mathbb{C}^4$, not a claimed quantum pulse circuit.

Two unchanged native events on an explicitly initialized binary spectator, occupied at ports $2, 3$ and repaired on $(0, 1)$, leave it fixed and supply two exact fair coins. Their retained records seed $I_4/4$ without an angle-error assumption. This uses the supplied simulator basis initialization; no hidden interacting-field preparation of that spectator is asserted. The working memory remains two bits, while the archive grows by at most five bits per tick, besides the two seed records and all $6^M$ source readouts. Stationarity traces these records; it never normalizes the successful field block or declares the growing archive stationary. The modular generator remains $-\gamma N - M \log a + \log 4$, $\gamma = \log(b/a)$, and every complete step intertwines the charge flow $e^{it\gamma N}$. No physical energy-temperature identification follows. Earlier local literal-source bounds remain uniform in volume and ideal adaptive history length; implemented per-step errors must still be added.

**A complete finite mixing certificate and its boundary.** On the invariant three-position highest-spin repair corner, with no joined Hamiltonian step inserted, the field/controller algebra is $\bigoplus_4 M_3$. Each E outcome has rank at most one. Exact integer enumeration finds at least one eight-tick history encountering E for every one of the $12^2$ input/output population pairs, with common probability denominator $2^8$. The corresponding rank-one Kraus histories prove

$$\Phi^8 = \frac{3}{64}\mathcal{D} + \frac{61}{64}\mathcal{R}, \qquad \mathcal{D}(X) = \mathrm{Tr}(X) I_{12}/12,$$

where $\mathcal{R}$ is completely positive, trace preserving and unital. Thus $I_{12}/12$ is the unique stationary density on this CQ algebra, and the archive-traced error contracts by $(61/64)^{\lfloor n/8 \rfloor}$, including arbitrary untouched quantum references and all field coherences. This is neither whole-Fock mixing nor mixing with interleaved Hamiltonian evolution. Indeed the full retained history recovers the initial corner basis state and memory. As a negative control, the recordwise reversible one-bit policy $c \mapsto c \,\mathrm{XOR}\, f(r)$ sends the uniform six-state input to $(3, 1, 2, 1, 3, 2)/12$, a trace distance $1/6$ from uniform. The two-register incoming-row balance is a substantive premise, not a consequence of memory reversibility alone.

**Exact energy Ward cells under coarse-graining.** The same complete-Fock Weyl/contact action supplies a conserved energy column on its standard three-dimensional lattice. With the declared half-bond convention $H = \sum_x h_x$, its range-two currents are $J_{xy} = i[h_x, h_y]$. Distributing each current equally over all shortest nearest-neighbour paths preserves its incidence $\delta_x - \delta_y$: intermediate vertices cancel on every path. The resulting oriented face current $F_{x,i}$ therefore satisfies $\dot{h}_x + \sum_i (F_{x,i} - F_{x-e_i,i}) = 0$ exactly. The rational Clifford certificate checks the actual $3 \times 3 \times 3$ action at $c = 1/8, U = 1$: all 351 site pairs, 81 nonzero currents, 54 faces and 27 local equations, on the complete 54-mode algebra. Its coverage of all $2^{54}$ Fock states is a symbolic operator identity, not execution of that many amplitudes. Independent complete small-Fock matrices attach all three hopping axes and the interaction to the existing source action.

For a finite time cell, the exact flux is $I_{x,i}(n) = \int_{t_n}^{t_{n+1}} F_{x,i}(t)\, dt$. Hence

$$h_x(t_{n+1}) - h_x(t_n) + \sum_{i=1}^{3} (I_{x,i}(n) - I_{x-e_i,i}(n)) = 0.$$

For any normalized input and untouched reference, setting $T_{00}(n, x) = \langle h_x(t_n) \rangle$ and $T_{i0}(n, x) = \langle I_{x-e_i,i}(n) \rangle$ supplies the $j = 0$ forward-difference Ward column of the finite tensor interface. These are cell energy and integrated flux, not pointwise continuum stress. Summing spatial blocks cancels internal faces, so $D_{\mathrm{coarse}} P_{\mathrm{faces}} = P_{\mathrm{sites}} D_{\mathrm{fine}}$. Successive block maps compose exactly; successive time cells telescope. All 27 rectangular width choices, including ragged blocks, pass the exact source certificate. Generic finite-sum divergence, continuity, block composition and time telescoping are also Lean-checked; the concrete CAR source binding remains the exact rational certificate.

The existing local source bounds control the approximation uniformly in ambient volume. With $A = U + 3c$, one has $\|h_x\| \le A$ and $\|F_{x,i}\| \le B = 14A^2$: a face receives absolute shortest-path weight at most seven. Its support fits a radius-three cube. If $L_h, L_F$ are the existing normalized locality tails and $\epsilon$ is the patch unitary error, an endpoint energy error is at most $A \min(2, L_h + 2\epsilon)$ and an integrated-face error at most $tB \min(2, L_F + 2\epsilon)$. Replacing the integral by $m$ midpoint samples adds the explicit bound $t^2 B\, 343(U + 6c)/(2m)$; endpoint sampling is not exact. A cell residual is bounded by two endpoint errors plus six face errors, with supplied readout errors added separately. The receipt evaluates these resource bounds without claiming to execute the large sample count. Input preparation, hardware errors and controller/failure-space energy remain separate. This closes a finite same-action energy-column attachment, not the three momentum columns, a symmetric Lorentz tensor, geometric modular weights, area stationarity, a continuum scaling theorem or the full Einstein premise register.

**Local momentum balance retains its force.** On the open Weyl/contact box, the three local central-difference momenta obey exact balances with kinetic face stress and quartic contact force. All spatial blocks and time cells retain that force. On the $3^3$ box at $U = 1$ its total-force Gram matrix is $(9/4)I_3$, so an internal closed divergence cannot absorb it. Conserved number-preserving

quadratic charges of the connected action reduce to number and hidden spin. These exact results in `model/lattice_momentum_balance/README.md` concern local quadratic candidates. The following periodic construction supplies conserved global momenta with a separate locality boundary.

**Global periodic momentum and the local question.** On a declared periodic box, graded translations $T_j$ commute with the full interacting action. Their principal Hermitian logarithms $P_j$ are exactly conserved. They differ from additive one-body momentum by integer many-body winding. On odd-prime rings every number-preserving translation logarithm has nonzero one-particle spin-trace kernels at all nonzero separations. For the principal branch, every range-$R$ density sum obeys $\|P_j - A_R\| \geq 1/(R+1)$ when $L > 2(R+1)$. These full-space statements, proved in `model/periodic_momentum/README.md`, leave controlled restricted state domains available. The following bounds quantify one such domain and retain its actual interaction leakage.

**A collision-suppressed local Ward bound.** The same periodic Weyl/contact action supplies a controlled local approximation beyond its nonlocal conserved momentum. Let $B$ be a spin-complete set of spatial Fourier modes, $f = |B|/V$, and let $\Omega$ restrict the initial state to that band and $N \leq N_{\max}$. Write $D = \sum_x n_{x\uparrow} n_{x\downarrow}$ and $d = \lfloor N_{\max}/2 \rfloor$. Band annihilators have norm $\sqrt{f}$, so $\Omega D \Omega \leq f \min(N_\uparrow, N_\downarrow) \leq fd$. Since $D^2 \leq dD$ on this number sector,

$$\|UD\Omega\| \leq Ud\sqrt{f}, \qquad \|(I-\Omega)e^{-itH}\Omega\| \leq \min\{1, |t|Ud\sqrt{f}\}.$$

The second bound uses free invariance of $\Omega$ and Duhamel's formula; the full interacting action is never projected. For two particles, the exact contact principal angles are $\lambda_K = |B \cap (K-B)|/V$, giving escape norm squared $\max_K \lambda_K(1-\lambda_K)$ before multiplication by $U^2$. Thus projecting the outgoing contact back into $B$ would miss its leading square-root leakage.

Let $s_j(x)$ be the local sine-momentum density. Write $\Pi_{xy,j}$ for the antisymmetric kinetic stress across a nearest-neighbor face. The exact identity is $i[H, s_j(x)] + \sum_y \Pi_{xy,j} = F_j(x)$, with $F_j(x) = i[UD, s_j(x)]$. Normal ordering its quartic contact terms gives $\|F_j(x)\| \leq U$ and $\|F_j(x)\Omega\| \leq 4Uf$. For preparation at zero and $0 \leq t_0 \leq t_1$, the actual integrated Ward residual therefore obeys

$$\left\| \left[ s_j(x,t_1) - s_j(x,t_0) + \sum_y \int_{t_0}^{t_1} \Pi_{xy,j}(t)\, dt \right] \Omega \right\|$$
$$\leq \min\left\{ U(t_1-t_0),\ 4Uf(t_1-t_0) + \frac{U^2 d\sqrt{f}}{2}(t_1^2 - t_0^2) \right\}.$$

The preparation clock is not restarted in later cells. Spatial blocks cancel their internal faces and sum the site bounds; time cells telescope. The complete 54-mode $3^3$ source certificate checks all 81 local equations, 81 kinetic faces per column and 27 rectangular coarse partitions per column. Finite-sum transport of the errors is also Lean-checked. This is symbolic full-Fock coverage, not execution of all its amplitudes.

The two-particle total sine momentum has multiplier $2\sin(K_j/2)\cos(k_j - K_j/2)$. The uniform contact vector gives the sharp complete-fiber force norm $\|i[UD, S_j]\| = \sqrt{2}U|\sin(K_j/2)|$. If initially $|k_j| \leq \kappa$ and $2\kappa < \pi$, total momentum stays within $|K_j| \leq 2\kappa$ even after relative-momentum scattering. Combining this conserved domain with the sine remainder gives

$$\|(P_j - S_j)e^{-itH}\Omega\| \leq \frac{\kappa^3}{3} + 4\kappa \min\{1, |t|U\sqrt{f}\}.$$

For a $64^3$ counting example with $B = \{-1,0,1\}^3$, $N_{\max} = 2$, $U = 1$ and $t = 1/8$, the certified leakage is below 0.00127, the local Ward defect below 0.000132, and the momentum error after

division by $\kappa$ below 0.0083. The separate eight-site exact-Hamiltonian diagnostic retains a nonzero unforced local defect, so the contact force has not been discarded. These are fixed-number dilute bounds. The critical band-edge scaling $f \sim a^3$, $U \sim a^{-2}$ instead makes $U\sqrt{f} \sim a^{-1/2}$, for which this estimate supplies no vanishing leakage. A finite-density or critically scaled local stress limit, Lorentz symmetry, physical units and the remaining Einstein premises remain separate. Proofs, finite domains and replay are in `model/dilute_momentum_ward/README.md`.

**Interacting lattice scattering and its zero threshold.** The same Weyl/contact action now admits an on-shell two-fermion scattering theorem. On the physical exchange subspace of $L^2(\mathbb{T}^3, \mathbb{C}^4)$ at total momentum $K$, its free multiplication operator $H_{0,K}$ is bounded self-adjoint and the contact perturbation is $U|d\rangle\langle d|$, with $\|d\| = 1$. The difference is trace class with norm $U$. The Kato–Rosenblum theorem therefore gives complete wave operators on the absolutely continuous subspaces and a unitary scattering operator.[5] With $g_+(E) = a(E) - i\pi\rho_0(E)$, its contact-channel eigenvalue is

$$S_c(E) = \frac{1 - Ua(E) - i\pi U\rho_0(E)}{1 - Ua(E) + i\pi U\rho_0(E)}.$$

The orthogonal reducing complement of the contact-generated subspace has identity scattering. Point spectrum, including the isolated pair and the free dark zero modes, remains outside the incoming scattering domain. Completeness here concerns two-body absolutely continuous scattering; it is not many-body or chiral-continuum completeness.

At $K = 0$, the singlet resolvent is $z/(z^2 - 4c^2|\sin k|^2)$. Coarea with all eight sine-map preimages gives, for $0 < |E| < 2c$,

$$\rho_0(E) = \frac{E^2}{4\pi^2 c^3} \sum_{n \geq 0} A_n \left(\frac{E}{2c}\right)^{2n}, \qquad A_0 = 1, \quad A_1 = \tfrac{1}{2}, \quad A_2 = \tfrac{11}{40}, \quad 0 < A_n \leq 1.$$

The coefficients are exact sphere moments of the binomial Jacobian. After degree $M$, the dimensionless remainder is at most $r^{2M+2}/(1 - r^2)$, $r = |E|/(2c)$; rational arithmetic and a Machin bound for $\pi$ give certified density intervals. Local analyticity and evenness give $g_+(E) \to 0$ at zero. Hence $\rho_U(E) = \rho_0(E)/|1 - Ug_+(E)|^2 \sim E^2/(4\pi^2 c^3)$. This supplies nontrivial scattering and a gapless contact spectral support on one interacting lattice action while retaining its isolated pair. Zero model energy is not an identified physical vacuum threshold, so no massless physical particle is assigned. The source-bound proof and imported scattering theorems are recorded in `model/contact_scattering/ONSHELL.md`.

**Interacting edge scattering and a moving bound pair.** The same two-body source family now has a controlled moving bound band. Set $c_a = \sqrt{3}/a^2$ and retain the repulsive coupling

$$U_a^{-1} = \frac{a^2 W_0}{2} - \frac{a^3\gamma}{\pi}, \qquad W_0 = \left\langle \frac{1}{\sum_j \cos^2 k_j} \right\rangle, \qquad \gamma > 0.$$

The normalized full-torus integral satisfies $2/3 \leq W_0 < 5$; $a\gamma \leq 1$ makes the coupling positive. The single global comparison is $J_a = 6/a^2 - H_a$, explicitly shifting energy and reversing time. Neither $\gamma$ nor this clock/energy choice is inferred from physical data.

[5] The imported trace-class and finite-rank scattering results are reviewed in https://arxiv.org/abs/1711.00528 and https://arxiv.org/abs/0902.3568; the attachment here verifies their premises for the stated Weyl/contact exchange fiber.

At zero total momentum the threshold estimate is

$$\left|\frac{g_0(6/a^2+z)-a^2W_0/2}{a^3}+\frac{\sqrt{z}}{\pi}\right|\le 19a|z|$$

for real $|a^2z|\le 1/8$, using the upper boundary square root; the exterior constant improves to 11. Coarea for all eight valleys, an inner ball of radius $1/2$ and its complementary measure give the respective Green remainders $18\eta$ and $36|\eta|$ before rescaling. Thus, for $0\le s\le S$, $a^2S\le 1/8$ and $836aS\le 7\gamma$,

$$S_J(s)\longrightarrow\frac{\gamma-i\sqrt{s}}{\gamma+i\sqrt{s}},\qquad \sup_{[0,S]}|S_J-S_{J,\mathrm{lim}}|\le\frac{1672aS}{7\gamma}.$$

This is the earlier nonidentity scalar contact-channel limit; its orthogonal free channels remain present. Lattice contact renormalization is an established mechanism, not a new particle identification; compare Koerber–Berkowitz–Luu (2019) and Winkler et al. (2006).

**Nonzero total momentum has a controlled pole.** Write $K=aP$, $q=k-K/2$, $d_j=\sin q_j\cos(K_j/2)$, $e_j=\cos q_j\sin(K_j/2)$ and $R^2=\sum_j\cos^2(K_j/2)$. For $|K_j|\le\pi/2$, concavity gives the exact complete free edge $B_a(K)=2c_aR$, attained at all eight $q_j=\pm\pi/2$. The four-spin singlet resolvent is

$$G_K(q,E)=\frac{E(E^2-4c_a^2e^2)}{E^4-4c_a^2(d^2+e^2)E^2+16c_a^4(d\cdot e)^2}.$$

At $E=B_a(K)+z$, put $t=E/(2c_a)$, $\eta=t^2-R^2$ and $V=\sum_j\cos K_j\cos^2 q_j$. Equivalently,

$$g_K(E)=\frac{t}{2c_a}\left\langle\frac{1-e^2/t^2}{V+\eta+(d\cdot e)^2/t^2}\right\rangle.$$

Let $\kappa=\max|K_j|\le 1/2$ and $h=a^2z\le 1/8$, with $z\ge 0$. Writing $Q=\sum_j\cos^2 q_j$, the denominator lies between $(1-\kappa^2/2)(Q+h+h^2/12)$ and $Q+h+h^2/12$; $e^2\le\kappa^2Q/4$ and $(d\cdot e)^2/t^2\le e^2$. The complete-torus integral and prefactor therefore give

$$|g_K(B_a(K)+z)-g_0(6/a^2+z)|\le 3a^2\kappa^2.$$

Combining this with the exterior threshold estimate proves

$$\left|\frac{g_K(B_a(K)+z)-a^2W_0/2}{a^3}+\frac{\sqrt{z}}{\pi}\right|\le a(11z+3P_{\max}^2),\qquad |P_j|\le P_{\max}.$$

Under $256a(\gamma^2+P_{\max}^2)\le\gamma$, the pole denominator has opposite signs at $z=\gamma^2/4$ and $z=4\gamma^2$. Strict monotonicity of the exterior Stieltjes transform supplies exactly one upper pole $E_a(K)=B_a(K)+\delta_a(K)$, uniformly on that momentum cube:

$$\frac{\gamma^2}{4}\le\delta_a(K)\le 4\gamma^2,\qquad |\delta_a(K)-\gamma^2|\le\frac{22}{7}a\gamma(132\gamma^2+9P_{\max}^2).$$

This does not invoke the older, stronger sufficient condition $U>B_a$.

**An explicit bound-band wavepacket comparison.** The same global $J_a$ has bound energy $\epsilon_a(P) = 6/a^2 - B_a(aP) - \delta_a(aP)$. Rationalizing $B_a$ and bounding the sine remainder gives

$$\sup_{|P_j|\le P_{\max}} \left| \epsilon_a(P) - \left( \frac{|P|^2}{4} - \gamma^2 \right) \right| \le \mathcal{E}_a := \frac{a^2 P_{\max}^4}{4} + \frac{22}{7} a\gamma(132\gamma^2 + 9P_{\max}^2).$$

In particular, subtracting $B_a(K)$ separately at each momentum would incorrectly remove the center-of-mass kinetic term.

Let $R_K = (E_a(K) - H_{0,K})^{-1}$ and $\psi_{a,K} = R_K d/\langle d, R_K^2 d\rangle^{1/2}$. Its positive contact overlap fixes its phase, and the pole equation makes it a normalized, measurable, exchange-antisymmetric eigenvector. With consistent measures $dK/(2\pi)^3$ and $dP/(2\pi)^3$, the formula

$$(W_a f)(K,k) = a^{-3/2} f(K/a)\psi_{a,K}(k)$$

defines an isometry on packets supported in the stated cube. Its range reduces the two-body source action; the remaining source states are retained. If $h_{\text{pair}}$ multiplies by $|P|^2/4 - \gamma^2$, then for arbitrary untouched references,

$$\|e^{-itJ_a} W_a - W_a e^{-ith_{\text{pair}}}\| \le \min\{2, |t|\mathcal{E}_a\}.$$

This proves coherent moving-pair propagation with an explicit rational spacing schedule. The limiting inertial mass is twice that of a single free edge in these model coordinates; neither an SI mass nor a relativistic rest energy is identified. The exact bounds, independent four-spin and torus checks, and wrong-subtraction/time-orientation controls are retained in `model/moving_bound_pair/README.md`.

The result concerns one composite sector. Nonzero-$K$ on-shell scattering, a physical vacuum/detector map, many-pair interactions, relativistic massive/massless sectors and Einstein dynamics remain separate. The earlier fixed-zero $c = v/a$ result also remains valid: $\int_{-M}^{M} |S - I|^2 dE \le 1024M^2/(63c)$ for arbitrary real finite $U(a)$ and $M \le c/4$, so every fixed energy-normalized packet becomes transparent while tuned shrinking resonances can persist. Its scaling and energy comparison differ explicitly from the bound-band construction.

**Identified model energies and the physical particle boundary.** The model energies and inertial mass are identifiable. The cube has eight exactly characterized interacting energies, each reached by the local probe with positive weight. The moving pair has $\epsilon(P) = |P|^2/4 - \gamma^2$ for the reflected generator $J_a = 6/a^2 - H_a$, hence binding energy $\gamma^2$ and inertial mass 2 in those model units. If physical energy and length units $E_{\text{u}}, L_{\text{u}}$ are independently attached and $p_{\text{phys}} = \hbar P/L_{\text{u}}$, then $M = 2\hbar^2/(E_{\text{u}} L_{\text{u}}^2)$ and $E_{\text{bind}} = E_{\text{u}}\gamma^2$, with $E_{\text{u}} T_{\text{u}} = \hbar$ for the common clock. These conversion formulas do not select the units or fix an absolute rest-energy offset. Binding energy is not total rest mass. For the cube, the following rounded displays have exact rational energy and weight intervals in the receipt, with the original contact unit $U = 1$:

| $H$ energy | $E - 3$ | Probe weight |
|---:|---:|---:|
| 1.816498920 | −1.183501080 | 0.002465366 |
| 1.860537501 | −1.139462499 | 0.005644624 |
| 1.955137632 | −1.044862368 | 0.005881219 |
| 2.108692846 | −0.891307154 | 0.007137664 |
| 2.167639734 | −0.832360266 | 0.004528123 |
| 3.015861346 | +0.015861346 | 0.421577939 |
| 3.030769653 | +0.030769653 | 0.415789141 |
| 3.044862368 | +0.044862368 | 0.136975924 |

Weights refer to the normalized centered density probe; the phase-kicked target has 15/16 times these weights plus an elastic weight 1/16 at energy 3. The lowest displayed line is not the $N = 6$ ground: $-3\sqrt{3}/8 \le E_{0,N=6} \le -\sqrt{3}/8$, by the free one-particle bound and a polarized zero-contact variational state. These are finite energy levels and signed transitions, not eight rest masses. The physical species assignment count remains zero. On the cube the density probe, its phase-kicked reference and all spectral components remain in $N = 6$: the transition operator is number-neutral and even under $\Gamma = (-1)^N$. An isometric dictionary intertwining this grading cannot map that response to a single odd fermion, since an odd creation operator changes parity. This does not exclude other odd model excitations or a fermion–antifermion response; even parity alone does not establish Bose exchange or Lorentz spin. Hidden spin and model number have not been identified with physical spin and electric charge. The physical target conventions are the PDG electron and quark quantum numbers and relativistic kinematics.[6]

The supplied cube reference also fails the vacuum ground-state spectral condition for its own $H$. Its positive density-probe spectral measure has five negative and three positive frequencies $\omega = z/16 - 1$; all eight weights are strictly positive by the exact Krylov certificate. In particular $z_- = 8 - 2\sqrt{19}$ has weight $(1 - 4/\sqrt{19})/14 > 0$ and gives an $N = 6$ eigenenergy $5/2 - \sqrt{19}/8 < 2$, below the reference energy 3. A number-preserving unitary swapping these two states lowers the energy, so the reference is not passive for the full field unitary algebra. This is a mathematical witness, not an executed extraction protocol: the actual local phase preserves the mean energy. These are frequencies of a positive greater correlation, not the two-sided frequencies of a retarded commutator. A scalar or common number chemical potential leaves the gaps unchanged, and time reversal still leaves mixed signs.

The moving-pair limit $\epsilon(P) = |P|^2/4 - \gamma^2$ is a genuine controlled nonrelativistic band. For any fixed affine energy map $E = \alpha\epsilon + \beta$, $\alpha \neq 0$, and fixed invertible linear momentum map $p = LP$, the residual $E^2 - c^2|p|^2 - m^2c^4$ has nonzero quartic coefficient $\alpha^2/16$ along a coordinate line. It cannot vanish on an open momentum set. Thus these maps do not turn this limit into an exact massive or massless Lorentz shell. A low-momentum approximation remains possible: after matching constant and quadratic terms, the difference between $E_0 + Ar^2$ and $\sqrt{E_0^2 + 2E_0Ar^2}$ is $A^2r^4/(E_0 + Ar^2 + \sqrt{E_0^2 + 2E_0Ar^2})$, for rest energy $E_0 > 0$ and $A > 0$.

`model/particle_identification` binds these scoped tests and the remaining vacuum, gauge-charge, spin, clock, detector and mass-shell requirements. The earlier dimer electron mass was a calibration input, not an independent mass prediction; its retrospective equality comparison does not identify the new cube or pair modes. No mass fitting, new eligible physical comparison, or whole-program exclusion is added.

The preceding identity audit leaves a relativistic construction open. The following extension constructs a massless finite-particle limit on the growing lattice. Its carrier and reference remain distinct from the fixed protected cube, and its limiting interaction vanishes.

**A controlled massless relativistic sector of the unchanged action.** The same Weyl/contact family admits a simultaneous eight-valley continuum comparison on every fixed finite-particle momentum domain. Set

$$H_a = d\Gamma(h_a) + U(a)D, \qquad h_a(k) = -\frac{v}{a}\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad D = \sum_x n_{x\uparrow}n_{x\downarrow},$$

[6] https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-quark-model.pdf; https://pdg.lbl.gov/2025/reviews/rpp2025-rev-kinematics.pdf.

with fixed $v > 0$ and finite $U(a) \geq 0$. This declares the scaling $c = v/a$; it adds no mass term, energy subtraction or time reversal. For $\nu = \pi b$, $b \in \{0,1\}^3$, let $C = [-\Lambda, \Lambda]^3$ and $a\Lambda < \pi/2$. With both Fourier measures normalized by $(2\pi)^{-3}$,

$$(W_a f)(\nu + ap) = a^{-3/2} f_b(p), \qquad W_a : \bigoplus_b L^2(C, \mathbb{C}^2) \longrightarrow L^2(\mathbb{T}^3, \mathbb{C}^2)$$

is an isometry, with zero output outside the eight disjoint patches. There is no additional factor $1/\sqrt{8}$. Its exterior powers give an isometry $W_a^{\leq N}$ on all particle numbers through a fixed $N$, with the vacuum and every valley retained.

Writing $s_j = (-1)^{b_j}$ and $\chi_b = -s_1 s_2 s_3$, the limiting blocks are

$$h_b(p) = -v \sum_j s_j \sigma_j p_j, \qquad E_\pm(p) = \pm v|p|, \qquad E^2 - v^2|p|^2 = 0.$$

All eight continuum masses are zero. Four blocks have each chirality. The proper frame $R_b = -\chi_b \operatorname{diag}(s)$ has a constant $SU(2)$ lift and spin $S_{b,j} = R_{b,jj}\sigma_j/2$, so $h_b = 2\chi_b v S_b \cdot p$. The positive-energy helicity is $\chi_b/2$; the negative-energy helicity is $-\chi_b/2$. At $p = 0$ the two spin states are degenerate and helicity is undefined. In these frames an opposite pair has $\alpha_j = \operatorname{diag}(-\sigma_j, \sigma_j)$. With off-diagonal identity $\beta$, the matrices $\gamma = (\beta, \beta\alpha_1, \beta\alpha_2, \beta\alpha_3)$ satisfy the Clifford relations for signature $(1,-1,-1,-1)$, checked exactly. Thus the limiting differential equation is massless Weyl/Dirac in coordinate $x^0 = vt$, not merely a fitted linear dispersion.[7] The fixed cutoff and the lattice regulator are not Lorentz invariant; neither the valley pairing nor this spin frame selects physical species.

**The full interacting evolution has an explicit comparison bound.** The sine remainder gives, uniformly on $C$,

$$\|h_a(\nu + ap) - h_b(p)\| \leq \frac{va^2}{6} \sum_j |p_j|^3 \leq \frac{va^2\Lambda^3}{2}.$$

For $a\Lambda \leq 1$ and $p \neq 0$, both spectral-projector errors are at most $a^2\Lambda^2/6$. The squared-energy defect lies between $-v^2a^2 \sum_j p_j^4/3$ and zero. No projector at the degenerate node is assigned.

Let $\Omega_a$ project onto the initial Fock band and $d = \lfloor N/2 \rfloor$. On the infinite lattice the site-diagonal band weight is exactly $f_a = 8(a\Lambda/\pi)^3$. The positive countable partial sums defining $D$ are bounded by $d$, hence converge strongly and satisfy $D^2 \leq dD$. On band states each annihilation has norm at most $\sqrt{f_a}$ and preserves the band for the remaining particles. Summing its squared norms gives $\Omega_a D \Omega_a \leq f_a d$, separately on each up/down number sector, and therefore $\|D\Omega_a\| \leq d\sqrt{f_a}$. This proves the infinite-volume attachment of the finite dilute-band estimate without substituting a continuum fraction for a finite count.

Duhamel's formula now yields

$$\|e^{-itH_a} W_a^{\leq N} - W_a^{\leq N} e^{-itd\Gamma(h_W)}\| \leq \min\{2, |t|\epsilon_a\},$$
$$\epsilon_a = \frac{Nva^2\Lambda^3}{2} + |U(a)|d\sqrt{8(a\Lambda/\pi)^3}.$$

The left evolution is the full source Hamiltonian: no band projection is inserted and every outgoing contact state remains present. The estimate also holds with arbitrary untouched references; it

[7] For the standard Weyl equation and helicity convention, see D. Tong, *Quantum Field Theory*, Sections 4.4 and 4.7, https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S4.html.

bounds half trace distance and individual probability errors, capped by one. For fixed $N, \Lambda, v$ and bounded times it tends to zero whenever $U(a) = o(a^{-3/2})$. At $N \leq 1$ the contact is exactly zero for every $U(a)$. Thus the proved limit is free; the critical pair scaling $U \sim a^{-2}$ and finite-density limits are outside this conclusion. For $N = 2$, $\Lambda = v = U = t = 1$ and $a = 1/64$, the complete-vector bound is less than 0.001237 and the projector bound is $1/24576$.

The finite source check retains all 128 one-particle Fourier columns on a periodic $4^3$ cube. An eight-valley two-particle state there has exact $\|D\psi\|^2 = 1/8$ and $\|(1-\Omega)D\psi\|^2 = 7/64$, so the contact escape is real. Propagation on all 8,128 two-particle states at $t = 1/8$ gives a numerical vector difference about 0.0441641, below the rigorous $\sqrt{1/8}/8$. These matrix diagnostics and exact source identities do not constitute an additional native preparation or detector execution.

**Resulting energies and the physical boundary.** At $v = 1$, $a = 1/64$, every valley has the following energies; both signs are retained, and only their common positive magnitude is displayed. The receipt gives rational enclosures for the rounded lattice values.

| $p$ | lattice $\lvert E_a\rvert$ | continuum $\lvert E\rvert$ | continuum mass |
|---|---|---|---|
| $(0,0,0)$ | 0 | 0 | 0 |
| $(1,0,0)$ | 0.999959310393 | 1 | 0 |
| $(1,1,0)$ | 1.414156018578 | $\sqrt{2}$ | 0 |
| $(1,1,1)$ | 1.731980331101 | $\sqrt{3}$ | 0 |

These are original-generator model energies, without an SI calibration. The finite-spacing mass-shell defect is a regulator error, not a tachyonic mass. This growing-lattice branch is distinct from the protected eight-site cube: neither its repair invariance nor the identity of its eight response lines is transferred to the valleys. The empty Fock state is not a ground state: already at finite spacing, a one-particle mode at $(\pi/2, 0, 0)$ has energy $-c$, with zero contact for every $U$. This compares different number sectors; it is not a number-preserving extraction protocol from the empty sector. A filled vacuum, its interacting stability and a physical antiparticle dictionary remain unconstructed. A single $2\times2$ constant Hermitian mass anticommuting with all three kinetic Pauli matrices is exactly zero; this limited obstruction does not exclude valley coupling or many-body mass generation. The package `model/relativistic_sector_audit` therefore adds a controlled massless relativistic dynamical sector, not massive physical species, a photon assignment, an interacting relativistic vacuum theory or Einstein dynamics.

The preceding continuum construction does not itself execute a detector. The following finite periodic construction attaches model flavor and energy labels to a source-compiled occupation readout, and compares the resulting massless sector with observed particle data.

**A detector identifies the model's propagating Weyl flavors.** The massless sector now has a concrete model-particle readout. On the periodic $8^3$ lattice at $c = 1/8$, retain all 1,024 physical CAR modes of the original Weyl/contact action. For each $b \in \{0,1\}^3$, choose momentum labels $\ell_b = (4b_x, 4b_y, 4b_z + 1)$, with $k_b = \pi\ell_b/4$. For component eigenvalue $r = \pm1$ of $\sigma_z$, the normalized mode is

$$\psi_{b,r}(x,s) = \frac{e^{ik_b\cdot x}}{\sqrt{512}}\delta_{s,(1-r)/2}, \qquad H\psi_{b,r} = E_{b,r}\psi_{b,r}, \qquad E_{b,r} = -r(-1)^{b_z}\frac{\sqrt{2}}{16}.$$

The contact term is exactly zero on the entire one-particle sector for every supplied $U \geq 0$. These sixteen modes are orthonormal, checked in Gaussian integers with amplitudes $(A+\sqrt{2}B)/32$. Their rank-one projectors satisfy $H\Pi_{b,r} = E_{b,r}\Pi_{b,r}$; the remaining one-particle projector commutes with

$H$. Thus a label gives an actual source-generator energy, not a fitted interpretation of a peak. All modes have $N = 1$, odd CAR parity and fermionic exchange statistics from the supplied CAR algebra.

The operational species are eight Weyl flavors indexed by measured valley momentum. Their common energy magnitude is $\sqrt{2}/16 \simeq 0.0883883476483$ in original model units; it alone does not distinguish the flavors. With $\chi_b = -(-1)^{b_x+b_y+b_z}$, the positive-energy branch has cone helicity $\chi_b/2$ and the negative branch has the opposite helicity.

| flavor $b$ | momentum labels $\ell_b$ | positive-branch component | positive cone helicity |
|---|---|---|---|
| 000 | $(0, 0, 1)$ | 1 | $-1/2$ |
| 001 | $(0, 0, 5)$ | 0 | $+1/2$ |
| 010 | $(0, 4, 1)$ | 1 | $+1/2$ |
| 011 | $(0, 4, 5)$ | 0 | $-1/2$ |
| 100 | $(4, 0, 1)$ | 1 | $+1/2$ |
| 101 | $(4, 0, 5)$ | 0 | $-1/2$ |
| 110 | $(4, 4, 1)$ | 1 | $-1/2$ |
| 111 | $(4, 4, 5)$ | 0 | $+1/2$ |

Here helicity uses the explicit proper valley frame $R_b = -\chi_b \operatorname{diag}((-1)^{b_j})$; it is $(-1)^{b_x+b_y} r/2$ on this family. The raw detector measures momentum and CAR component, so bare $\sigma_z/2$ is not silently identified with this cone spin or with physical particle spin. The total directional current $J_z(k) = -c \cos k_z \, \sigma_z$ has eigenvalue $E_{b,r}$ here, with nonzero current $\pm\sqrt{2}/8192$ on every forward z bond. Current is stationary on these selected modes; $[H, J_z] \neq 0$ at generic momenta. The witness is propagating flux, even for a uniform-density plane wave.

**The Fourier readout is compiled with complete native errors.** The detector applies the fermionic extension of $F[k, x] = e^{-2\pi i k \cdot x/8}/\sqrt{512}$ separately to each component, then copies all 1,024 occupations to fresh pointers. Its output includes every binary occupation string, the vacuum, all other one-particle addresses, multiple occupations and every export abort. On the chosen sixteen effects it measures the energies above; the other 1,008 one-particle records remain present without that energy assignment. On higher-number inputs these same records measure mode occupations, not the total interacting energy.

An explicit three-axis FFT uses 4,608 two-mode butterflies and 768 bit-reversal transpositions. Each butterfly is a mode phase followed by the native hopping word with coefficient $-i$ and duration $\pi/4$. All nonadjacent operations use graded FSWAP routes, including their double-occupation sign. Exact arithmetic in $\mathbb{Z}[\zeta_8]/(\zeta_8^4 + 1)$ proves the 64 entries and 64 Gram sums of the one-axis transform; tensoring gives the full mode map. One transform requests 1,260,800 adjacent FSWAPs and 32,909,824 native pulses. Native excitation from the vacuum, inverse-transform preparation, forward-transform readout and all occupation pointers total 65,822,721 formal pulses. Original-$H$ propagation is a separate specified model-law service, not an uncounted synthesized field word.

The complete literal-source error for preparation and detection is

$$\epsilon_{\text{source}} < 8.31 \times 10^{-9}.$$

The estimate retains all first-export failures and is valid with untouched references. It does not assume that literal native maps are exactly Gaussian or number preserving. Unitary original-$H$ propagation on the full Fock space does not amplify preparation error. Therefore, at every model time, a prepared catalog member produces its correct native label with unconditional probability

at least $1 - \epsilon_{\text{source}}$; all wrong, outside, vacuum, multiple-occupation and abort outputs together have probability at most $\epsilon_{\text{source}}$.

The replay executes the ideal Gaussian comparison on all 1,024 one-particle rows and all sixteen inputs. At time eight its complete output-vector discrepancy from the exact spectral prediction is about $1.94 \times 10^{-15}$, separate from the rigorous native bound. A seeded 65,536-record illustration retains all sixteen label counts and every zero outside count for the exact uniform input mixture. These are model-simulated counts, not a literal $2^{1024}$-row native execution or laboratory data. No survival normalization is used.

**Mass readout and comparison with measured particles.** For the periodic family $L = 8 \cdot 2^m$, $a = 2^{-m}$ and $c = (1/8)/a$, the model period stays eight and the momentum offset stays $p_z = \pi/4$. The exact branch energies satisfy

$$E_{\pm}(a) = \pm \frac{\sin(a\pi/4)}{8a} \longrightarrow \pm \frac{\pi}{32}, \qquad m_{\text{continuum}} = 0.$$

This is a direct finite-torus consequence of the original spectrum. The finite-spacing mass-shell defect is a regulator correction, not a tachyonic mass. No SI scale is assigned.

A separate retrospective comparison uses fixed PDG 2025 targets, which never enter the source or detector calculation:

| physical target | measured mass in MeV/$c^2$ | present continuum mass |
|---|---|---|
| electron | $0.51099895000 \pm 0.00000000015$ | 0 |
| muon | $105.6583755 \pm 0.0000023$ | 0 |
| tau | $1776.93 \pm 0.09$ | 0 |

No common positive unit conversion maps zero to these positive masses; equal finite-momentum energies are not a rest-mass hierarchy.[8] The photon has spin one, so direct spin-preserving identification with these cone-helicity-1/2 modes also fails. Three such massless flavors give zero mass-squared differences, outside the nonzero ranges of the specified three-neutrino fit in PDG Table 14.7.[9] A single massless neutrino or a composite photon in another sector is not excluded. The model's Weyl flavors are now operationally identified; direct identifications with measured physical species remain zero. The empty reference is not a ground state, and the negative branch is retained rather than renamed as a physical antiparticle. Interacting vacuum stability, mass generation, physical charges and the empirical dictionary remain additional work.

The missing ground-reference and mass-generation steps admit further constructions with explicit domains. The original cube has a global interacting ground, and its one-dimensional restriction has an interaction-generated canonical charge gap. A specified refinement of that restriction also permits a controlled relativistic mass calculation. The following results retain the reference and coupling choices rather than combining these distinct domains into an assumed physical sector.

**A global interacting ground on the original cube.** The unchanged open $2 \times 2 \times 2$ Weyl/contact source, with $c = 1/8$ and $U = 1$, now admits a certified comparison across its entire 65 536-dimensional Fock space:

$$E_G = -\frac{\sqrt{3}}{4}, \qquad N_G = 4, \qquad \dim \ker(H - E_G) = 5.$$

[8] PDG 2025 listings: https://pdg.lbl.gov/2025/listings/rpp2025-list-electron.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-muon.pdf; https://pdg.lbl.gov/2025/listings/rpp2025-list-tau.pdf.

[9] The photon listing is https://pdg.lbl.gov/2025/listings/rpp2025-list-photon.pdf. The selected normal-ordering fit gives $\Delta m^2_{21} \in [6.92, 8.05] \times 10^{-5}\,\text{eV}^2/c^4$ and $\Delta m^2_{32} \in [2.388, 2.531] \times 10^{-3}\,\text{eV}^2/c^4$; these are reported $3\sigma$ fit ranges, not exact bounds: https://pdg.lbl.gov/2025/reviews/rpp2025-rev-neutrino-mixing.pdf.

The five states form one hidden-spin $S = 2$ multiplet. No chemical potential, energy recentering, mass term, or replacement Hamiltonian is introduced. This is a finite interacting ground reference, not a claim of a unique or Lorentz-invariant infinite-volume vacuum.

The existing onsite unitary frame sends the original hopping to the spin-independent signed cube adjacency $A/16$, where

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \quad BB^T = 3I_4, \quad K = \begin{pmatrix} \sqrt{3}I_4 \\ -B^T \end{pmatrix}, \quad K^*K = 6I_4.$$

Filling the four negative orbitals $K/\sqrt{6}$ with hidden spin up gives an explicit normalized state $G$. Its occupation coefficients belong to $\mathbb{Q}(\sqrt{3})$, it has no doublons, and the original physical-frame eigenvalue equation is checked exactly. With $\lambda = \sqrt{3}/16$, the $N = 4$ bound $H \geq -4\lambda$ is saturated precisely by negative-band states annihilated by every local pair annihilator. On the full 70-dimensional four-particle negative-band space, the stacked contact map has spin-weight block dimensions $1, 16, 36, 16, 1$ and ranks $0, 15, 35, 15, 0$. Nonzero integer minors modulo101 certify the rank lower bounds; five explicit spin-lowered vectors certify the matching upper bounds. This proves the complete five-dimensional ground kernel in that number sector.

For $N \leq 3$, positivity gives $H \geq -N\lambda > -4\lambda$. For $N \geq 9$, the occupancy bound $D \geq N - 8$ gives $H \geq -N\lambda + N - 8 > 0$. For $N = 5, 6, 7, 8$, hidden $SU(2)$ symmetry ensures that every eigenvalue occurs in the balanced spin weight. Those block dimensions are $1568, 3136, 3920, 4900$. A numerical factorization proposes an integer matrix $Q$; only the following subsequently checked integer identity enters the proof:

$$H = 2^{-40}QQ^T - \frac{27}{64}I + R, \qquad \|R\|_2 \leq \|R\|_\infty < \frac{1}{512}.$$

All integer product and row-sum overflow bounds are checked before evaluation. Since $QQ^T$ is positive semidefinite for every proposal, the argument requires no trusted floating eigenvalue, factorization error estimate, or nonzero-pivot premise. The exact comparison

$$\frac{\sqrt{3}}{4} > \frac{441}{1024} > \frac{217}{512}$$

then proves that every $N \neq 4$ lies more than $7/1024$ above $E_G$. This last bound concerns charged sectors; no numerical lower bound for neutral excitations inside $N = 4$ is asserted. Any density matrix supported on this finite ground space is passive for the stated $H$.

**Positive charged spectral data above this ground.** At the origin the onsite frame is the identity, so the probes below are the original physical CAR modes. Their exact local spectral data are

| probe or component | $N$ | weight | $\omega = E - E_G$ |
|---|---|---|---|
| $c_\uparrow G$ | 3 | 1/2 | $\lambda$ |
| $c_\uparrow^\dagger G$ | 5 | 1/2 | $\lambda$ |
| $c_\downarrow G$ | 3 | 0 | absent |
| $S = 5/2$ component of $c_\downarrow^\dagger G$ | 5 | 1/10 | $\lambda$ |
| $\eta = 5/2$ component of $c_\downarrow^\dagger G$ | 5 | 1/10 | $1 + \lambda$ |

Thus the resolved model gaps are approximately 0.108253175473 and 1.108253175473 in source energy units. They are finite-volume gaps, not relativistic rest masses. The remaining down-addition weight is $4/5$, with strictly positive spectral support and nonzero variance.

More explicitly, set $v = c_{\downarrow}^{\dagger} G$, with $\|v\| = 1$. The exact vectors

$$v_s = \frac{1}{5} S^- S^+ v, \qquad v_\eta = \frac{1}{5} \eta^+ \eta^- v, \qquad v_r = v - v_s - v_\eta$$

are mutually orthogonal and have squared norms $1/10, 1/10, 4/5$. The first two obey the original Hamiltonian eigenvalue equations at $-3\lambda$ and $1 - 3\lambda$. Consequently the complete greater addition measure satisfies

$$\mu_\downarrow = \frac{1}{10}\delta_\lambda + \frac{1}{10}\delta_{1+\lambda} + \frac{4}{5}\mu_r, \qquad \operatorname{supp} \mu_r \subset (7/1024, \infty).$$

The displayed $1/10$ values are exact symmetry-resolved contributions; without further resolution of $\mu_r$, coincident residual atoms are not excluded. All moments below are checked over $\mathbb{Q}(\sqrt{3})$:

$$\mathbb{E}\omega = \frac{1}{2}, \qquad \mathbb{E}\omega^2 = \frac{131}{256}, \qquad \operatorname{Var}(\omega) = \frac{67}{256},$$

$$\mathbb{E}_r\omega = \frac{1}{2} - \frac{\sqrt{3}}{64}, \qquad \mathbb{E}_r\omega^2 = \frac{131}{256} - \frac{\sqrt{3}}{64}, \qquad \operatorname{Var}_r(\omega) = \frac{1069}{4096} > 0.$$

The contact expectation in $v$ is $1/2$; deleting that interaction changes its mean gap from $1/2$ to zero. This channel therefore probes the actual contact interaction. Addition and removal carry model number changes $+1$ and $-1$ and odd CAR grading relative to $G$; electric charge and physical Lorentz spin do not follow from these labels.

**Attachment to the declared repair process.** The existing four-dimer repair completion has24 active columns, all entirely in $N = 6$. Its complete130-outcome dictionary therefore acts on $N = 3 \oplus N = 4 \oplus N = 5$ as zero for every active Kraus outcome and as identity for the outside projector. This6748-dimensional statement retains cross-number coherences and arbitrary reference systems. Since the original $H$ preserves number, it also survives every finite interleaving with the ordinary field evolution. This is passive outside-service protection, not active correction or autonomous preparation of the ground.

The new finite ground removes the negative-frequency objection that applied to the earlier excited reference. It does not retroactively turn that earlier preparation into ground-state preparation. A native source and detector for $G$, the unresolved charged spectrum, a controlled growing-volume interacting vacuum, relativistic massive poles, physical dictionaries and units, and the geometric Einstein premises remain separate obligations. The exact certificate and negative controls are retained in `model/interacting_vacuum`.

**An interaction-generated charge scale in the original one-dimensional law.** There is a positive many-body charge threshold without inserting a Dirac mass. On the unchanged periodic Weyl/contact geometry $(L, 1, 1)$, $L = 4m$, the onsite frame $G_x = (-1)^x (i\sigma_x)^x$ satisfies $G_x (i\sigma_x) G_{x+1}^{\dagger} = -I$ and has period four. Its CAR lift fixes local doublons, so the original action is exactly the repulsive Hubbard chain with $t = c/2 = 1/16$ and $U = 1$. Four exact local frame identities, source-matrix checks through $L = 16$, and an independent 256-state full-Fock witness bind this join. Unlike the earlier logarithmic-reference alternative Hamiltonian, no coupling is reassigned.

For the *half-filled canonical ground sector*, the imported Lieb–Wu charge-gap theorem gives

$$\Delta_c = \lim_{L\to\infty} [E_L(L+1) + E_L(L-1) - 2E_L(L)] = U - 4t + 8t \sum_{k\geq 1} \binom{1/2}{k} (2t/U)^{2k-1} \eta(2k-1).$$

Here the series is used only for $2t/U < 1$. Twelve binomial terms and 96-term rational Euler bounds for each Dirichlet eta value enclose $\Delta_c = 0.7715517157380375\ldots$ with width below $1.67\,10^{-25}$. The binomial tail is bounded by $8t|\binom{1/2}{K+1}|(2t/U)^{2K+1}/[1-(2t/U)^2]$; the nonnegative Euler coefficients give tail $2^{-96}$. Floating quadrature is an independent check only.

Particle–hole symmetry fixes the charge costs for the original $H$:

$$E_+ = \frac{U+\Delta_c}{2} = 0.885775857869\ldots, \qquad E_- = \frac{\Delta_c - U}{2} = -0.114224142131\ldots.$$

Thus half filling is a specified canonical reference, not the original $\mu = 0$ grand-canonical vacuum. The symmetric threshold $\Delta_c/2 = 0.3857758578690188\ldots$ uses the explicitly centered $K = H - (U/2)N$; $\Delta_c$ itself is invariant under any common $-\mu N$. The spin sector remains gapless, contact still vanishes in the one-particle sector, and no preparation or complete-repair preservation of this many-body reference is established by the gap theorem.

**The fixed-interaction continuum boundary.** The equivalent positive integral for the charge gap and the elementary $K_1$ integral bounds imply, for every $t, U > 0$,

$$\Delta_0 \le \Delta_c \le \Delta_0 \frac{1+3/(8A)}{1-e^{-2A}}, \qquad A = \frac{2\pi t}{U}, \qquad \Delta_0 = \frac{8}{\pi}\sqrt{Ut}\,e^{-A}.$$

Consequently fixed $U$ with $t = v/(2a)$ has $\Delta_c \to 0$ as $a \to 0$: this lattice gap does not itself supply a nonzero continuum mass. A sufficient additional running family is

$$a(A) = \frac{4ve^{-A}}{M\sqrt{2\pi A}}, \qquad t(A) = \frac{v}{2a(A)}, \qquad U(A) = \frac{\pi v}{a(A)A},$$

for a *chosen* $M, v > 0$. It gives $\Delta_0/2 = M$ exactly and $1 \le \Delta_c/(2M) \le [1+3/(8A)]/[1-e^{-2A}] \to 1$. This is a quantitative charge-threshold limit, not a derivation of $M$ from the fixed source, a complete relativistic scaling theory, or a $3+1$-dimensional massive species. The receipt `model/mass_generation_audit/results.json` keeps these reference, coupling-family, and dimensional premises explicit.

The preceding charge-threshold result alone does not prove a mass shell. The next construction supplies energy and momentum together, with a controlled relativistic limit along an explicitly chosen coupling family.

**A controlled relativistic charge mass on an explicit running family.** The preceding charge gap can be promoted to a relativistic dispersion *in one spatial dimension on a specified coupling trajectory*. The source-bound half-filled Hubbard reference and centered generator $K = H - (U/2)N$ remain explicit. The staggered particle–hole map $d_{x,\downarrow} \mapsto (-1)^x d^\dagger_{x,\downarrow}$, with $d_{x,\uparrow}$ unchanged, preserves hopping and reverses the centered interaction. A complete 256-state CAR witness checks this duality and the translation identity $PTP^\dagger = -(-1)^{N_\downarrow}T$. Consequently the attractive spin dispersion becomes the repulsive charge dispersion, with the appropriate momentum-carrier shift. Here $p$ is relative to that low-energy branch; the onsite source frame also carries its specified uniform internal rotation. The even-ring SO(4) selection rules are retained: an elementary thermodynamic quasiparticle is not automatically an isolated finite-ring state or a local-CAR pole.

Writing $A = 2\pi t/U$, $\theta = A\sin k$, $|k| < \pi/2$, and $v = 2ta$, the imported exact dressed dispersion has the source-normalized form

$$E = \frac{8t}{\pi}\sum_{\substack{m\ge 1\\ m\ \mathrm{odd}}} \frac{K_1(mA)}{m}\cosh(m\theta), \qquad vp = \frac{8t}{\pi}\sum_{\substack{m\ge 1\\ m\ \mathrm{odd}}} \frac{K_0(mA)}{m}\sinh(m\theta).$$

On $a(A) = 4ve^{-A}/[M\sqrt{2\pi A}]$, $t = v/(2a)$, $U = \pi v/(aA)$, the leading Bessel coefficient is exactly $M$. For $A \geq 1$ and $|\theta| \leq \Theta < A$, put

$$q = e^{-2(A-\Theta)}, \qquad r_E = \frac{1 + 3/(8A)}{1-q} - 1, \qquad r_p = \max\{1/(8A), q/(1-q)\}.$$

The Gamma-integral tangent bounds $1 - 1/(8z) \leq K_0(z)/b(z) \leq 1$ and $1 \leq K_1(z)/b(z) \leq 1 + 3/(8z)$, with $b(z) = \sqrt{\pi/(2z)}e^{-z}$, and the positive odd-mode tail prove

$$|E/M - \cosh\theta| \leq r_E \cosh\Theta, \qquad |vp/M - \sinh\theta| \leq r_p \sinh\Theta,$$

$$\left|(E^2 - v^2p^2)/M^2 - 1\right| \leq \cosh^2\Theta(2r_E + r_E^2) + \sinh^2\Theta(2r_p + r_p^2) \;\longrightarrow\; 0.$$

At $\Theta = 1$, the last bound is below 0.034 for $A = 64$ and 0.017 for $A = 128$. Thus the limiting charge-doublet dispersion has $E = M\cosh\theta$, $p = (M/v)\sinh\theta$, and the calculated rest mass is $m = M/v^2$. Exact rational finite-parameter enclosures and independent Bessel/integral controls are retained in `model/hubbard_relativistic_sector/results.json`.

For a concrete trajectory through the original point, take $A_0 = \pi/8$, $v = 1/8$ and $M_0 = e^{-\pi/8}/\pi$. Then $a(A_0) = 1$, $t(A_0) = 1/16$, $U(A_0) = 1$, and

$$M_0 = 0.2149329913552644\ldots, \qquad m_0 = \frac{64e^{-\pi/8}}{\pi} = 13.75571144673692\ldots \quad \text{in model mass units.}$$

This is the mass of the *chosen running trajectory*, not a unique mass selected by the original bare point. Indeed, for any $M > 0$, multiplying $a_M(A)$ by $1 + (M/M_0 - 1)e^{-(A-A_0)}$ gives the same starting point and limiting rest energy $M$. The initial lattice half-gap is different, and the fixed-$U$ continuum family has zero limiting charge mass. No native ground preparation, repair-invariant charged pole, $3+1$-dimensional massive species, or empirical particle-mass identification follows from this one-dimensional result.

**Other finite interfaces.** Separate Weyl/contact benchmarks give finite-range CAR commutator bounds and local history estimates uniform in volume, yielding norm-Cauchy thermodynamic dynamics at fixed lattice spacing. They do not establish a chiral spatial continuum. For partition expectations $E_c$ and positive rates,

$$L = \sum_c \gamma_c(E_c - I), \qquad \mathrm{Tr}(X^*LX) = -\sum_c \gamma_c \|X - E_cX\|_{\mathrm{HS}}^2$$

identifies $\ker L$ with the intersection of the partition commutants, without assuming commuting pinchings or independent no-cancellation. Applying this theorem requires the stated generator. Equivariant rational chain maps and homotopies on the supplied finite support tower are homology maps, not quantum channels or a dynamical refinement limit.

**What the negative controls exclude.** Several finite benchmarks constrain possible identifications. A supplied Weyl/contact action has an isolated moving bound-pair band, complete contact residues and an optical identity, but its positive contact section makes the band topologically trivial. Its certified curvature fails the rest/kinetic mass relation with the same speed parameter. Contact interactions also couple proper Weyl momentum pockets to their complements. The lattice scattering theorem, finite thermal positivity and KMS identities do not remove those obstructions or supply physical asymptotic detectors, a chiral continuum or calibrated cross sections.

The centered repulsive dimer gives a sharper family-wide exclusion. For $t = c/2 > 0$, $U \geq 0$, $\mu = U/2$ and $d = \sqrt{U^2 + 16t^2}$, its two charge-one doublets have gaps $g_\pm = d/2 \pm t$, hence

$$1 < g_+/g_- \leq 3.$$

No common positive linear conversion yields the charged-lepton hierarchy or three distinct same-charge families there. Allowing another chemical potential can evade the ratio bound but does not create a third doublet. The fixed complete charged-frequency catalogue also fails its broader lepton-ratio comparison. The periodic four-site extension fails its declared $E(k)^2 = m^2 + c^2 \sin^2 k$ benchmark at $c = 1/8$; a two-parameter fit to two distinct momentum values does not rescue that test. These exclude stated finite dictionaries, not every larger action or a renormalized continuum limit. Electron/positron masses set by one common measured calibration, and retrospective parameter comparisons, provide no independent mass prediction.

Finite current response also has a precise scope. An open-bond phase is pure gauge and measures no magnetic moment; the contact kernel is not a Coulomb kernel. On the complete periodic ring, a genuine loop phase has gauge-invariant current and total curvature,

$$E_0' = \langle \partial_\phi H \rangle, \qquad E_0'' = \langle \partial_\phi^2 H \rangle - 2 \sum_{n \neq 0} \frac{|\langle n | \partial_\phi H | 0 \rangle|^2}{E_n - E_0},$$

under the isolated-ground hypothesis. Degenerate free and atomic controls lie outside that formula. The calculation retains all 256 Fock states but does not identify a photon or physical Lorentz spin.

Clock and memory exclusions are similarly conditional. A frozen real-operation grammar has a normalized trace-distance lower bound $1/2$ for a specified nontrivial modular action of its nontracial reference. Finite independent copies of the original native reference also fail the required field modular-gap ratios. These do not exclude the explicitly enlarged clock and control laws above. The earlier pair/dark-reference encoding covers a fixed corner, not the whole spatial field. Finite stationary observer references concern source/buffer marginals at the stated repair rate and on their recurrent support; delayed variants need their own construction, and growing exported histories are not made stationary by these marginal references. Actively corrected coherent memories can have flat logical energy and a fixed core, so memory lifetime alone does not establish a moving massive particle. Properties proved on distinct carriers require an explicit intertwiner before they can be combined.

**Physical target and remaining obligations.** The target is an interacting quantum field theory with gapless photons and experimentally matched massive sectors, not a positive global mass gap. A gapless photon rules out such a global gap. An exact nontrivial positive-energy conformal theory in a flat-space vacuum representation with unbroken unitary dilations also cannot have a positive flat-space vacuum mass gap: for a spectral value $E > 0$ and proposed gap $g > 0$, scale covariance also puts $g/2$ in the spectrum. This does not exclude a gap in scaling dimensions or discrete finite-radius cylinder levels; the cylinder radius supplies an energy scale. Conformal behavior may describe a critical or short-distance regime, but a deformation or symmetry-breaking mechanism responsible for massive flat-space sectors must then be specified and implemented.

The finite results leave concrete physical joins. A localized defect must carry the protected charge and the same measured field observables. Spatial refinement must control the additional lattice modes and yield the required Lorentz propagation, while keeping the interactions and reference state. The electromagnetic field and gauge–Higgs interactions must be dynamical, with charge constraints, detector coupling and their infrared behavior treated on that same theory. Massive excitations need a common energy and clock normalization, including the relation $E''(0) = c^2/E_0$

for a relativistic branch with $E_0 > 0$, together with the appropriate stress and proper-time relations. The neutrino sector and other measured particle properties require their own matching tests. Neither a successful finite source compilation nor a calibrated spectrum alone discharges these obligations.

## This paper's hypothesis and place in the tower

This paper is layer 2 of the tower: the carrier and its repair law. Its hypothesis is the following.

> *The substrate is a federation of twelve-port icosahedral carriers whose thirty internal seams and routed cross-carrier ports are repaired by the native C/E seam law. One carrier supports the local observer, clock and geometry constructions; federations of* 20*,* 80 *and* 320 *carriers on the nested icosahedral screens serve larger domains and refinement.*

**What is proved.** Unconditional finite theorems on the declared carrier: the counting-and-incidence selector (Theorem 2.3); the characterization of the classical seam rule and the five-multiplier record family (Section 2.2); the rank-three slow band of the uniform repair mean (Proposition 2.6); angular-rank isotropy and the rank-six residue (Section 3); the exact seam incidence, Green, action and stability results (Section 4); the stationary repair covariance (Section 8); and the central-record and checkpoint theorems (Section 9). Conditional implications: the continuum Maxwell limit takes a supplied Euclidean domain and reversible dynamics; the dispersion coefficients and photon threshold take physical position, clock and sector attachments; the edge-center scalar register (Appendix A) takes completeness and reserve premises.

**Open joins.** The twelve-port architecture, the seam schedule and the record-law covariances are premises; closure does not select them. Cross-carrier routing on the federations is a supplied convention, and a record-derived refinement-natural routing that preserves seam adjacency is unconstructed (Section 5). The rank-three band is a dimensionless response metric on twelve ports; physical position, an extended three-dimensional space, a physical clock and an energy unit require further maps.

**What this layer feeds upward.** Layer 3 executes observers on these carriers [22]; layer 4 reads their clocks and the screen generator; layer 5 places stable matter in localized defects of overlap transport. That transport must live on the seams and routed ports of this layer. The seam graph has nineteen independent holonomies, but the oriented face complex has vanishing first cohomology and native load contrasts are exact, so a protected native defect needs transport data that the repair law must supply (Section 4.1).

**Refutation.** The carrier branch fails if an observable depends on hidden carrier coordinates after the full visible signature is fixed (Claim 1.4), if no admissible routing of the federations is refinement-natural and adjacency preserving, or if the repair laws on this architecture admit no localized defect sector protected under admitted repairs. The tested native double-load candidate is transient, and the existing port-ancestry map fails adjacency; both are facts about the tested law.

### Matter operators and gravitational response

The finite response has an explicit conditional matter and interaction readout. A declared spin lift and trace redistribution act on $\Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$. Charge and Casimir projectors recover the five internal multiplets of one Standard Model generation. With an independently declared physical Weyl algebra, one scalar doublet admits the three charged-fermion interaction contractions. The optional neutrino interaction is dimension five. These exact current identities select no Yukawa coefficients, Higgs vacuum or observed masses. Identifying both family factors with the same unbroken irreducible source triplet instead forces scalar intertwiners and degenerate family masses.

The conditional Lorentzian scalar action also admits a common local tetrad dictionary for scalar, spinor and Maxwell principal symbols. A finite CAR construction shares its ideal spatial operator and incident-count clock. Completing a separate tetrahedral cochain control removes edge-cycle zero modes introduced by truncation and preserves its scalar action block. These statements keep their distinct global domains, boundary conditions and reference states; they establish no complete physical chiral field net.

For coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$, the induced screen area is $dA = \mu\, d^2x$, while proper spatial volume is $dV = \mu N\, dr\, d^2x$, with $N^2 = \mu\sqrt{\det a}$. Counting stationarity consequently requires a further identification with fixed-volume geometric entropy stationarity. The static radial-product metric has vanishing radial null Einstein contractions. Its allowed scalar excitation can have positive relative null stress, so a fixed background, with the reference and other sources fixed, cannot satisfy both Einstein balances. A responsive geometry must supply the corresponding area focusing. This is a scoped restriction of that field/metric construction, independent of the validity of the conditional Einstein theorem.

The exact finite tensors and complete observation controls are given in `model/observer_particles/`. Their representation, field-operator and geometric witnesses instantiate finite hypotheses on their stated branches. They do not remove the separate source-selection, physical-quantity or controlled-refinement hypotheses.

## Carrier signature, screens and observers

The tower panel above orders the layers. Three conventions are shared by every layer that uses the carrier.

### Three meanings of screen

The word "screen" is used for three related objects that are identified only through an explicit map.

1. The *local carrier boundary* is the twelve-port oriented interface of one icosahedral carrier. Its incidence has $(V, E, F) = (12, 30, 20)$.

2. The *federation screen* is the routed system of interfaces, records, repairs and checkpoints of many carriers at finite cutoff.

3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal $S^2$ used for caps, collars, modular flow and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. Identical local carriers can be routed as a path, a cycle, a higher-genus complex or a spherical complex. The map from routed carriers to a support-visible spherical nerve is a physical bridge with its own premises.

### Presentation invariance and architecture sensitivity

OPH is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the icosahedral branch that signature is

$$\mathcal{C}_{i,r} = (\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I^{\mathrm{or}}_{i,r}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \mathsf{Chk}_{i,r}, \mathsf{Resp}_{i,r}, c_{sr}),$$

with accessible algebra, state, port set, oriented incidence, record algebra, repair or feedback interface, checkpoints, visible response law and refinement lineage. Hidden coordinates, port names, worker partitions, materials and wiring presentations are silent when an isomorphism preserves this tuple and its error model. A change of port number, incidence, orientation, accessible algebra, response, repair law, clock or refinement lineage can change the admitted physics. A cube and an icosahedron are different carrier contracts even when built from the same material.

### When a carrier is an observer

A carrier body realizes an observer when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls and checkpoint continuation. One carrier may pass that test, or a connected subfederation may pass it instead; no theorem fixes the primitive observer size by counting carrier bodies. Equal-frequency phase locking can synchronize an interface; it makes the interface an observer only together with these record and feedback clauses.

### How the layers compose

Consensus, geometry and gravity are composable branches of one carrier tower: accepted repair produces the quotient-visible normal form, the exact port response supplies the rank-three slow-response carrier, and the support screen feeds the modular and Lorentz constructions of the spacetime layer. The gauge layer uses the same carrier architecture, where the complete twelve-port response and endogenous holonomy force the abstract local Lie algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. The matrix current and matter action are conditional realizations; a common source construction joining them to the geometric tower is the matter layer's open join. Matching dimensions or symmetry labels does not supply that join.

## Contents


**1 Observer patches and their carriers** **37**

**2 The twelve-port carrier and its native repair law** **38**
- 2.1 The echosahedral carrier and its selector . . . 38
- 2.2 The native seam rule and its record laws . . . 40
- 2.3 An alternative variational selector . . . 41
- 2.4 The integer seam-current record . . . 42
- 2.5 Face-corner fibres . . . 44

**3 Angular-rank isotropy and the seam symbol** **45**
- 3.1 The complete seam symbol . . . 47

**4 Exact seam electromagnetism on the carrier** **51**
- 4.1 Overlap transport and the defect obligation . . . 59

**5 Federations, screens and refinement** **59**
5.1 The multiresolution regulator chart . . . 60
5.2 Sphere fold and the carrier-to-support bridge . . . 62
5.3 The finite federation operator domain . . . 63

**6 Repair, equilibrium and modular structure on the carrier** **65**
6.1 Finite modular and register theorems . . . 65

**7 Patch federation and overlap synchronization** **69**

**8 Native repair histories and finite scale transfer** **70**
8.1 Native stationary repair and absolute record power . . . 70
8.2 Locally sampled attempt records . . . 71
8.3 Temporal filtering and fixed-area normalization . . . 72
8.4 Conditional scale transfer in the large settled configurations . . . 73
8.5 Port transport, internal response and metric sensitivity . . . 74
8.6 Absolute refinement covariance and nested records . . . 75
8.7 What a fixed-geometry repair history identifies . . . 75

**9 Records, observers and checkpoints** **77**

**10 Central records and measurement** **79**
10.1 Record-conditioned modular cap responses . . . 79
10.2 A finite Bell candidate . . . 80

**11 Edge sectors and the compact-gauge layer** **80**

**12 Finite quotient ensemble compatibility** **81**

**13 Conclusion** **86**

**A The edge-center scalar register** **86**
A.1 Rank-one scalar-slot completeness . . . 86
A.2 Edge-center algebra, scalar activation and the protected reserve . . . 87
A.3 Coherent-matter scalar-source forcing . . . 89
A.4 The oriented register and the reserve mean . . . 90
A.5 Finite-thickness collar survival . . . 90

# 1 Observer patches and their carriers

An observer patch is first an operational algebraic object,

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, $\rho_i$ is the accessible state, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the exact central record algebra or a declared approximate record algebra, $\mathcal{I}_e$ is an overlap-visible interface algebra, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the visible restriction map, $\mathcal{U}_i$ is the allowed family of update and repair instruments, and $\mathrm{Chk}_i$ is checkpoint data sufficient for the target reconstruction problem.

**Definition 1.1** (Abstract observer patch)**.** *An abstract observer patch is the tuple $\mathsf{O}_i$ above, considered up to isomorphism of accessible algebras, record statistics, visible restrictions, repair instruments, and checkpoint continuation.*

**Definition 1.2** (Support patch)**.** *A support patch is a geometric chart for an abstract observer patch on a branch with reconstructed geometry. Examples are a cap $P_i \subset S^2$, a collar, or a causal diamond. The support patch supplies chart data for modular flow, entropy variation, and overlap comparison.*

**Definition 1.3** (Carrier patch)**.** *A carrier patch is a physical or digital implementation satisfying $\mathrm{Realize}_\varepsilon(H_i) \to \mathsf{O}_i$: the implementation induces the declared accessible algebra, visible interface statistics, record readout, repair instruments, and checkpoint continuation within error $\varepsilon$.*

The observer identity lives in the quotient data exposed by $\mathsf{O}_i$. The tuple does not impose equal primitive observer size: fixed-cutoff OPH requires bounded accessible algebras and declared interfaces, and an observer-supporting object may be a variable finite federation of carriers. A theorem selecting isomorphic fixed-capacity elementary carriers would need homogeneous cellulation, isomorphic local algebras, equivariant overlaps, refinement preservation of the cell type, and a proof that one cell or a fixed block of cells is the primitive carrier.

**Claim 1.4** (Presentation invariance and carrier sensitivity)**.** *Two carrier patches are physically equivalent for the declared OPH observables when a quotient-visible isomorphism preserves their accessible algebras and states, interface algebras, record statistics, repair maps, checkpoint continuation, response laws, and refinement lineage. On the twelve-port branch it must also preserve the oriented incidence and its antipodal pairing. A material, coordinate, or wiring change satisfying this contract is silent. A change of the preserved carrier signature is outside the equivalence claim and may change the physical branch.*

The claim is theorem-level on a branch that supplies the required quotient isomorphism and error bound. It fails if a predicted observable depends on hidden carrier coordinates after the full visible signature has been held fixed. It says nothing about a cube and an icosahedron, whose visible incidence structures differ. OPH is therefore presentation-invariant and carrier-sensitive: port incidence, orientation, response, repair, clock and refinement data can change the physical branch.

**Definition 1.5** (Operational observer test)**.** *A candidate physical system realizes an observer patch when it supplies a bounded accessible interface, durable re-readable records, self-read or internal state estimation, record-conditioned future behavior, boundary prediction better than shuffled-record controls, and checkpoint continuation within a declared error.*

The formal finite receipt implements seven corresponding algebraic clauses at one regulator. Two operational observers have unequal owner-region records whose typed restrictions agree on a nonzero common section accessible to both; a one-corner mutation breaks the agreement. This is a bounded overlap-readable observer receipt, without a physical instrument, source realization, refinement naturality, or higher-overlap coherence.

**Definition 1.6** (Federated patch carrier)**.** *A fixed-cutoff federated patch carrier is a tuple*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}_{i\in V}, \{\mathcal{I}_e\}_{e\in E}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\}),$$

*where $V$ is a finite set of patches, $E$ is a finite overlap graph, $\mathcal{A}_i$ is the local finite algebra of patch $i$, $\mathcal{I}_e$ is the finite interface algebra on overlap $e = \{i, j\}$, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the declared visible restriction, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the patch record algebra, and $\mathcal{U}_i$ is the allowed local update and repair interface. The carrier is observer-facing when every physical claim is made through the visible restrictions, record algebras and quotient-local observables; hidden coordinates may exist but are not directly physical.*

The smallest instance is one twelve-port carrier. In the executable observer construction a single carrier supports overlapping observer regions, native repair and local event clocks, and joined carriers serve larger regions, distant interventions and refinement [22]. A fixed-cutoff simulator keeps three operators on the physical quotient $Q_r$ distinct: the strict-descent accepted-repair normalizer $\mathrm{Rep}_r$, a separately declared equilibrium or Euclidean relaxation $e^{-tL_r^{\mathrm{eq}}}$, and the reversible modular automorphism group $\sigma_t^{\rho}$. If an accepted move lowers a well-founded repair measure, its reverse cannot be another accepted descending step, so the accepted relation cannot serve as a reversible equilibrium law. A common local proposal menu may feed all three constructions without identifying them.

# 2 The twelve-port carrier and its native repair law

## 2.1 The echosahedral carrier and its selector

**Definition 2.1** (Echosahedral carrier)**.** *An* echosahedral carrier*, or echosahedron, is a finite patch object*

$$\mathcal{E} = (\mathcal{A}, \{P_a\}_{a=0}^{11}, \{\rho_a\}_{a=0}^{11}, \mathcal{R}, \mathcal{M}, \mathcal{U}, \mathcal{G}_{\mathcal{E}}),$$

*where $\mathcal{A}$ is the internal finite algebra, $P_0, \ldots, P_{11}$ are twelve labeled overlap ports, $\rho_a : \mathcal{A} \to \mathcal{I}_a$ are port readout maps, $\mathcal{R} \subseteq Z(\mathcal{A})$ is the observer-accessible record algebra, $\mathcal{M}$ is a finite mismatch-score family on exposed port packets, $\mathcal{U}$ is a finite family of local update and repair instruments, and $\mathcal{G}_{\mathcal{E}}$ is a declared finite symmetry group of the port arrangement. Its ports are the vertices of an oriented icosahedron and its thirty internal seams are the edges.*

**Definition 2.2** (Certified echosahedral counting realization)**.** *A certified local lineage augments the quotient-visible echosahedral patch by:*

(i) *twelve primitive orthogonal port-center atoms $e_{r,p}$, each of normalized trace $1/12$;*

(ii) *the oriented 12-vertex, 30-edge, 20-face triangular incidence packet, with degree-five vertices and five-cycle links;*

(iii) *a separately declared integer atom-counting grammar with total-charge fiber $\mathcal{Q}_r = \{q \in \mathbb{Z}^{12} : \sum_p q_p = 12\}$, and the normalized central-readback Hilbert–Schmidt cost $H_r(q) = \sum_p q_p^2$;*

(iv) *an append-only twelve-register event packet in which writes and retractions carry signs* $+1$ *and* $-1$*, together with conservative whole-unit seam repairs and minimum-move settling cost. An accepted repair orients a seam from* $i$ *to* $j$ *only when* $d = q_i - q_j \geq 2$*, then replaces* $(q_i, q_j)$ *by* $(q_i - 1, q_j + 1)$*; uphill and equalizing-reversal moves are not repair steps;*

(v) *every refinement carries an orientation-preserving incidence isomorphism. It preserves the port centers and readback packet, and these isomorphisms obey the refinement cocycle; and*

(vi) *a source manifest that rejects Standard Model, product-adjoint, gauge, coupling, particle, measured-target, and fitted-coordinate fields, is closed to undeclared fields, and encodes trace fractions exactly.*

*The signed-event packet realizes integer loads and its separate dynamic settling cost. The total-charge fiber, quadratic cost normalization, and positive readback scale are named realization data. An arbitrary carrier satisfying A1–A3 need not select them.*

**Theorem 2.3** (Echosahedral counting-and-incidence selector theorem)**.** *On every lineage of Definition 2.2, the declared realization packet canonically determines:*

(i) *the twelve coefficient lines* $\mathbb{R}e_{r,p}$ *and the unique all-one defect allocation, with* $H_{\min} = 12$*, next floor* $14$*, and exact gap* $2$*;*

(ii) *integer loads generated by the signed atomic events, termination of every accepted settling path by strict descent of* $V(q) = \sum_p q_p^2$*, and a minimum settling-move cost natural under every carrier rotation and declared refinement;*

(iii) *the unique fixed-point-free inverse pairing by graph distance three and the resulting six axes;*

(iv) *the full incidence group* $A_5 \times C_2$ *and its orientation-preserving subgroup* $A_5$*;*

(v) *the exact incidence Gram matrix*

$$G_{pq} \in \{1, 1/\sqrt{5}, -1/\sqrt{5}, -1\} \quad \text{at distances } 0, 1, 2, 3,$$

*with*

$$G^2 = 4G, \qquad \operatorname{tr} G = 12, \qquad \operatorname{rank} G = 3,$$

*whose oriented factorization is the regular six-axis icosahedral frame up to* $SO(3)$*; and*

(vi) *refinement naturality and arbitrary simultaneous port-relabeling equivariance of all these objects.*

*Thus this realization gives the exact twelve-line unit allocation and gap, integer settling dynamics, inverse pairing, proper* $A_5$ *action, and rank-three Gram frame stated above.*

*Proof.* Primitive central atoms give the intrinsic line family. On the total-charge fiber,

$$H(q) = 12 + \sum_p (q_p - 1)^2,$$

so the all-one allocation is unique and the least nonzero integral zero-sum deviation has squared norm two. A seam repair with oriented mismatch $d \geq 2$ changes the load square by $-2(d-1)$, which proves termination. Every declared rotation and refinement is an isomorphism of the repair graph and therefore preserves shortest-path cost. The explicit full-pile witness settles in eighteen

moves. The exact incidence distance profile is $(1, 5, 5, 1)$, giving a unique distance-three involution. Enumeration gives 120 incidence automorphisms and 60 orientation-preserving ones; conjugation on the five Klein-four subgroups identifies the latter faithfully with all even permutations of five objects, hence with $A_5$. Exact multiplication in $\mathbb{Q}(\sqrt{5})$ gives $G^2 = 4G$, and the trace fixes rank three. Every declared refinement or complete relabeling preserves the source formulas. The exact implementation and countermodels are carried by the echosahedral selector certificate and its accompanying analysis. □

*Remark* 2.4 (Load-bearing hypotheses). Without primitive atoms the line split is continuous; without equal trace or total charge twelve the all-unit minimizer is not unique; a linear cost has $\binom{23}{11}$ minimizers; twelve labels without incidence have 10,395 fixed-point-free pairings; without orientation the selected group is $A_5 \times C_2$; and non-incidence or noncocyclic refinement maps fail naturality. The theorem derives the selector from the declared carrier packet. It does not prove that bare Euler incidence or every possible OPH carrier selects that packet, and it constructs no physical gauge-current algebra. The total load modulo twelve is a necessary consensus obstruction. The family $\{\frac{1}{2}e_{r,p}\}$ rescales event values, seam readback and repair threshold together; it is the same event mechanism in other readback units. For a separately declared uniform reference $\tau_p = 1/12$, a feasible perturbation $\rho = \tau + \varepsilon v$ with $\sum_p v_p = 0$ has $D(\rho\|\tau) = 6\varepsilon^2 \sum_p v_p^2 + O(\varepsilon^3)$; this infinitesimal curvature supplies neither the discrete cost $H_r$ nor the minimum settling-move cost, and exact separating witnesses distinguish all three objects.

## 2.2 The native seam rule and its record laws

The executable observer construction [22] fixes one repair law on this carrier. For an oriented seam $e = (a, b)$ with loads $x_a, x_b$ and a fair coin $c \in \{0, 1\}$, set $d = x_a - x_b$ and

$$j = \tfrac{1}{2}\big[d - (2c - 1)(|d| \bmod 2)\big], \qquad x \longmapsto x + j\,(\delta_b - \delta_a).$$

The endpoints become the floor and the ceiling of their mean; when their total is odd the coin decides which endpoint receives the extra unit. The rule conserves the total and never increases $\sum_p x_p^2$. At difference one it waits or swaps, so the balanced binary shell keeps moving instead of settling pathwise. Accepted whole-unit repairs at difference at least two are the strict-descent moves of Theorem 2.3.

**Proposition 2.5** (The classical seam rule is characterized)**.** *Let a stochastic seam rule (i) conserve the seam total, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be symmetric under exchange of the two endpoints. Then it is the rule above with a fair coin.*

*Proof.* By (i) and (ii) the outputs are the floor and ceiling of the mean. An even total has one output. An odd total has two, and by (iii) endpoint $a$ receives the ceiling with a constant probability $q$. Exchange of the endpoints maps $q$ to $1 - q$, so (iv) gives $q = 1/2$. □

Each requirement does work. Without (ii), partial and idle moves are admitted. Without (iii), the rule that gives the extra unit to the larger endpoint satisfies the others and freezes every binary configuration. On the executed total-two sector the seam difference is at most two, so there the requirements fix only the tie-break coin. On classical diagonal states of the binary shell the rule is the lazy symmetric simple exclusion process with two particles; the twelve collision states of the 78-configuration carrier are transient.

A quantum seam event may also keep or destroy coherence between configurations that move the same way. Require of a one-seam channel seam locality, the exact classical law, local number-phase covariance (commutation with $\exp(i\sum_p t_p N_p)$ for every real $t$) and endpoint-swap covariance; both covariances are declared premises. Every admissible channel then has the form

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j(\kappa(x), \kappa(x'))\, |x + jD\rangle\langle x' + jD|,$$

where $\kappa(x) = (x_a, x_b)$ is the seam key, $D$ moves one load from $a$ to $b$, and each $G_j$ is positive semidefinite with diagonal equal to the classical probability of current $j$. On the total-two carrier these laws form one convex family, classified exactly by five coherence multipliers $k, g, h, m, n$, with $|k| \le 1$, $|g|, |h|, |n| \le 1/\sqrt{2}$, $h = g$ when $k = 1$, and $m$ real with $|m| \le 1/2$ [22]. The executed law C records coin and current ($k = 1$, $g = 1/2$); the comparison law E also records the seam parity ($k = 1$, $g = 0$) and is idempotent. Forgetting a label afterwards does not undo its dephasing, so C is not a classical coarse-graining of E. Under a common definite-current record interface all members share the classical current laws, and one coherent interference experiment reads one coherence multiplier directly.

**Proposition 2.6** (The uniform repair mean selects the rank-three Gram band)**.** *In a balanced binary sector under independent unit-rate seam attempts, the native rule acts on centered loads with mean generator $-L/2$, where $L = 5I - A$ is the Laplacian of the thirty internal seams (Theorem 8.1). The positive eigenvalues of $L$ are $5 - \sqrt{5}, 6, 5 + \sqrt{5}$, with multiplicities $3, 5, 3$. The slowest band, eigenvalue $5 - \sqrt{5}$, is the range of the incidence Gram matrix $G$ of Theorem 2.3, and its orthogonal projector is $G/4$.*

*Proof.* Write $G = UU^{\mathsf{T}}$, whose rows $u_p$ are the unit port directions of the oriented frame; adjacent ports have $u_p \cdot u_q = 1/\sqrt{5}$. The order-five rotation about $u_p$ permutes the five neighbours of $p$, so their sum is parallel to $u_p$ with component $5/\sqrt{5} = \sqrt{5}$. Hence $AU = \sqrt{5}\,U$ and $LU = (5 - \sqrt{5})U$. The range of $G$ has dimension three, the multiplicity of $5 - \sqrt{5}$, and $G^2 = 4G$ makes $G/4$ its orthogonal projector. □

The band is a dimensionless response geometry on twelve ports. Native records reconstruct it from currents and clock fluctuations, and all thirty seams form its shortest shell [22]. It is distinct from the spectral dimension two of the refining screen generator (Section 8.5), and its rank three does not derive three physical spatial dimensions.

## 2.3 An alternative variational selector

**Theorem 2.7** (Alternative variational icosahedral screen sieve)**.** *Assume the fixed-cutoff OPH screen branch is represented by a quantum-link triangulation of $S^2$, with finite Hilbert spaces on links, Gauss constraints at vertices, boundary-gauge-invariant physical algebras, and a refinement-stable quotient ensemble locally six-valent away from curvature defects. Let*

$$q_v = 6 - \deg(v).$$

*Assume integer charges, a feasible twelve-unit configuration, and an additive cost with $h(0) = 0$, $h(1) > 0$, $h(k) \ge h(1)|k|$, strictly for $|k| \ge 2$. Let edge-center collars expose each unit defect as one central port, with a fixed-point-free inverse-port involution giving six axes $u_i$. Assume the minimum-cost locus contains a six-axis configuration with $F_1 = 2I_3$ and $F_2 = (4/5)I_5$. If the source selector first minimizes the additive cost and, among those minimizers, maximizes*

$$\det F_1 \det F_2, \qquad F_1 = \sum_i u_i u_i^{\mathsf{T}}, \quad F_2 = \sum_i |Q_i\rangle\langle Q_i|, \quad Q_i = u_i u_i^{\mathsf{T}} - I_3/3,$$

*then the screen has twelve unit ports on the regular icosahedral orbit. An equal-weight additive invariant load $X$ is read locally as $X/12$.*

*Proof.* For a triangulated sphere with $V$ vertices, $E$ edges, and $F$ faces, Euler and triangular incidence give

$$V - E + F = 2, \qquad 3F = 2E, \qquad E = 3V - 6.$$

Therefore

$$\sum_v q_v = \sum_v (6 - \deg(v)) = 6V - 2E = 12.$$

Thus $\sum_v q_v = 12$. The cost satisfies

$$\sum_v h(q_v) \geq h(1) \sum_v |q_v| \geq 12h(1).$$

Equality requires nonnegative unit charges, and feasibility supplies the twelve-unit minimum. The selector's first stage therefore returns exactly twelve unit defects. The collar rule turns them into twelve central ports. For six axes, $\operatorname{tr} F_1 = 6$ and $\operatorname{tr} F_2 = 4$. Determinant AM–GM gives $\det F_1 \leq 2^3$ and $\det F_2 \leq (4/5)^5$, with equality exactly at $F_1 = 2I_3$ and $F_2 = (4/5)I_5$. The assumed feasible isotropic frame attains both bounds, so the selector's second stage does too. The six equal-norm $Q_i$ then form a regular simplex in $\operatorname{Sym}_0(3)$, so $(u_i \cdot u_j)^2 = 1/5$ for $i \neq j$. The associated Seidel matrix satisfies $S^2 = 5I_6$; after switching one vertex is isolated and the remaining negative-edge graph is a five-cycle. This is the unique switching class and gives the six axes of a regular icosahedron, whose proper rotation group $A_5$ is transitive on the twelve ports. Equal-weight additivity gives the local read $X/12$. □

*Remark* 2.8 (Scope of the variational selector). The identification $X = \log(N/\pi)$ of the load is imported from the companion capacity relation. The D-optimal vector/quadrupole functional is a branch premise; a strictly completely monotonic pair cost is another sufficient selector by universal optimality. Neither is needed on the certified lineage of Theorem 2.3. Here $\sum_v (6 - \deg v) = 12$ follows from spherical incidence, while integer charge typing and the strict additive cost are assumptions. The cell value $\bar{\ell}_{\text{shared}} = P/4$ and the $\mathbb{Z}_6$ class equidistribution of Theorem A.24 have separate premises; the port and slot counts imply neither.

## 2.4 The integer seam-current record

Give each of the thirty undirected seams a bookkeeping orientation and let an integer current carry the corresponding signed incidence boundary on the twelve ports. Pair each port with its graph-distance-three inverse and take the antipodal-odd part of that boundary. The result is a six-coordinate integer record.

**Theorem 2.9** (Exact seam-current image and response completion)**.** *The image of the complete integer seam-current map is exactly*

$$D_6 = \left\{ z \in \mathbb{Z}^6 : \sum_{i=1}^{6} z_i \equiv 0 \pmod 2 \right\}.$$

*The six-by-thirty integer map has Smith invariants* $(1, 1, 1, 1, 1, 2)$*. Its residual cokernel is one parity bit, and an explicit six-seam section reaches every element of* $D_6$*. Pull back the response-selected Gram metric of Theorem 2.3 to* $D_6$*. The resulting metric image is injective and dense in the same three-dimensional Euclidean response carrier, so its abstract metric completion is canonically isometric to that carrier. Cumulative* $D_6$*-records act simply transitively by isometries on the integer record carrier, and this action extends to translations of the completion.*

*Proof.* Exact integer incidence has rank eleven on the twelve-port boundary space and cycle rank nineteen. Composing it with the antipodal-odd readout gives an even coordinate sum. Conversely, the explicit six-seam section reconstructs an arbitrary even-sum vector, which proves the image statement. The parity map supplies the two cokernel cosets and the Smith form. The response Gram map is injective on the integer controls. The golden-ratio coordinate lines are dense, and doubled control vectors belong to $D_6$, so the restricted image remains dense in the rank-three response carrier. Record addition and the pullback metric give the stated homogeneous isometric action. All finite incidence, section, parity, density, and action identities are checked in Lean [18]. The thirteen finite covariance/isometry table identities in the proper-action subpacket use `native_decide`; those subreceipts therefore trust Lean's native compiler and runtime as well as its kernel and are not kernel-only proofs. □

**Theorem 2.10** (Finite seam continuity and Ward-promotion boundary)**.** *Let $q, q', s \in \mathbb{Z}^{12}$ be stored load, updated load, and source, and let $J \in \mathbb{Z}^{30}$ be a current on the bookkeeping-oriented seams. Adopt the net-inflow incidence convention. If*

$$q' - q = s + \partial J,$$

*then for every port region $R$,*

$$\sum_{p \in R} (q'_p - q_p) = \sum_{p \in R} s_p - \Phi_{\partial R}(J),$$

*where $\Phi_{\partial R}$ is signed outward flux. Internal seams cancel. For the complete port set, a zero-total source preserves total load.*

*After scalar extension to $\mathbb{Q}$, a port load $z$ has a solution of $\partial J = z$ exactly when $\sum_p z_p = 0$. A declared port-zero spanning-tree section constructs one solution; the full solution fibre is its translate by $\ker \partial$, which has dimension nineteen. This choice is explicit and carries no naturality claim.*

*Separately, let $\Phi$ and $\Phi^\dagger$ be real linear maps on a finite state space that are dual for the expectation pairing. A charge expectation is preserved for every signed state exactly when $\Phi^\dagger Q = Q$. Channel covariance is insufficient: uniform averaging on two states commutes with their swap and erases the nonzero swap-odd charge. Antisymmetric transfer, action symmetry, channel covariance, and charge conservation are therefore distinct statements.*

*Proof.* Regional summation of the oriented incidence map pairs every internal seam with opposite endpoint signs. Rank-nullity for the connected twelve-vertex, thirty-edge graph gives incidence rank eleven and cycle rank nineteen, and the declared tree section proves surjectivity onto the neutral hyperplane. Finite expectation-pairing nondegeneracy proves the dual fixed-point criterion by testing signed point masses. The two-state averaging computation is an exact negative control. The regional, Gauss-fibre, duality, and control statements are checked in Lean [18]. □

The theorem assumes the pointwise finite update and assigns no physical meaning to load, current, update order, or charge. A continuum Ward identity requires a source-derived family of these finite witnesses, directed cofinal refinement with a positive scale tending to zero, realization as components of one stress tensor, weak convergence on a separating test-field class, a finite residual declared on the continuity witness and independently derived from it, convergence of that residual, chart covariance, and one inhabited common tower. The Lean `WardLimitManifest` requires a not-all-zero finite datum and a nontrivial residual and derives residual vanishing at the candidate limit by uniqueness of limits; residual convergence must follow from an independent compatibility estimate and cannot assume the target Ward equality.

The metric in Theorem 2.9 is the response Gram metric, not the Euclidean metric of the integer tuple $z \in \mathbb{Z}^6$. The theorem constructs an abstract three-dimensional completion of records, with no identification with physical position, no ruler and no global observer gluing.

## 2.5 Face-corner fibres

**Theorem 2.11** (Icosahedral face-corner bundle)**.** *On the icosahedral screen-sieve branch, the twenty outward-oriented triangular faces form the homogeneous orbit*

$$\mathsf{F}_{20} \simeq A_5/C_3.$$

*The stabilizer of a face acts regularly on its three corners. Hence the face orbit carries an $A_5$-associated bundle of local three-dimensional permutation fibers with cyclic shift $R^3 = 1$. A fiberwise Hermitian operator $C$ that is transported $A_5$-equivariantly across the face orbit and commutes fiberwise with this shift has the circulant form*

$$C = a1 + bR + \bar{b}R^2,$$

*and its unordered real spectrum is independent of the chosen face representative. Reversing the face orientation exchanges $R$ and $R^2$ and preserves that unordered spectrum.*

*Proof.* The icosahedron has $(V, E, F) = (12, 30, 20)$. The orientation-preserving icosahedral group $A_5$ is transitive on the outward-oriented faces. By orbit–stabilizer the face stabilizer has order $60/20 = 3$, hence is $C_3$, and its nontrivial rotations cyclically permute the three corners. The commutant of the regular cyclic shift on one fiber is the circulant algebra; Hermiticity gives the displayed coefficients. The stated $A_5$-equivariance transports that operator to every other face by conjugation and therefore preserves the spectrum. Without this equivariance, independent coefficients could be chosen on different faces. Orientation reversal complex-conjugates the transported presentation and again preserves the unordered real eigenvalues. □

*Remark* 2.12 (Face-carrier boundary). The theorem supplies local geometric $C_3$ fibers, not one canonical global three-dimensional matter-family space. The sixty face-corner flags form a free transitive $A_5$-set, whose linearization is the regular sixty-dimensional $A_5$ representation. A physical family claim therefore requires a quotient-visible section, connection, or intertwiner from this bundle to the relevant matter multiplicity space. Cyclic covariance alone also leaves $a$, $|b|$, and $\arg b$ free. In particular this theorem does not emit a charged-family phase, determinant normalization, or mass coordinate.

**An independent golden character-field control.** The standard $H_4$ coordinate families give 120 unit quaternions over $\mathbb{Q}(\sqrt{5})$. Direct enumeration checks their group law, inverses and center $\{1, -1\}$, and the canonical quaternion-to-matrix map gives a faithful determinant-one doublet with character norm one. Symmetric powers, exact character inner products, constituent subtraction and field conjugation recover the complete irreducible system, with dimensions $1, 2, 2, 3, 3, 4, 5, 6, 4$ and squared-dimension sum 120. Tensoring with the doublet gives a nine-vertex, eight-edge tree whose dimension vector is a 2-eigenvector, isomorphic to an independently encoded affine $E_8$ tree. A second implementation constructs the carrier by closure from two generators. Its nonrational trace generates the character field $\mathbb{Q}(\sqrt{5})$, and the matrix coefficient field is $\mathbb{Q}(\sqrt{5}, i)$. Conjugating $\sqrt{5}$ while fixing $i$ gives a distinct faithful irreducible doublet whose fusion graph also has affine $E_8$ type, so this graph type does not select between $\varphi$ and $1-\varphi$. Comparison with a separate $\mathrm{SL}(2, \mathbb{F}_5)$ reconstruction confirms these finite invariants [18]. This exact executable corroboration supplies

no explicit isomorphism between the two group carriers, no Lean proof of finite character theory, and no physical embedding selector.

## 3 Angular-rank isotropy and the seam symbol

The twelve-port orbit fixes the directional content of every icosahedrally invariant cosine kinetics. These identities are the mathematical basis of the frozen dispersion comparison; its physical attachments are premises of that comparison.

**Theorem 3.1** (Exact multipole fixed point of the twelve-port orbit)**.** *Let $u_1, \dots, u_{12}$ be the twelve unit port directions of the regular icosahedral orbit and let $P_l$ denote the degree-$l$ Legendre polynomial. Then for every unit vector $n$,*

$$\sum_{i=1}^{12} P_l(u_i \cdot n) = 0, \qquad l = 1, \dots, 5,$$

*and the normalized rank-six invariant*

$$\mathcal{I}_6(n) = \frac{25}{132} \sum_{i=1}^{12} P_6(u_i \cdot n)$$

*takes the exact values $1$ on the twelve port directions, $-5/16$ on the thirty edge midpoints, and $-5/9$ on the twenty face centers. Its critical set on the sphere is exactly the sixty-two axis points: twelve maxima at the ports, thirty saddles at the edge midpoints, and twenty minima at the face centers, every tangent Hessian eigenvalue nonzero in exact arithmetic, with Euler count $12 - 30 + 20 = 2$. On the multiplicity-free band split $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ of Theorem 6.6, the lowest surviving multipole ranks are $0, 1, 3, 2$, the antipodal partner of each port is derived from adjacency as its unique distance-three port, and the blind antipodal response $R = -J$, with $J$ that derived antipodal permutation, acts with sign vector $(-1, +1, +1, -1)$ in band order $(\mathbf{1}, \mathbf{3}, \mathbf{3}', \mathbf{5})$.*

*Proof.* The icosahedral invariant dimension at angular rank $L$ is

$$m_L = \frac{1}{60}\Big[(2L+1) + 15\,U_{2L}(0) + 20\,U_{2L}(\tfrac{1}{2}) + 12\,U_{2L}(\tfrac{\varphi}{2}) + 12\,U_{2L}(\tfrac{\varphi-1}{2})\Big]$$

with $U$ the Chebyshev polynomials of the second kind; the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$. Each orbit sum $\sum_i P_l(u_i \cdot n)$ is an invariant of rank $l$, hence vanishes for $l = 1, \dots, 5$, and the rank-six invariant line is one-dimensional. The special values, the sixty-two-point Morse census, the derived antipode, and the band signs are certified in exact arithmetic by the machine-readable certificate [18]. □

**Proposition 3.2** (Quantitative stability bounds for the equal-weight cosine member)**.** *Let $Q_x(n) = \sum_{i=1}^{12}[1 - \cos(x u_i \cdot n)]$, where $n, u_i \in S^2$, and let $0 < x \le 1$. After removal of the isotropic part, the terms through eighth order are*

$$A(x)\mathcal{I}_6(n), \qquad A(x) = \frac{2x^6(30 - x^2)}{118125} > 0.$$

*Writing the complete $x^{10}$-and-higher angular tail after division by $A(x)$ as $E_x$, exact bounds are*

$$\|\nabla E_x\| \le \frac{6875}{101152} < \frac{7}{100}, \qquad \|\operatorname{Hess}_{S^2} E_x\| \le \frac{383125}{562658} < \frac{7}{10}.$$

*The 62 critical directions of $\mathcal{I}_6$ form 31 projective axes, with minimum squared-sine separation $1/2 - \sqrt{5}/6$.*

*Proof.* Exact reduction on the unit sphere gives

$$\sum_i (u_i \cdot n)^6 = \frac{12}{7} + \frac{64}{175}\mathcal{I}_6(n), \qquad \sum_i (u_i \cdot n)^8 = \frac{4}{3} + \frac{256}{375}\mathcal{I}_6(n).$$

The sixth- and eighth-order terms of the cosine series therefore have anisotropic coefficients $4x^6/7875$ and $-2x^8/118125$, which combine to the stated positive amplitude. For every term of order $2m \geq 10$, the spherical gradient is bounded by $12x^{2m}/(2m-1)!$. The Euclidean Hessian is bounded by $12x^{2m}/(2m-2)!$, and the intrinsic Hessian adds at most the gradient bound through the radial correction. On $x \leq 1$, the successive ratios are at most $1/110$ and $1/90$, while $A(x) \geq 58x^6/118125$. Summing the two geometric majorants and reducing the rational numbers gives the displayed bounds. Exact enumeration of the three carrier orbits gives the axis count and separation. The machine receipt replays these reductions independently [18]. □

**Theorem 3.3** (Exact persistence of the equal-weight cosine fingerprint)**.** *Under the assumptions of the preceding proposition, the full kernel $Q_x$ has exactly sixty-two stationary directions for every $0 < x \leq 1$. They are the twelve port maxima, twenty face-center minima, and thirty edge-center saddles of $\mathcal{I}_6$. Every stationary point is Morse with the same index as the corresponding rank-six critical point.*

*Proof.* Equal port weights make $Q_x$ invariant under the proper carrier rotations. The order-five, order-three, and order-two stabilizers of the vertex, face, and edge axes have no fixed tangent vector. Every one of the sixty-two axis directions is therefore stationary for the full cosine kernel.

After division by the positive amplitude $A(x)$, three exact projective charts cover the sphere modulo antipodes. A deterministic dyadic subdivision into 2,624 leaves proves that the tangent-gradient norm of $\mathcal{I}_6$ exceeds the complete tail bound away from the disjoint neighborhoods $\sin^2\theta \leq 1/4096$ of the 31 axes. No further stationary direction occurs there.

For one representative of each axis orbit, normalized local coordinates give the exact factorization $G(y) = C(y)y$, with $G(0) = 0$. Exact interval bounds on $C(y) - C(0)$, using $\|E\|_2 \leq \sqrt{\|E\|_1\|E\|_\infty}$, leave a strictly positive smallest-singular-value margin after the tail Hessian is included. The local chart obeys $\|J_y\| \leq 1$ and $\mathrm{dist}(0, y) = \arctan\|y\| \leq \|y\|$, so the intrinsic $C_2$ tail bound is the required coordinate-gradient bound. Each neighborhood contains its axis center as its unique stationary direction. Separate exact Hessian margins preserve the two definite signatures and the indefinite edge signature. Carrier symmetry transports the three local certificates to all 31 axes. The interval arithmetic and an independent implementation are recorded in the machine receipt [18]. □

**Theorem 3.4** (Spin-six residue universality on the invariant carrier class)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1 - \cos(a\,k \cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in $a$, and every directional term below angular rank ten is one multiple of $\mathcal{I}_6$. The rank-six Legendre coefficient of $t^{2m}$ is $13\,(2m)!/((2m-6)!!\,(2m+7)!!)$, strictly positive for every $m \geq 3$. For each fundamental orbit the sixth-moment $\mathcal{I}_6$ multiple is nonzero: $64/175$ on the twelve port directions, $-64/189$ on the twenty face directions, and $-2/7$ on the thirty edge directions. Whenever the weighted sixth-moment coefficient is nonzero, the first directional artifact appears at order $a^4k^6$ and one binary refinement step suppresses it by exactly $1/16$ at that order. The exact same-radius positive-weight member with weight $1$ on the twelve port directions and $27/25$ on the twenty face directions cancels the $\mathcal{I}_6$ residue at every order. It is isotropic through $k^8$ and first populates angular rank ten at order $a^8k^{10}$. A separate control, with port*

*radius* 1 *and weight* 1, *face radius* 2 *and weight* 27/1600, *cancels the sixth-moment* $\mathcal{I}_6$ *term while retaining the raw eighth-moment* $\mathcal{I}_6$ *multiple* $-256/125$. *Rank six is the least symmetry-allowed angular rank in the class, although mixtures can cancel it.*

*Proof.* The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $L \le 2m$; the invariant table of Theorem 3.1 forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The positivity of the level-six coefficients, the per-orbit sixth-moment multiples, the refinement factor, the same-radius angular-rank-ten control, and the two-radius eighth-moment control are certified in exact arithmetic [18]. □

On the primitive twelve-port propagation branch, whose premises are stated in the flagship synthesis [15], these identities give the dispersion coefficients $C_4 = -a^2/20$, $B_0 = a^4/840$, and $B_6 = 2a^4/7875$, hence $B_6/C_4^2 = 32/315$, $B_0/C_4^2 = 10/21$, and $B_6/B_0 = 16/75$, with the unique rank-six icosahedral shape oriented in $SO(3)/A_5$. The scalar or polarization-independent sector bridge, coherent frame transport, and exclusivity are declared physical premises of that branch. Here spin six denotes spherical-harmonic angular rank, not particle spin.

For positive shell mixtures, the quartic coefficient instead reads $C_4 = -(a^2/20)\mu_4/\mu_2$, with $\mu_m = \sum_s w_s|O_s|r_s^m$. A bound therefore constrains $a_{\text{eff}} = a\sqrt{\mu_4/\mu_2}$; a bound on the coordinate $a$ requires independent control of the moment ratio. Long-wavelength numerical evaluation uses $1 - \cos x = 2\sin^2(x/2)$, the analytic zero-mode limit, and the small-argument series for derivative corrections, so cancellation does not turn the continuum mode into a spurious zero-frequency mode.

The finite repair operator acts on thirty internal seams, not on spatial translations, and no finite artifact joins a complete twelve-port translation operator to a physical readout of that same operator. Equal-weight stencils on the twelve vertex, twenty face and thirty edge direction orbits have distinct rank-six coefficient rays; proper-carrier transitivity fixes weights within a chosen orbit and does not choose among these spatial alternatives.

### 3.1 The complete seam symbol

**Theorem 3.5** (Conditional homogeneous Dirichlet generator)**.** *Suppose the admissible directed-seam move laws and their objective are natural under the proper carrier action, and suppose the normalized admissible minimizer is strictly positive and unique. The weight on each of the sixty directed seams is then* 1/60. *Let*

$$(Pf)(x) = \frac{1}{60}\sum_e f(x + v_e), \qquad L = I - P,$$

*where* $v_e$ *is the exact response-coordinate translation of a directed seam. Then* $P$ *preserves constants and positivity,* $L$ *is translation invariant, and*

$$2f(x)(Lf)(x) - L(f^2)(x) = \frac{1}{60}\sum_e \big(f(x + v_e) - f(x)\big)^2 \ge 0.$$

*Plane waves diagonalize* $L$. *If* $a_{\text{edge}}^2 = 2 - 2/\sqrt{5}$ *denotes the exact squared response-Gram norm of a seam translation, then*

$$\frac{6}{a_{\text{edge}}^2}\lambda_L(k)$$

*is exactly the normalized thirty-edge cosine character in the response coordinates. A supplied nonzero coordinate dilation* $a$ *gives*

$$\Lambda_a(k, n) = \frac{1}{5a^2}\sum_{j=1}^{30}\left[1 - \cos(ak\, w_j \cdot n)\right], \tag{1}$$

*where the $w_j$ are the unit edge directions. The dilation is a coordinate choice until a physical position map fixes its meaning.*

*Proof.* Transitivity and uniqueness make the selected weight constant on the directed seam orbit, and normalization fixes it to $1/60$. Reindexing the directed orbit gives one positive and one negative translation for each of the thirty seams. The square identity follows by expanding $L = I - P$. The plane-wave identity is the exact character calculation for those translation pairs. The response Gram theorem gives the displayed seam norm. Exact source incidence, selection, translation, and Fourier identities are machine-checked in the finite proof chain [18]. □

The naturality and unique-minimizer clauses in Theorem 3.5 are explicit premises. The finite source fixes the orbit and its response metric; it does not prove that this homogeneous internal action is a physical field action.

**Theorem 3.6** (Global contraction of a complete positive cosine auxiliary frequency)**.** *Let $\{(\rho_i, v_i)\}_{i\in I}$ be a finite weighted family in the Euclidean carrier chart with $\rho_i > 0$, and suppose that*

$$\sum_{i\in I} \rho_i (v_i \cdot x)^2 = t|x|^2 \qquad \text{for every } x,$$

*where $t > 0$. For $a \neq 0$, set*

$$\Lambda_a(k) = \frac{2}{ta^2} \sum_{i\in I} \rho_i [1 - \cos(a v_i \cdot k)]$$

*and define the sine-feature coordinates*

$$\Phi_{a,i}(k) = \sqrt{\frac{4\rho_i}{a^2 t}} \sin\left(\frac{a v_i \cdot k}{2}\right).$$

*Then*

$$\Lambda_a(k) = \|\Phi_a(k)\|^2, \qquad \|\Phi_a(k) - \Phi_a(p)\| \le |k - p|.$$

*Consequently the nonnegative auxiliary frequency $\Omega_a(k) = \|\Phi_a(k)\|$ satisfies*

$$|\Omega_a(k) - \Omega_a(p)| \le |k - p| \qquad \text{for all } k, p.$$

*The complete primitive-port support has $t = 4$ and prefactor $1/(2a^2)$. The complete seam support has $t = 10$ and prefactor $1/(5a^2)$, reproducing Equation (1).*

*Proof.* The half-angle identity gives $\Lambda_a(k) = \|\Phi_a(k)\|^2$. The global sine-difference bound controls each coordinate difference by $\sqrt{\rho_i/t}\,|v_i \cdot (k - p)|$. Summing the squared bounds and applying the tight-frame identity gives the contraction. The reverse triangle inequality for norms gives the frequency bound. The two support constants follow from their exact second-moment identities. The feature identity, contraction, support constants, and symbol bindings are machine-checked, and an independent exact verifier reconstructs the two finite supports [18]. □

The unit constant is a certified upper bound. No optimality theorem is used. The variables in Theorem 3.6 belong to the selected Euclidean carrier chart. A physical reading requires position and frequency maps, a clock, a field sector, wave-packet or signal- front dynamics, a frame and boost law, a finite physical scale, source-lag and detector models, and a frozen comparison contract.

**Proposition 3.7** (Exact unit-domain remainder)**.** *Put $q = ak$, let $-5/9 \le \mathcal{I}_6(n) \le 1$, and write*

$$\widehat{\Lambda}(q,n) = \frac{1}{5}\sum_{j=1}^{30}[1-\cos(qw_j\cdot n)].$$

*The exact eighth moment is*

$$M_8(n) = \frac{10}{3} - \frac{8}{15}\mathcal{I}_6(n).$$

*Define*

$$P_6(q,n) = q^2 - \frac{q^4}{20} + \left(\frac{1}{840} - \frac{\mathcal{I}_6(n)}{12600}\right)q^6,$$
$$P_8(q,n) = P_6(q,n) + \left(-\frac{1}{60480} + \frac{\mathcal{I}_6(n)}{378000}\right)q^8.$$

*Uniformly for $0 \le q \le 1$,*

$$|\widehat{\Lambda} - P_6| \le \frac{7}{388800}q^8, \qquad 0 \le \widehat{\Lambda} - P_8 \le \frac{7}{34992000}q^{10}, \qquad \frac{19}{20}q^2 \le \widehat{\Lambda} \le q^2.$$

*In the convention*

$$\Lambda_a(k,n) = k^2 + C_4k^4 + [B_0 + B_6\mathcal{I}_6(n)]k^6 + O(a^6k^8),$$

*the low-momentum coefficients obey*

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = -\frac{a^4}{12600}.$$

*Proof.* The exact edge-orbit moments through order eight reduce to the isotropic line and the unique rank-six invariant. The alternating Taylor bounds for cosine, together with the exact range of $\mathcal{I}_6$, give the two remainder inequalities and the positive quadratic bounds. The moment identity and the remainder arithmetic are independently replayable in exact arithmetic and checked by the Lean theorem chain [18]. □

**Proposition 3.8** (Exhaustive fixed-design recovery calibration)**.** *Consider the twelve design rows formed by*

$$q \in \left\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right\}, \qquad \mathcal{I}_6 \in \left\{1, -\frac{5}{9}, -\frac{5}{16}\right\},$$

*and the three-column model for $C_4/a^2$, $B_0/a^4$, and $B_6/a^4$. Enumerate all $2^{12} = 4096$ independent sign-error vectors with amplitude $1/200$. Exact ordinary least squares and exhaustive $19/20$ interval calibration give coordinate coverage counts $3904/4096$, $3892/4096$, and $3892/4096$. The corresponding nonzero-signal detection counts are $4096/4096$, $234/4096$, and $195/4096$. The joint higher-order pair has coverage $3892/4096$ and detection count $209/4096$. The exact remainder of Proposition 3.7 cannot erase the leading-coefficient detection margin in this design.*

This is a finite synthetic calibration with no sampling error. The sign-error law is a declared dimensionless stress model. The result gives no continuous-direction coverage, detector response, nuisance model, tail calibration at discovery significance, or experimental sensitivity claim. It rejects zero for the leading coefficient throughout this design and does not determine the linked higher-order pair.

One conditional metric theorem records the exact relation available if a single physical similarity metric is attached to the normalized port frame, the primitive vertex action, and the seam-current action. Let $a_{\mathrm{cell}}$ be one physical cut-area element, let $s^2 > 0$ be the squared physical length represented by one response-metric unit, and suppose a port-dual sector obeys $12a_{\mathrm{cell}} = 4\pi s^2$. If the vertex action has squared step $a_v^2 = s^2$ and the seam action uses the same metric, then

$$a_v^2 = \frac{3}{\pi} a_{\mathrm{cell}}, \qquad a_e^2 = \frac{3}{\pi}\left(2 - \frac{2}{\sqrt{5}}\right) a_{\mathrm{cell}}, \qquad \frac{a_e^2}{a_v^2} = 2 - \frac{2}{\sqrt{5}}.$$

With the additional definition $a_{\mathrm{cell}} = P\ell^2$, where $\ell > 0$,

$$\frac{a_v^2}{\ell^2} = \frac{3}{\pi} P, \qquad \frac{a_e^2}{\ell^2} = \frac{3}{\pi}\left(2 - \frac{2}{\sqrt{5}}\right) P.$$

No source theorem constructs this joint attachment or identifies $\ell$ with a physical length. Equal normalized port weights and seam weights do not imply it.

A direct arrival-time comparison uses one-particle propagation and needs a physical position map, clock, field sector and propagation equation, a controlled frequency remainder, a frame and boost law, cosmological transport, emission-time and detector models, nuisance treatment and an exclusive readout. A pair-production threshold or opacity comparison additionally needs physical electron and positron dispersion, shared energy-momentum conservation, interactions, a cross section, background-radiation transport and source composition. The finite seam theorems supply none of these attachments; the next proposition records the exact kinematics available once they are declared.

**Proposition 3.9** (Conditional photon-decay exclusion and electron-positron threshold)**.** *Let the nonnegative square root of* (1) *be attached to a physical photon frequency in the Euclidean carrier metric, with nonzero a. For* $\mathbf{k} = kn$, $k \geq 0$, *and* $|n| = 1$, *the exact finite symbol satisfies*

$$0 \leq \Omega_\gamma(k, n)^2 = \Lambda_a(k, n) \leq k^2.$$

*Suppose energy and momentum compose additively in the declared frame and the electron and positron have positive mass and Lorentz-invariant positive-energy dispersion. A photon cannot decay into an electron-positron pair. For two incoming photons, any declared final-state energy within the seam-current incoming-energy budget does not exceed the Lorentz-invariant incoming-energy budget at the same momenta. The exact upper bound is global for nonzero a; the separately certified Taylor-remainder interval uses* $0 \leq |a|k \leq 1$.

*For a separate leading ultrarelativistic calculation, use* $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2} E_i^4$ *through the retained order, with* $i \in \{\gamma, +, -\}$, $m_\gamma = 0$, $m_+ = m_- = m_e$, $E$ *the hard-photon energy, and* $\epsilon$ *the soft-photon energy. Treat the soft background photon as Lorentz invariant at leading order and allow independent dimension-six coefficients* $\delta_{\gamma,2}, \delta_{+,2}, \delta_{-,2}$. *Head-on incoming momenta, collinear outgoing momenta, additive conservation, and a fixed positron energy share* $0 < x < 1$ *give*

$$\left[\delta_{\gamma,2} - x^3\delta_{+,2} - (1-x)^3\delta_{-,2}\right] E^4 + 4\epsilon E - \frac{m_e^2}{x(1-x)} = 0.$$

*At* $x = 1/2$, *the visible coefficient is*

$$\delta_{\gamma,2} - \frac{\delta_{+,2} + \delta_{-,2}}{8}.$$

*This map has rank one and a two-dimensional coefficient fiber. On the Lorentz-invariant charged-lepton branch, $\delta_{+,2} = \delta_{-,2} = 0$, and for $m_e > 0$, $0 < x < 1$,*

$$\frac{m_e^2}{x(1-x)} - 4m_e^2 = \frac{m_e^2(2x-1)^2}{x(1-x)}.$$

*Hence equal sharing is the unique global maximizer of the leading head-on, collinear threshold residual on this branch.*

*On the separate universal negative-coefficient branch $\delta_{\gamma,2} = \delta_{+,2} = \delta_{-,2} = -d$, $d > 0$, put $u = x(1-x)$. The same residual vanishes exactly when*

$$\epsilon = \frac{m_e^2}{4Eu} + \frac{3dE^3}{4}u, \qquad 0 < u \le \frac{1}{4}.$$

*Every such value is bounded below by the reciprocal-linear envelope. Equal sharing minimizes in the endpoint regime $3dE^4 \le 16m_e^2$; above the transition $E_t = 2\sqrt{m_e}(3d)^{-1/4}$, the interior balance $u_* = m_e/(\sqrt{3d}\,E^2)$ gives $\epsilon_{\mathrm{env}} = (\sqrt{3}/2)m_e E\sqrt{d}$. A soft energy below this envelope cannot solve the declared leading residual at any open share. At and above the transition the balanced equality witness is itself realized by an open share, so the envelope is the constrained physical-share minimum there.*

*Proof.* The elementary inequality $1 - \cos y \le y^2/2$, together with the exact second moment of the thirty normalized seam directions, gives $\Lambda_a(k,n) \le k^2$ in the same Euclidean metric used by the carrier completion. A positive-mass lepton pair with Lorentz-invariant positive-energy dispersion has energy strictly greater than the magnitude of its total momentum. The triangle inequality gives the decay exclusion and the incoming-energy-domain inclusion. Expanding the three dispersion relations through first order in the dimension-six coefficients and enforcing the declared head-on momentum balance gives the displayed threshold equation. Its equal-share row is $(1, -1/8, -1/8)$, which has rank one and a two-dimensional kernel. The displayed square identity proves the Lorentz-invariant-lepton optimizer. The common-coefficient cubic identity reduces the coefficient to $-3dx(1-x)$. The bound $0 < u \le 1/4$, an exact square-factor AM–GM identity, its balanced witness, the endpoint comparison, transition, and no-solution statement prove the universal-branch envelope; a separate existence theorem closes physical-share attainment above the transition. The use of $E_i^4$ in this threshold equation is the retained first-order, ultrarelativistic rewrite of the carrier's $p_i^4$ correction, not an exact identity of the finite dispersion. The complete identities and metric typing are checked in Lean, with deterministic receipt calculations for the coefficient algebra and representative roots [18]. □

The conditional kinematic theorem supplies no interaction, rate or detector content for physical photons or charged leptons. It supplies neither the general independent-coefficient share optimization nor the full anisotropic minimization over outgoing three-momentum, does not identify the three dispersion coefficients separately, and gives no vertex, opacity, shower or empirical exclusion. The particle layer uses its leading window [16].

## 4 Exact seam electromagnetism on the carrier

The seam symbol, the incidence complex and the oriented faces carry an exact finite counterpart of each ingredient of a Maxwell theory. One composed machine-checked surface (`LightSignalAdequacySurface`) takes the equal source-counting measure, an auxiliary oscillator lift and a physical-frequency identification as declared inputs and re-exports the forced weight, the

dimensionless generator, the rank-two transverse completion with its exact zero mode, and a second-order wave-shaped evolution as one package; its companion (`LightSignalMaxwellComposition`) carries the supplied mode dynamics onto the opposite-sign curl pairing of Proposition 4.2. The forced counting measure is orientation-balanced across the double cover of the thirty seams, and its expected signed port boundary vanishes on the same incidence object that carries the Gauss receipts below. Seam connections modulo port gauge are classified by their nineteen chord holonomies (`SeamU1HolonomyClassification`): the first cohomology of the seam graph with circle coefficients is $U(1)^{19}$, matching the Gauss cycle rank. That connection is a finite kinematic object with no curvature dynamics or physical field identification.

**Proposition 4.1** (Conditional basis-free transverse oscillator)**.** *For the nonzero coordinate dilation of Theorem 3.5 and nonzero $k \in \mathbb{R}^3$, let*

$$T_k = \{v \in \mathbb{R}^3 : k \cdot v = 0\}.$$

*Then $\dim_{\mathbb{R}} T_k = 2$, and the basis-free orthogonal projector onto $T_k$,*

$$\Pi_T(k)v = v - \frac{k \cdot v}{k \cdot k} k,$$

*is idempotent. The scalar symbol $\Lambda_a(k)$ preserves $T_k$ and acts equally on its two dimensions. If one declares the canonical oscillator generator*

$$(A_T, V_T)' = (V_T, -\Lambda_a(k) A_T),$$

*then*

$$A_T'' + \Lambda_a(k) A_T = 0, \qquad \omega_a(k)^2 = \Lambda_a(k).$$

*The spatial symbol vanishes at $k = 0$, hence its nonnegative square-root branch satisfies $\omega_a(0) = 0$. The algebraic first variation of*

$$E_k = \frac{1}{2}\left(|V_T|^2 + \Lambda_a(k)|A_T|^2\right)$$

*vanishes on that generator.*

The proposition chooses the spatial symbol as an oscillator stiffness. It constructs no physical dynamics, covariance, clock, or particle content. The zero mode is an exact property of the declared mathematical oscillator. A physical photon statement requires the missing sector, action, clock, position, frame, quantization, and pole attachments.

There is nevertheless one exact first-order factorization of this modal oscillator that is useful for locating the remaining electromagnetic gap. Complexify $T_k$, and, for $k \neq 0$, define

$$D_{a,k} v = i \, \frac{\omega_a(k)}{\sqrt{k \cdot k}} \, k \times v.$$

The coefficient is the committed nonnegative square root $\omega_a(k)^2 = \Lambda_a(k)$; it is not a new kinetic symbol.

**Proposition 4.2** (Exact modal Maxwell-shaped factorization and boundary)**.** *For every nonzero $k$, $k \cdot D_{a,k} v = 0$. On the complexified transverse fibre,*

$$D_{a,k}^2 v = \Lambda_a(k) v.$$

*Consequently the opposite-sign paired generator*

$$\mathcal{G}_{a,k}(E,B) = (D_{a,k}B, -D_{a,k}E)$$

*obeys*

$$\mathcal{G}_{a,k}^2(E,B) = -\Lambda_a(k)(E,B),$$

*and preserves both modal transversality constraints. The same-sign mutation* $\mathcal{G}_{a,k}^+(E,B) = (D_{a,k}B, D_{a,k}E)$ *instead squares to* $+\Lambda_a(k)(E,B)$*; whenever the spatial action on the tested amplitude is nonzero, it cannot obey the displayed oscillator equation.*

*Proof.* The vector triple-product identity gives $k \times (k \times v) = -|k|^2 v$ on $T_k$, while $k \cdot (k \times v) = 0$. Multiplication by $i\omega_a(k)/|k|$ therefore gives $D_{a,k}^2 = \omega_a(k)^2 = \Lambda_a(k)$ on that fibre. Squaring the two off-diagonal sign choices proves the paired statements. The universal identities, scalar-extension bridge, transversality receipts, and mutation control are checked in Lean; an independent off-axis Gaussian-rational replay checks both sign and normalization mutations [18]. □

This is an exact pointwise Fourier-modal, hence generally pseudodifferential, factorization. The factor $\omega_a(k)/|k|$ is momentum dependent, so the proposition does not prove a local position-space curl operator. It supplies neither a reality pairing between the $k$ and $-k$ modes nor an assembled real field; it identifies neither amplitude as a physical electric or magnetic field. It also supplies no source-produced dynamics, U(1) potential or gauge quotient, Maxwell action, bridge from the rational seam-incidence Gauss receipts to this complex modal divergence, conserved physical current or source coupling, Lorentz covariance, continuum control, or laboratory readout. Propositions 4.4 through 4.6 commit exact finite seam-carrier counterparts of the gauge quotient, a local face-curvature action with conserved-source coupling, and the canonical Gauss solution; the construction supplies no bridge from those receipts to this complex modal divergence or any physical attachment listed above. The result depends on every stated premise and implies no empirical prediction. The construction supplies no physical local-field, gauge-action, source-current, continuum, or readout attachment.

**Theorem 4.3** (Continuum Maxwell limit of the complete seam symbol)**.** *Supply one Euclidean domain* $\mathbb{R}^3$ *with Lebesgue measure, the complete equal-weight seam symbol* $\Lambda_a$ *at every* $a > 0$*, and the reversible curl-pair generator above. Write* $r = |k|$*,* $\omega_a = \sqrt{\Lambda_a}$ *and* $P_T(k)$ *for the orthogonal transverse projection on both field components. Use the convention* $P_T(0) = 0$*. For* $r > 0$ *let* $\mathcal{J}(k) = \mathcal{G}_{0,k}/r$*, where* $\mathcal{G}_{0,k}(E,B) = (ik \times B, -ik \times E)$*; set* $\mathcal{J}(0) = 0$*. On the full six-component space define*

$$U_a(t,k) = I + (\cos(t\omega_a) - 1)P_T + \sin(t\omega_a)\mathcal{J}.$$

*At* $k = 0$ *this is the identity. These multipliers define strongly continuous unitary groups on* $L^2(\mathbb{R}^3;\mathbb{C}^6)$ *preserving real fields and both divergence-free constraints when those constraints hold initially. For every* $k$ *and* $a > 0$*,*

$$0 \le r^2 - \Lambda_a(k) \le a^2 r^4/20, \tag{2}$$

$$0 \le r - \omega_a(k) \le a^2 r^3/20, \tag{3}$$

$$\|U_a(t,k) - U_0(t,k)\| \le \min\{2, |t|a^2 r^3/20\}. \tag{4}$$

*Here* $U_0$ *is the local Maxwell group, with longitudinal components fixed. In particular, for* $F \in H^3(\mathbb{R}^3;\mathbb{C}^6)$*,*

$$\sup_{|t|\le T} \|(U_a(t) - U_0(t))F\|_2 \le \frac{Ta^2}{20}\,\||D|^3 F\|_2, \qquad T \ge 0,$$

*where $|D|^3$ has Fourier multiplier $r^3$. For every $F \in L^2$ the left side tends to zero as $a \to 0$, without an imposed momentum cutoff.*

*For the same initial datum $F$ and the same prescribed forcing $f \in L^1([0,T]; H^3)$ on every scale, the mild solutions satisfy*

$$\sup_{0\le t\le T} \|F_a(t) - F_0(t)\|_2 \le \frac{a^2}{20}\left(T\,\||D|^3 F\|_2 + \int_0^T (T-s)\,\||D|^3 f(s)\|_2\,ds\right).$$

*Strong convergence also holds for $F \in L^2$ and $f \in L^1([0,T]; L^2)$. For $f = (-j, 0)$ and a prescribed charge obeying $\partial_t \rho + \nabla \cdot j = 0$, the initially satisfied constraints $\nabla \cdot E = \rho$ and $\nabla \cdot B = 0$ propagate, in the distributional sense, at every scale and in the limit.*

*Proof.* The complete source support has $\sum_e (k \cdot w_e)^2 = 10r^2$ and $\sum_e (k \cdot w_e)^4 = 6r^4$. The global inequalities

$$1 - x^2/2 \le \cos x \le 1 - x^2/2 + x^4/24$$

give (2) after summing the full cosine symbol. The upper cosine inequality follows by integrating $\cos x \ge 1 - x^2/2$ twice from zero and using evenness; it requires no small-argument restriction. Positivity of $\Lambda_a$ is termwise. For $r > 0$, factor $r^2 - \omega_a^2 = (r - \omega_a)(r + \omega_a)$ and use $r + \omega_a \ge r$ to obtain (3); at zero both frequencies vanish. These universal inequalities and their sine/cosine propagator consequences are checked in the Lean module `SeamMaxwellContinuum`.

The Fourier curl is Hermitian, so $\mathcal{J}^* = -\mathcal{J}$, $\mathcal{J}^2 = -P_T$, and $\mathcal{J}$ vanishes on the longitudinal sector. The displayed formula is therefore $\exp(t\omega_a \mathcal{J})$, is unitary, and obeys the group law. Its transverse eigenvalues are $e^{it\omega_a}$ and $e^{-it\omega_a}$. Thus its difference from $U_0$ has norm $2|\sin(t(\omega_a - r)/2)|$, proving (4). The evenness of $\omega_a$ and $P_T$, and $\mathcal{J}(-k) = \overline{\mathcal{J}(k)}$, preserve $\widehat{F}(-k) = \overline{\widehat{F}(k)}$ and hence real fields. The zero mode is fixed, not divided by its frequency. Pointwise time continuity and the bound two give strong continuity by dominated convergence.

Apply Plancherel to (4) for the $H^3$ bound. For arbitrary $L^2$ data, its square is dominated by $4|\widehat{F}(k)|^2$ and tends to zero uniformly for $|t| \le T$ at each fixed $k$. Dominated convergence proves the uniform strong limit. Apply the same bound inside Duhamel's integral for the forcing estimate; the $L^1_t L^2_x$ result follows with the bound $2\|f(s)\|_2$. Finally, $k \cdot (k \times v) = 0$ gives $\partial_t \nabla \cdot E = -\nabla \cdot j$ and $\partial_t \nabla \cdot B = 0$; the stipulated charge continuity gives Gauss propagation. The field assembly and functional-analytic limit are the paper argument just given, rather than additional Lean declarations. Independent matrix-exponential controls test the full propagator, constant forcing, reality, zero mode, high momenta, and sign and normalization mutations. □

The theorem recovers a local continuum field equation from the complete source-shaped operator after specifying a common spatial domain and reversible dynamics. The scale maps act on that same domain; they are not refinements of authenticated source logs. The finite-scale curl is generally pseudodifferential and no exact finite propagation cone is inferred from its frequency bound. The Markov generator also available from the seam action does not select the reversible curl-pair dynamics. The supplied current is not generated by observer repair, and no equality with the finite seam-incidence Gauss/action/readout packet is proved. Physical position, operational time, charge units, detector records, and an empirical realization of this field family require additional maps.

The two seam-carrier packages are premise-free exact finite mathematics on the committed thirty-seam, twelve-port incidence table. Write $\partial$ for the committed rational seam boundary, $d$ for its exact transpose, $\langle x, y\rangle = \sum_e x_e y_e$ for the equal-weight seam pairing, and $\|F\|^2 = \langle F, F\rangle$ for the seam energy.

**Proposition 4.4** (Exact rational Coulomb Green function of the committed seam Laplacian)**.** *The vertex Laplacian $L = \partial d$ of the committed incidence gives every port degree five, and an explicit symmetric rational Green matrix $G$, with integer entries over the common denominator* 180*, satisfies exactly*

$$LG = GL = I - \tfrac{1}{12}\,\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad G^{\mathsf{T}} = G, \qquad G\mathbf{1} = 0,$$

*and the kernel of $L$ is exactly the constant loads, proved along the committed spanning tree through the energy identity. For every neutral rational port load $\rho$, the seam field $F_\rho = d(G\rho)$ solves the committed discrete Gauss problem $\partial F_\rho = \rho$; it is the unique Gauss solution orthogonal to every source-free cycle, and it is the unique Gauss solution of minimal seam energy, by the exact Thomson decomposition*

$$\|F\|^2 = \|F_\rho\|^2 + \|F - F_\rho\|^2 \qquad \text{for every solution } F \text{ of } \partial F = \rho.$$

*This supplies the canonical choice absent from the committed discrete Gauss layer of Theorem 2.10: each nineteen-dimensional solution fibre carries exactly one cycle-orthogonal point, equivalently exactly one energy-minimal point. The exact adjacent-port dipole receipt across seam zero has potential classes indexed by the pair of graph distances to the two poles, taking exactly the values*

$$\frac{11}{60}, \qquad \frac{1}{20}, \qquad \frac{1}{60}, \qquad 0$$

*and their negatives, with the displayed list strictly decreasing, and the single-source Green values by graph distance are*

$$\frac{7}{36}, \qquad \frac{1}{90}, \qquad -\frac{7}{180}, \qquad -\frac{1}{18},$$

*strictly decreasing. Finally, the committed uniform seam-repair operator acts on real port loads as $1 - \frac{1}{60}L$ for the real scalar extension of this same Laplacian, and its fixed loads are exactly the constants, that is, exactly the kernel of $L$, exactly the Gauss-obstruction directions.*

*Proof.* The Laplacian matrix is derived from the committed table alone, and the Green identities are kernel-checked integer matrix identities over the denominator 180, cast exactly to $\mathbb{Q}$ and $\mathbb{R}$. The kernel characterization chains constancy along the eleven committed tree seams through the energy identity. The Thomson clauses compose the committed Gauss-fibre description with the displayed orthogonal energy decomposition. The dipole and single-source values are re-derived from the committed incidence by a two-step hop classification proved to coincide with graph distance, and the repair bridge is an exact operator identity against the committed uniform repair. The chain is machine-checked in `DiscreteCoulombGreen` with no native decision procedure [18]. □

The same Green matrix assembles a global real Hodge-projection package on the committed seams. Because the Green matrix is global, this is not a local curvature action.

**Proposition 4.5** (Global Hodge-projector action and separate Coulomb solution)**.** *Let real position-space seam fields be real-valued functions $A$ on the thirty committed seams, and let a real port gauge function $\chi$ act by $A \mapsto A + d\chi$ through the real cast of the committed incidence. The field-strength projector $P = 1 - dG\partial$ is idempotent and self-adjoint for the equal-weight seam pairing; its kernel is exactly the image of $d$, and its range is exactly the kernel of $\partial$, the cycle space, of real dimension nineteen. The projected cycle representative $PA$ is gauge invariant and determines $A$ exactly up to gauge, and the quotient of position fields by gauge is linearly equivalent to the cycle space, in exact dimension agreement with the committed finite $U(1)^{19}$ chord-holonomy*

*classification, of which it is the real Lie-algebra/dimension companion rather than an identification with the compact torus. For every source $J$ on the seams, the sourced quadratic action*

$$S_J(A) = \frac{1}{2}\,\|PA\|^2 - \langle J, A\rangle$$

*is gauge invariant exactly when $\partial J = 0$, in both directions; it obeys the exact quadratic expansion*

$$S_J(A+h) = S_J(A) + \langle PA - J,\, h\rangle + \frac{1}{2}\,\|Ph\|^2$$

*with no remainder; its first variation at $A$ vanishes identically exactly when $PA = J$; the stationarity equation is solvable exactly when $J$ is conserved, and then $A = J$ is a solution; any two solutions differ exactly by a gauge; and every solution is a global minimum, with equality exactly on the gauge orbit. Separately, for a neutral rational port load, the real cast of the Coulomb field of Proposition 4.4 is the unique minimal-energy solution of the Gauss problem. It is a coboundary and hence obeys $PE_\rho = 0$: it is not a stationary electrostatic sector of the displayed sourced Hodge action.*

*Proof.* Idempotence, self-adjointness, and the kernel and range identifications follow from the Green identities and the adjointness of $d$ and $\partial$; rank-nullity through the projector gives dimension nineteen, agreeing with the chord count and the rational cycle rank. The gauge-invariance equivalence and the expansion and variational clauses are exact for the global Hodge action, and $PE_\rho = 0$ pins the non-join with the Gauss/Thomson statement of Proposition 4.4. The chain is machine-checked in `PositionSpaceMaxwellAction`, whose composed theorem `positionSpaceMaxwellCoulomb_receipt` carries all clauses as one typed conjunction; an independent exact-rational verifier replays the Green, Coulomb, Thomson and repair values with different linear algebra [18]. □

The committed oriented twenty-face incidence supplies the local operator that the Green-built Hodge projector does not. Let $C : \mathbb{R}^{30} \to \mathbb{R}^{20}$ be its signed face–seam matrix. Each face row has three entries, each seam occurs in two faces with opposite orientations, and the boundary-of-boundary identity is $Cd = 0$.

**Proposition 4.6** (Local face-curvature action and separate scalar Coulomb sector)**.** *On the committed oriented face complex, $C$ has rank nineteen and*

$$\ker C = \operatorname{im} d.$$

*The local kinetic operator $H = C^{\mathsf{T}}C$ has exactly five nonzero entries in every seam row, 150 nonzero entries in its $30 \times 30$ matrix, and diagonal entry two. For a seam current $J$,*

$$S_J^{\mathrm{face}}(A) = \frac{1}{2}\|CA\|^2 - \langle J, A\rangle$$

*is gauge invariant under $A \mapsto A + d\chi$ exactly when $\partial J = 0$. Its stationarity equation $HA = J$ is solvable exactly for those conserved currents; stationary points are global minimizers and are unique modulo gradients.*

*Independently, for a port load $\rho$, the scalar action*

$$S_\rho^{\mathrm{scal}}(\phi) = \frac{1}{2}\|d\phi\|^2 - \langle \rho, \phi\rangle$$

*has stationarity equation $L\phi = \rho$. For neutral $\rho$, the canonical potential $\phi = G\rho$ is stationary and globally minimizing, uniquely modulo constants. The direct-product static action therefore combines a local vector face-curvature sector and the exact scalar Coulomb sector while retaining $J$ and $\rho$ as different typed sources.*

*Proof.* The rank calculation is exhibited by an explicit dual-tree section of the codimension-one face-sum space. Rank–nullity and $Cd = 0$ then identify the kernel with the eleven-dimensional gradient space. The local support counts, adjoint identity, range characterization, gauge equivalence, stationarity, and minimum statements are checked in `LocalFaceMaxwellAction`. An independent exact-rational replay reconstructs both incidences, their ranks and nullspaces, the complete support patterns, the Green solve, and the characteristic factors of $CC^{\mathsf{T}}$ and $\partial\partial^{\mathsf{T}}$. Erasing the signed orientation of one face boundary produces a machine-checked nonzero boundary-of-boundary control. □

**Proposition 4.7** (Conditional temporal identities and continuity on the committed carrier)**.** *Index seam potentials $A_n \in \mathbb{R}^{30}$ and port potentials $\phi_n \in \mathbb{R}^{12}$ by a declared evolution step $n \in \mathbb{N}$, and set*

$$E_n = -(A_{n+1} - A_n) - d\phi_n, \qquad B_n = CA_n,$$

*so that $E$ sits on half steps and $B$ on integer steps. The leapfrog Ampère law $E_{n+1} - E_n = C^{\mathsf{T}}B_{n+1} - J_n$ for a seam-current history $J$ is the unit-step instance of the declared discrete action (Proposition 4.8). Then Faraday's law $B_{n+1} - B_n = -CE_n$ holds identically. Along the declared evolution, the Gauss constraint $\partial E_n = \rho_n$ propagates from step $n$ to step $n+1$ exactly when the discrete continuity equation*

$$\rho_{n+1} - \rho_n + \partial J_n = 0$$

*holds, in both directions; the port load $\rho$ and the seam current $J$ remain distinct types, and this equivalence is their only bridge. The staggered quadratic form $\mathcal{E}_n = \frac{1}{2}\|E_n\|^2 + \frac{1}{2}\langle B_n, B_{n+1}\rangle$ satisfies the exact per-step balance $\mathcal{E}_{n+1} = \mathcal{E}_n - \frac{1}{2}\langle E_n + E_{n+1}, J_n\rangle$, hence is exactly conserved for $J = 0$. No positivity claim is made. The time-dependent gauge transformation $A_n \mapsto A_n + d\chi_n$, $\phi_n \mapsto \phi_n - (\chi_{n+1} - \chi_n)$ leaves $E$, $B$, the evolution law, and $\mathcal{E}$ invariant. A constant-in-time solution with $J = 0$ has $B_n = 0$, a forced-neutral load, and $E_n$ exactly the canonical Coulomb field $dG\rho$; conversely every neutral $\rho$ admits such a static solution. Zero-current evolution obeys the exact discrete wave law*

$$A_{n+2} - 2A_{n+1} + A_n + C^{\mathsf{T}}CA_{n+1} + d(\phi_{n+1} - \phi_n) = 0.$$

*Proof.* The Faraday, continuity-equivalence, Gauss-propagation, quadratic-form, gauge, static-join, and wave clauses are checked in `TemporalMaxwellEvolution` as one composed receipt from a single typed antecedent bundle; the neutral-load static existence and an explicit nonstatic inhabitant are separate theorems of the same module, the inhabitant being zero initial data kicked by the boundary cycle of one committed face, with magnetic field zero at step zero and three on that face at step one. Faraday's law and the Gauss propagation use $Cd = 0$ and its exact transpose $\partial C^{\mathsf{T}} = 0$; the balance identity uses the adjointness of $C$ and $C^{\mathsf{T}}$ under the committed equal-weight pairing together with the staggered magnetic cross term $\langle B_n, B_{n+1}\rangle$, which makes the balance exact. This conservation law is not a stability theorem at unit step: the committed exact spectrum contains 5 and $3 + \sqrt{5}$, both above the unit-step leapfrog threshold 4, so growing modes exist. The timestep-scaled continuation with the sharp threshold is the next proposition. □

**Proposition 4.8** (Action origin and sharp stability of the step-scaled evolution)**.** *Fix a real step $h > 0$ and set $E_n = -(A_{n+1} - A_n)/h - d\phi_n$, $B_n = CA_n$. Let $S_h$ be the window sum of $\frac{h}{2}\|E_n\|^2 - h\,\mathcal{S}_{\mathrm{loc}}(J_n, A_{n+1}) + h\langle\rho_n, \phi_n\rangle$, with $\mathcal{S}_{\mathrm{loc}}$ the local sourced face action above. The exact expansion of $S_h$ has a linear part and an explicit quadratic remainder; stationarity under variations of $A$ vanishing at both window endpoints is equivalent to $E_{n+1} - E_n = h\,(C^{\mathsf{T}}B_{n+1} - J_n)$ at every interior step, and stationarity under variations of $\phi$ is equivalent to $\partial E_n = \rho_n$ at every window step. The form $\mathcal{E}_n = \frac{1}{2}\|E_n\|^2 + \frac{1}{2}\langle B_n, B_{n+1}\rangle$ obeys $\mathcal{E}_{n+1} = \mathcal{E}_n - \frac{h}{2}\langle E_n + E_{n+1}, J_n\rangle$, equals $\frac{1}{2}\|E_n\|^2 + \frac{1}{8}\|C(A_n +$*

$A_{n+1})\|^2 - \frac{h^2}{8}\|CE_n\|^2$, *and, whenever* $\|Cv\|^2 \le \Lambda\|v\|^2$ *with* $h^2\Lambda < 4$ *and* $J = 0$, *bounds* $\|E_n\|^2 \le 8\mathcal{E}_0/(4 - h^2\Lambda)$ *and* $\|B_n\|^2 \le 16\mathcal{E}_0/(4 - h^2\Lambda)$ *at every step. On the committed carrier the sharp constant is* $\Lambda = 3 + \sqrt{5}$*: the five spectral projectors of* $CC^{\mathsf{T}}$ *are explicit rational matrices whose identities are kernel-checked, and the golden sector obeys* $\|(N - 3)u\|^2 = 5\|u\|^2$. *Conversely, an eigenvector* $v$ *of* $C^{\mathsf{T}}C$ *with eigenvalue* $\lambda$ *and* $h^2\lambda > 4$ *gives the zero-current solution* $A_n = r^n v$ *with* $r < -1$ *and unbounded* $\|E_n\|^2$. *At* $h^2\lambda = 4$, *the generalized mode* $A_n = (-1)^n nv$ *has electric seam energy proportional to* $(2n + 1)^2$. *On the committed carrier, for every nonzero* $h$, *uniform boundedness of electric seam energy for all zero-current data is therefore equivalent to the strict inequality* $h^2(3 + \sqrt{5}) < 4$*; within that window the magnetic face energy is uniformly bounded as above. No bound on the gauge potential* $A$ *follows, since kernel histories may shear. Explicit eigenvectors with eigenvalues* $5$ *and* $3 + \sqrt{5}$ *are exhibited, and the unit step is unstable. The kinetic term and the step are declared; the step index is not physical time; the construction supplies no source, frame, continuum, or readout attachment.*

*Proof.* Every clause is a theorem of `ScaledMaxwellStability`: the action expansion and the two stationarity equivalences by summation by parts with a delta variation at each interior node, the balance and the energy identity by the adjointness of $C$ and $C^{\mathsf{T}}$ under the committed pairing, the bounds by conservation and the energy identity, the sharp constant by five explicit rational spectral projectors of $CC^{\mathsf{T}}$ checked by kernel decision on integer tables together with a Cauchy–Schwarz transport through $C^{\mathsf{T}}$, and the unbounded modes by explicit eigenvectors and the real root $r < -1$ of $r^2 - (2 - h^2\lambda)r + 1 = 0$. The composed receipt `scaledMaxwellStability_receipt` carries the clauses from one typed bundle, with an explicit inhabitant at $h = 1/2$. □

The stability layer carries one committed certified step. The declared step $h = 4/5$ satisfies $(4/5)^2(3 + \sqrt{5}) \le 84/25 < 4$, through $\sqrt{5} < 9/4$, with rational margin $16/25$; at that step the staggered form is nonnegative for every history, conserved for zero current, obeys the balance with literal coefficient $2/5$, and every zero-current solution obeys $\|E_n\|^2 \le \frac{25}{2}\,\mathcal{E}_0$ and $\|B_n\|^2 \le 25\,\mathcal{E}_0$ at every step. Any positive step strictly below the threshold yields the same certified package, and a second inhabitant at $h = 1/2$ shows the committed choice is a declared selection, not forced. A typed design-only handoff lists what a preregistered instrument run must bind: the certified instrument, the carrier constant with its Courant proof, a declared readout map, declared control histories, and a nonempty rational kill band. No run, seed, or frozen prediction exists, and the committed step also inhabits the antecedent bundle above on a nonstatic demonstration history (`CertifiedScaledStepInstrument`).

The frozen dispersion prediction of the ladder carries a schema-level arming checklist. Each prose clause of its comparison protocol, the source-derived homogeneous edge position action with equal weights, the continuous-field same-operator sector attachment, cofinal gluing, the declared finite carrier scale, coherent frame and boost transport, the frozen nuisance and coverage readout rules, and the positive exclusivity lower bound, is one named placeholder field and one atomically registered premise. The committed results prove the coefficient identities as functions of an unfilled positive scale and the control-support certificate. The module also constructs stipulated mock records by setting proposition fields to true and choosing arbitrary scale, lower-bound, and natural-number digest labels. Their nonemptiness and packaging equivalence are not physical eligibility, externality, or custody theorems. No source-derived, artifact-bound evidential inhabitant is present, but a future source scale, lower bound, or attachment is not ruled out. No comparison data enters and the frozen bytes are untouched (`DispersionArmingInterface`).

These propositions are exact finite mathematics on the committed carrier, not a physical Maxwell theory. The equal-weight pairing coincides with the equal seam-counting selection. Port

loads are not charges, seam values are not fields or fluxes, the Green matrix is not a laboratory potential, and "Coulomb" names the canonical discrete Gauss solution of the finite table. The evolution step index is a declared parameter, not physical time; Ampère names the declared discrete update and Faraday the proved backward-difference identity. No Lorentz covariance, continuum control, physical source production, calibrated propagation or laboratory readout is constructed, and no empirical prediction follows.

### 4.1 Overlap transport and the defect obligation

The matter hypothesis of layer 5 places stable massive matter in localized defects whose boundary transport carries a holonomy that admitted repairs cannot erase. On one carrier that transport lives on the seams, and the finite results above fix what the native complex supplies.

**Proposition 4.9** (First cohomology of the carrier complexes)**.** *The seam graph has cycle rank nineteen, and its flat $U(1)$ seam connections modulo port gauge form $U(1)^{19}$. The oriented twenty-face complex has vanishing first cohomology over $\mathbb{R}$ and one-dimensional second cohomology. Every real seam field with zero face curvature is a port gradient, and every port-load contrast $d\phi$ has zero face curvature.*

*Proof.* The first statement is the chord-holonomy classification and the rank-nineteen Gauss fibre of Theorem 2.10. By Proposition 4.6, $\ker C = \operatorname{im} d$, so the first cohomology $\ker C/\operatorname{im} d$ vanishes, and $C$ has rank nineteen on twenty faces, leaving a one-dimensional cokernel. The last clause is $Cd = 0$. □

The face complex is a triangulated sphere, so flat transport on it is gauge trivial, and each of the nineteen seam holonomies is carried by face fluxes. The consensus layer proves that on a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch [13]. That theorem supplies no update of the offsets, localizes no moving defect and identifies no residual with excitation energy. Declared $\mathbb{Z}_6$ links and punctures allow a separate flux experiment, but they are additional state and update premises, and unrestricted local link changes can erase the pair.

The one local native candidate tested directly is transient. Under the unchanged native C/E law at total load two, the twelve double-load classes feed one closed irreducible binary class of 66 states. Under independent uniform choice of the thirty seams, a double load survives $n$ attempts with probability $(5/6)^n$, a mean lifetime of six seam attempts, and both complete C/E instruments satisfy $\sum_h K_h^* P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$ on the collision projector, with no binary-to-collision feed. Removing the five seams at one port gives binary closed classes of sizes 11 and 55; that changes the law and does not exhibit a sector the unchanged law protects [22]. A native defect sector therefore needs overlap-transport data with nonzero curvature, or a state and update rule for it, that the repair law must supply. Its construction is the open join between this layer and layer 5.

## 5 Federations, screens and refinement

A spherical screen is the observer-facing chart of an accessible cut: a cellulated sphere encodes finite capacity, caps, collars, edge centers and cuts, while the microscopic carrier is the finite federation. Caps on $S^2$ are the support regions whose modular flows become Lorentz boosts in the controlled scaling branch, through the classical isomorphisms $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$; these are imported and not source-derived. Attaching the chart to a populated $3+1$-dimensional

event spacetime is conditional on the physical-causality, manifoldlikeness, dimension and volume certificates of the spacetime layer [20].

**Federations and cross-carrier routing.** The executable construction joins elementary carriers on the nested screens with 20, 80 and 320 carriers [22]. Federations join the same elementary factors and introduce no different kind of elementary observer. One carrier suffices for the local observer, clock and finite geometry results; joined carriers serve larger regions, distant interventions and refinement. The local carrier Gram is record-derived. Cross-carrier routes use a supplied cell-centre and local-tangent-frame convention, and the cell-ancestry map between levels is not record-derived. That port-ancestry map fails to preserve seam adjacency: from level 0 to level 1, 74 of 2,520 fine seams map to non-seams and 2 fine seams collapse to one coarse port; from level 1 to level 2, 280 of 10,080 fine seams map to non-seams and 4 collapse. A record-derived, refinement-natural routing is therefore an open join of this layer. Exact native-hop balls give finite dimensionless distance data, with roughly quadratic growth over part of the tested range; this establishes no three-dimensional physical bulk and no path-independent global frame.

**Executed federations.** The integer nearest-agreement law has been executed on federations with one declared port-to-neighbour gluing [17]. On the level-six icosahedral cell tower, 81,920 carriers with 2,580,480 seams settle under sixteen shuffled schedules from one seeded load to the balanced class, with no accepted move raising $V = \sum_i N_i^2$, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals; a standalone verifier reconstructs the multisets and hashes from the primitive arrays. Level-five to level-ten receipts extend the same law to 20,971,520 carriers and 660,602,880 seams, settling in 84 to 100 sweeps. The gluing is a declared convention whose selection and refinement naturality are underived. A separate closure-loop archive recovers, from event logs alone under the declared probe and operation grammar, the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1 + 3 + 8$, with the invariant vector preserved under re-instantiation for two iterations; tetrahedral and octahedral controls recover their distinct specifications.

### 5.1 The multiresolution regulator chart

**Definition 5.1** (Reference multiresolution carrier)**.** *Choose the nested geodesic icosahedral subdivisions of $S^2$. A regulator is*

$$r = (m, L, b),$$

*where $m$ is the subdivision depth, $L$ is the finite support or volume scale of the exported carrier, and $b$ is a boundary/phase label. Every added cell, collar, edge-center, or observer-visible channel $j$ carries a finite algebra*

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

*and a faithful detail state $\tau_j$. At regulator $r$,*

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

*A finite-depth local presentation circuit $W_r$ maps these multiresolution coordinates to the declared patch, port, collar, and gauge coordinates:*

$$M_r = \mathrm{Ad}(W_r)(\widetilde{M}_r), \qquad \widehat{\omega}_r = \widetilde{\omega}_r \circ \mathrm{Ad}(W_r^*).$$

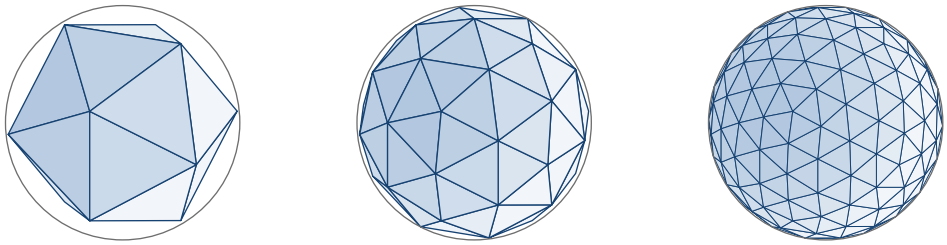
*For $r \preceq s$, refinement adds detail factors. The refinement embedding and coarse-graining are*

$$\iota_{rs}(A) = W_s\big[(W_r^* A W_r) \otimes \mathbf{1}\big]W_s^*,$$

*and*

$$Q_{sr}(X) = W_r\big[(\mathrm{id} \otimes \tau_{s\setminus r})(W_s^* X W_s)\big]W_r^*.$$

*The embedded map $E_{sr} = \iota_{rs}Q_{sr}$ is the canonical state-preserving conditional expectation.*



depth 0: 20 cells depth 1: 80 cells depth 2: 320 cells

Figure 1: Nested geodesic icosahedral subdivisions of $S^2$ at depths $m = 0, 1, 2$, with $20{\cdot}4^m$ triangular cells: the reference multiresolution carrier of Definition 5.1. The executable construction's observer federations use screen levels with 20, 80 and 320 elementary carriers.

**Theorem 5.2** (Fixed-cutoff regulator certificate)**.** *The maps in Definition 5.1 are unital, completely positive, composition-compatible, and preserve the faithful reference states. The refinement embeddings are unital injective $*$-homomorphisms, the $E_{sr}$ are faithful conditional expectations, and the reference modular groups restrict exactly along the tower. The inductive limit has canonical finite-regulator renormalization maps given by these conditional expectations; their martingale tails are the finite-volume and lattice-spacing Cauchy errors used by the compact and main papers.*

*Proof.* In bare coordinates all statements reduce to tensoring an observable with the identity and averaging the detail factors against the faithful product state. Composition follows from product-state associativity. Conjugating by the presentation circuits gives the physical patch coordinates without changing the algebraic identities. □

*Remark* 5.3 (Reference state, ground state, and vacuum). The state $\widehat{\omega}_r$ is the finite-regulator reference state. It may be called a finite-volume ground state only after a positive self-adjoint transfer or Hamiltonian has been constructed and shown to have that state as its ground state. The term continuum vacuum is reserved for the GNS/OS limit after positive energy and the declared vacuum-sector uniqueness or superselection condition have been proved.

**Screen cells and capacity.** Write $P_\star = a_{\mathrm{cell}}/\ell_\star^2$ and $N_\Lambda = A_{\mathrm{screen}}/(4\ell_\star^2)$, with $N_\Lambda = 3.31 \times 10^{122}$ the $\Lambda$-located working capacity, a measured-side comparison display that is not an input to the closure. An equal-area cellulation has $K_{\mathrm{cell}} = A_{\mathrm{screen}}/a_{\mathrm{cell}} = 4N_\Lambda/P_\star \simeq 8.12{\times}10^{122}$ canonical screen cells, and $8.66 \times 10^{122}$ at the conditional bridge capacity. This count assigns no independent Hilbert factor to a cell: $P_\star/4 \simeq 0.408$ nats, about 0.588 bits, is not the logarithm of an integer-dimensional autonomous cell algebra. The area law lives on shared cuts, edge centers and constrained overlap data, and a carrier patch, graph vertex or worker-local shard is not one canonical screen cell without a separate measure-realization theorem.

## 5.2 Sphere fold and the carrier-to-support bridge

**Definition 5.4** (Sphere fold)**.** *Fix a regulator $r$. Let $\Sigma_r$ be the finite presentation space of a patch federation, $\Gamma_r$ its presentation-redundancy groupoid, $\pi_r : \Sigma_r \to Q_r = \Sigma_r/\Gamma_r$ the physical quotient map, $n_r : Q_r \to Q_{r,\mathrm{nf}}$ the accepted repair normal-form map, and $\chi_{S,r} : Q_{r,\mathrm{nf}} \to \mathsf{Chart}_S(r)$ the support-visible spherical screen chart, whose outputs are caps, collars, cuts, edge-sector records, boundary data and geometric readouts. The sphere fold of $s \in \Sigma_r$ is $\mathrm{Fold}_{S,r}(s) := \chi_{S,r}(n_r(\pi_r(s)))$. Two presentations have the same fold exactly when their repaired quotient normal forms induce the same support-visible caps, collars, edge-sector records and screen readouts.*

The support chart also exposes a derived cap-normal readout with output $\mathsf{CapChart}_r = (C_r, \partial C_r, \mathbf{c}_r, \alpha_r, n_{C,r}, \varepsilon_{\mathrm{round},r}, \varepsilon_{\mathrm{inc},r})$; on the exact round-cap branch $n_{C,r} = (\cot \alpha_r, \csc \alpha_r\, \mathbf{c}_r)$, and a finite fitted branch is an approximate round-cap chart unless a rigidity or convergence certificate bounds it. The normal is support-chart data derived from the repaired quotient, not a carrier coordinate or hidden port label. It determines a geodesic half-space in $H^3$; selecting an observer point in $H^3$ requires a separate frame, tetrad or clock. The spacetime layer produces the cap chart from the support-visible incidence complex of the repaired normal form [20].

**Definition 5.5** (Carrier-to-support screen bridge)**.** *A carrier-to-support screen bridge at cutoff $r$ consists of a source-bound map from routed carrier ports and seams to the support projections of the repaired observer-patch net, together with: (i) unital interface-algebra homomorphisms on every seam; (ii) coherent recovery maps on higher overlaps; (iii) a quotient-visible nerve whose support incidence is computed from the normal form; and (iv) refinement maps commuting with routing, quotienting, incidence, records, and support projection. A spherical bridge additionally certifies that the global nerve satisfies the spherical-incidence, cap-mesh, oriented cross-ratio, and normalization receipts consumed by the spacetime and Einstein paper. Local carrier coordinates are excluded from the support output unless this map derives them as quotient-visible data. The nerve retains every supported higher simplex. A spherical surface certificate cannot replace a supported solid simplex by its boundary. Refinement topology requires a certified subdivision or homeomorphism; an endpoint-valued projection from a midpoint subdivision is only homotopic to the homeomorphism. Cap orientation needs an inside witness or equivalent oriented data, and angular radii require a declared conformal observer frame.*

**Proposition 5.6** (Local Icosahedral incidence does not select the global support)**.** *The certified twelve-port incidence of every local carrier does not determine the topology or conformal structure of the federation support.*

*Proof.* Fix one certified local carrier and make one copy at every vertex of an arbitrary finite graph. Route selected port pairs along that graph while leaving each local port algebra, oriented incidence, antipode, and $A_5$ action unchanged. A path, a cycle, a tree, and a cellulation of a closed surface therefore have identical local carrier certificates and different federation nerves. The local certificate cannot distinguish them. Full interface maps, higher-overlap coherence, and the global incidence/refinement receipt are necessary parents of a support-screen claim. □

**Finite bridge theorem.** The source-derived incidence nerve uses the twelve ports as carrier charts, the thirty edges as seam algebras, and the twenty oriented faces as nonvacuous triple restrictions. It verifies every interface homomorphism and face cocycle. On the same bound artifact, phase readback produces accepted confluent seam repairs, and the operational observer passes record, feedback, prediction, and checkpoint controls. The carrier nerve maps refinement-naturally to an oriented $S^2$ support limit. This closes the finite carrier, federation, observer, and support

bridge of Definition 5.5. It does not produce an $H^3$ frame, physical source-causal continuum, BW/KMS clock, physical scale, laboratory attachment, or continuum fields.

**Local ports and the spherical support.** A degree-one federation bridge and each carrier's local twelve-port boundary are separate data. In the declared source construction, the local oriented icosahedral boundary $K$ and the level-zero support $S_0$ have the same ordered face table. Let $s_0 : S_0 \to N_0$ be the designated section of the nerve projection $b_0 : N_0 \to S_0$, and set $z_0 = (s_0)_*[S_0]$.

**Proposition 5.7** (Oriented seed attachments)**.** *The maps $h_0 = s_0 \circ p$, where $p : K \to S_0$ ranges over orientation-preserving simplicial isomorphisms, form a set of sixty elements with a free transitive $A_5$ action. Every such map satisfies*

$$(h_0)_*[K] = z_0$$

*as an integer chain. For the fixed source row with fibre sizes $(2, 1, \ldots, 1)$, the designated section and the process link within the double fibre, the proper support relabelings preserving this decoration form $C_5$. Their action on the sixty maps has twelve orbits.*

*Proof.* The shared oriented face table supplies an isomorphism, so every other one is its composite with a proper automorphism of the icosahedron. These automorphisms form $A_5$, of order sixty. Composition acts freely and transitively. Each automorphism permutes the twenty oriented triangles, giving $p_*[K] = [S_0]$ exactly; composing with the section proves the chain identity. The double fibre singles out one support vertex. Its stabilizer consists of the five rotations around that vertex and preserves the section and process link. This subgroup acts freely on the seed maps, so every orbit has five elements and there are $60/5 = 12$ orbits. □

The twelve orbits refer to this fixed marked presentation. They do not classify physical alternatives or equivalence after jointly transporting the source decoration and local port charts. Exact enumeration and a separate independent reconstruction in the public repository check the sixty maps and their chain images. The support checks also verify $(b_r)_* z_r = [S_r]$ and the section/coarsening squares at four stages with $12, 42, 162, 642$ support vertices.

This seed classification does not supply a source-selected attachment. The analyzed source definitions prescribe each local boundary independently of the federation nerve. The local carrier construction scales barycentric coordinates within a single face, whereas the support tower uses midpoint subdivision and fibre coarsening. Those data do not specify a common local incidence tower and a refinement-commuting family $h_r$.

The two local Galois radial maps have degrees one and seven. A comparison with the degree-one support bridge requires a source-bound attachment and continuous realizations as maps between the same oriented $S^2$, with the same orientation convention. Under these hypotheses degree classifies their homotopy classes: the degree-one radial map is homotopic to the support composite, and the degree-seven map cannot. The equality of seed tables alone supplies neither those hypotheses nor a physical direction, metric or scale.

The fold adds no independent dynamics. Repair compares overlaps, updates records and reaches the accepted normal form; folding names the spherical chart presentation of that repaired quotient state.

## 5.3 The finite federation operator domain

**Proposition 5.8** (Finite signed-seam operator realization)**.** *On the observer-visible seam complex of* 8,662 *carriers and* 11,816 *seams, the declared reversing transport frustrates all* 38 *triangles. Two coordinate routes through the same $\mathbb{F}_2$ rank implementation agree with the exact rank identity and*

*give lift-ambiguity rank* 3,117. *Scalar, chiral, and gauge sections carry a sign-twisted derivative of support radius one. In the finite receipt, the unit-counting measure is a declared realization choice. Exact integer evaluations on the declared deterministic sections test the adjoint and kinetic identities, gauge covariance, observer-cover gluing, subcomplex restriction naturality, and Dirichlet boundary restriction. The signed-graph rank theorem gives twisted-kernel dimension zero.*

*Proof.* The seam holonomies and both boundary-rank eliminations are evaluated over $\mathbb{F}_2$. The kernel statement reduces to the signed-graph consistency theorem; the other operator checks use deterministic integer-valued sections. The declared transport and unit-counting measure define this finite realization. Independent replay checks its source artifacts and exact operator identities [17]. □

An authenticated semantic log defines a finite locally finite poset [1, 2] whose proved meaning is authenticated read-from dependence. Its canonical source height, zero at roots and one plus the maximum direct-parent height otherwise, is the longest authenticated-parent-chain length and strictly increases on generated ancestry. Independently, the exact rank-three positive source Gram quotient and a real axis define

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x),$$

a four-dimensional ambient target carrier with Lorentz inertia $(1,3)$. Every finite strict partial order compiles exactly into an abstract authenticated semantic log, so this is a grammar-expressivity result, not a dynamics-selection theorem. A supplied spatial readback satisfying the edge speed bound sends every generated ancestry into the future cone; equal-height spatial separation and strict spacelikeness of unsupported increasing-height pairs give exact order–cone equivalence on the placed image. This is not an intrinsic dimension estimate for the finite poset. A physical reading needs one source-selected refinement family with exact two-way order–cone agreement, dense isotropic link directions, calibrated $\#I/\rho \to \mathrm{Vol}(I)$, independent dimension, interval, manifoldlikeness and topology tests, and a unique Lorentzian limit [3, 4, 5, 6, 8, 9]. The deterministic log is not a Poisson sprinkling [7], public capacity is not event number, and event number is not spacetime volume without calibration. The controlled flat causal construction, its retrospective count clock and the operational cone criterion belong to the spacetime layer [20].

**Corollary 5.9** (Exact positive finite-domain signed-seam gap)**.** *With the declared unit-counting measure, the sign-twisted kinetic Hamiltonian is integer symmetric and positive semidefinite. Its kernel is trivial on the connected frustrated visible seam complex, so its smallest eigenvalue is strictly positive. The determinant argument gives an exact positive floor. A separate sparse computation gives* $0.1175367033\ldots$ *with relative residual below* $1.5\times10^{-14}$*; this decimal is a numerical refinement rather than the certificate.*

**Corollary 5.10** (Declared rank-forty-five tensor extension)**.** *The declared fifteen-state Standard Model charge table has vanishing mixed gravitational, weak, color, and cubic abelian anomaly sums, even weak-doublet parity, and the diagonal* $\mathbb{Z}_6$ *action fixes every state. Its chiral charge multiset is disjoint from its conjugate. Under the separate complete-band and cost-order premises, tensoring the selected rank-three response band with this table gives a conditional complex rank-*45 *extension. The local-domain receipt tests a separately declared operator* $D_\sigma \otimes I_{45}$*. It has support radius one, zero kernel, and the same exact positive finite-domain gap as the signed seam operator, with multiplicity* 45*. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node operator domain have no certified source, domain, or transport bridge. The receipt contains no Yukawa coefficient, particle mass, or pole datum [17].*

# 6 Repair, equilibrium and modular structure on the carrier

**Theorem 6.1** (Finite rate reconstruction on the reversible layer)**.** *Let $G = (X, E)$ be a connected finite state graph with positive rates $q^{\mathrm{eq}}_{u\to v}$ in both directions of every edge. Define*

$$a_{u\to v} := \log \frac{q^{\mathrm{eq}}_{v\to u}}{q^{\mathrm{eq}}_{u\to v}}.$$

*The following conditions are equivalent:*

(i) *every oriented cycle has zero affinity;*

(ii) *there is a potential $K : X \to \mathbb{R}$, unique up to an additive constant, with $a_{u\to v} = K_v - K_u$;*

(iii) *$\pi_x \propto e^{-K_x}$ satisfies detailed balance $\pi_u q^{\mathrm{eq}}_{u\to v} = \pi_v q^{\mathrm{eq}}_{v\to u}$.*

*The rates belong to the reversible equilibrium layer. They are not rates of the accepted strict-descent relation.*

*Proof.* Condition (ii) telescopes around every cycle. Under (i), fix a root and define $K_x$ by summing $a$ along any path from the root to $x$; zero cycle affinity gives path independence. Exponentiating $K_v - K_u = a_{u\to v}$ gives detailed balance, and taking logarithms of detailed balance gives (ii). □

The potential $K$ is a classical Gibbs potential. On the commutative state algebra its modular automorphism is trivial, so rate reconstruction creates no nontrivial modular clock. The finite potential-theory core, with condition (i) read over closed walks, is machine-checked in the Lean thermodynamics library (`KolmogorovCriterion`) [18]; the source-derived rates and the channel realization are outside that check.

## 6.1 Finite modular and register theorems

The finite equilibrium layer supports exact statements before any continuum or physical-current interpretation, and they fix which data a simulator has to produce.

**Theorem 6.2** (Exact modular coarse graining)**.** *Let $c : X_s \to X_r$ be a surjective coarse-graining map between finite sets and let $\mu_s(x) > 0$. The pushed-forward law and its modular potential are*

$$\mu_r(y) = \sum_{c(x)=y} \mu_s(x), \qquad K_r(y) = -\log\left(\sum_{c(x)=y} e^{-K_s(x)}\right), \qquad K_s = -\log \mu_s.$$

*Thus modular potentials coarse-grain by a log-sum-exp rule, up to the common additive normalization convention. More generally, if $\mu_s(x) = Z_s^{-1} m_s(x) e^{-S_s(x)}$ and $m_r(y) > 0$ is the declared coarse reference weight, then*

$$S^{\mathrm{eff}}_r(y) = -\log\left(\sum_{c(x)=y} m_s(x) e^{-S_s(x)}\right) + \log m_r(y)$$

*gives the exact pushed-forward law. For a finite quantum channel $\Phi_{sr}$, the corresponding statement is $\rho_r = \Phi_{sr}(\rho_s)$ and $K_r = -\log \rho_r$ whenever the output is faithful. An arithmetic average of $K_s$ is not the coarse modular potential in general.*

*Proof.* Both classical formulas follow by substituting $\mu_s = e^{-K_s}$ or $\mu_s = Z_s^{-1} m_s e^{-S_s}$ into the definition of the pushforward. The quantum formula is the definition of the modular Hamiltonian of the pushed-forward faithful state. □

**Proposition 6.3** (A faithful equilibrium law does not select its generator)**.** *Let $G = (X, E)$ be a connected finite graph and let $\mu_x > 0$. For every collection of positive symmetric edge conductances $c_{xy} = c_{yx}$,*

$$q_{x\to y} = \frac{c_{xy}}{\mu_x}$$

*defines a continuous-time Markov generator that is reversible with stationary law $\mu$. Distinct conductance families give distinct generators with the same state. A geometry, MaxEnt state, or repair normal form therefore selects no equilibrium dynamics unless an additional rate or conductance rule is supplied.*

*Proof.* The identity $\mu_x q_{x\to y} = c_{xy} = \mu_y q_{y\to x}$ is detailed balance. The diagonal entries are fixed by row-sum zero. Varying any conductance varies the generator while preserving the same stationary law. □

**Theorem 6.4** (Weighted graph Hodge split and entropy production)**.** *Choose an orientation of a connected finite graph, positive vertex and edge weights, and let $D$ be the vertex-to-edge coboundary. Every real edge one-cochain has the unique orthogonal decomposition*

$$a = DK + h, \qquad D^* h = 0,$$

*with $K$ unique up to an additive constant. For an irreducible stationary Markov law $\pi$ whose transition graph is $E$, assume the rate support is bidirectional: $q_{x\to y} > 0$ and $q_{y\to x} > 0$ for every $\{x, y\} \in E$. Define on each chosen edge*

$$J_{xy} = \pi_x q_{x\to y} - \pi_y q_{y\to x}, \qquad F_{xy} = \log \frac{\pi_x q_{x\to y}}{\pi_y q_{y\to x}}.$$

*Then the steady entropy-production rate is*

$$\sigma = \sum_{\{x,y\}\in E} J_{xy} F_{xy} \geq 0,$$

*and it vanishes exactly when detailed balance holds on every edge. The bare logarithmic rate ratio omits the stationary weights and is therefore not, by itself, the thermodynamic force.*

*Proof.* Finite-dimensional orthogonal projection gives the Hodge split, since $(\operatorname{im} D)^\perp = \ker D^*$. Each entropy-production summand has the form $(A - B)\log(A/B) \geq 0$ for positive $A, B$. Equality holds exactly when $A = B$ edge by edge. □

**Theorem 6.5** (Finite KMS-symmetric jump generator)**.** *Let $\rho = Z^{-1} e^{-K}$ be faithful on a finite matrix algebra. Suppose a Heisenberg-picture Lindblad generator has jump operators $V_{\omega,a}$ satisfying*

$$[K, V_{\omega,a}] = -\omega V_{\omega,a}, \qquad V_{-\omega,a} = V_{\omega,a}^*, \qquad \gamma_a(-\omega) = e^{-\omega}\gamma_a(\omega),$$

*with positive Kossakowski matrices in every frequency sector. Suppose also that its Lamb-shift Hamiltonian commutes with $K$, and that the generator is self-adjoint for the declared symmetric GNS inner product*

$$\langle A, B\rangle_{\rho,1/2} = \operatorname{Tr}(\rho^{1/2} A^* \rho^{1/2} B).$$

*Then the semigroup is completely positive and unital, preserves $\rho$, commutes with the modular flow $A \mapsto e^{itK} A e^{-itK}$, and respects the displayed frequency-sector decomposition. The sign in the KMS rate relation is tied to the displayed commutator convention.*

*Proof.* Positive Kossakowski matrices give complete positivity. Unitality and GNS self-adjointness imply $\mathrm{Tr}(\rho L(A)) = \langle \mathbf{1}, L(A)\rangle_{\rho,1/2} = \langle L(\mathbf{1}), A\rangle_{\rho,1/2} = 0$, hence stationarity. The eigenoperator relation, the paired KMS rates, and $[K, H_{\mathrm{LS}}] = 0$ make every term modular covariant and preserve its frequency sector. □

**Theorem 6.6** (Exact $A_5 \times \mathbb{Z}_2$ register commutant)**.** *For the twelve-port permutation module,*

$$\mathbb{R}[\mathsf{P}_{12}] \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$$

*is multiplicity-free. The icosahedral adjacency operator has four distinct eigenvalues* $5, \sqrt{5}, -\sqrt{5}, -1$ *on these summands. Consequently its real* $A_5$*-commutant is* $\mathbb{R}[A_{\mathrm{ico}}]$*, represented by polynomials of degree at most three. On the oriented register* $\mathsf{R}_{24} = \mathbb{R}[\mathsf{P}_{12}] \otimes \mathbb{R}[\mathbb{Z}_2]$*, every self-adjoint operator commuting with* $A_5$ *and orientation reversal is specified by one real scalar on each irreducible summand in each parity sector, eight scalars in total.*

*Proof.* Schur's lemma gives one scalar on each real irreducible summand. The four distinct adjacency eigenvalues make the spectral projectors degree-three interpolation polynomials in $A_{\mathrm{ico}}$. Diagonalizing the commuting reversal involution gives two parity copies of the same multiplicity-free decomposition. □

**Proposition 6.7** (No equivariant directed 24-cycle)**.** *Regard the oriented register as the transitive* $(A_5 \times \mathbb{Z}_2)/(C_5 \times \{1\})$*-set. It has no* $A_5 \times \mathbb{Z}_2$*-equivariant successor permutation that is a single directed 24-cycle.*

*Proof.* Its equivariant automorphism group is

$$N_{A_5 \times \mathbb{Z}_2}(C_5 \times \{1\})/(C_5 \times \{1\}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

because $N_{A_5}(C_5) \cong D_{10}$. Every equivariant permutation therefore has order at most two, whereas a 24-cycle has order 24. □

**Definition 6.8** (Source-derived oriented-channel realization)**.** *Let* $E^{\mathrm{or}}$ *be the oriented transitions of the reversible quotient-state layer and set*

$$\mathcal{H}_E = \ell^2(E^{\mathrm{or}}), \qquad F_E|e\rangle = a_e|e\rangle, \qquad a_e = \log \frac{q_{\bar{e}}^{\mathrm{eq}}}{q_e^{\mathrm{eq}}}.$$

*A source-derived* 24*-channel realization is a full-rank map*

$$C : \mathcal{H}_{24} := \mathbb{C}[\mathsf{P}_{12}] \otimes \mathbb{C}^2_{\mathrm{or}} \longrightarrow \mathcal{H}_E$$

*whose coefficients, aggregation rule, and provenance are emitted by the physical proposal ledger. After replacing* $C$ *by* $C(C^*C)^{-1/2}$*, assume* $C^*C = \mathbf{1}$*. Edge reversal is denoted* $R_E|e\rangle = |\bar{e}\rangle$*, and register reversal by* $R_{24}$*. The state set* $X$*, transition set* $E^{\mathrm{or}}$*, and label map into the* 24 *channel types are distinct objects.*

**Theorem 6.9** (Oriented-register modular compression)**.** *For a source-derived realization* $C$*, the register is invariant under the edge modular grading exactly when*

$$\epsilon_{\mathrm{ch}} := \big\|(\mathbf{1} - CC^*)F_E C\big\|_{\mathrm{HS}} = 0.$$

*Equivalently, there is a unique operator* $\Omega_{24}$ *with*

$$F_E C = C\Omega_{24}, \qquad \Omega_{24} = C^* F_E C.$$

*Suppose in addition that the edge rates are $A_5$-equivariant, $C$ intertwines the edge and register actions of $A_5$ and reversal, and hence $R_E F_E R_E = -F_E$. Then*

$$\Omega_{24} \cong \bigoplus_{\lambda \in \{1,3,3',5\}} \mathbf{1}_{V_\lambda} \otimes M_\lambda, \qquad \operatorname{spec}(M_\lambda) = \{-\omega_\lambda, +\omega_\lambda\},$$

*for four nonnegative source-derived values $\omega_1, \omega_3, \omega_{3'}, \omega_5$. Thus the paired frequencies have degeneracies $1, 3, 3, 5$. The symmetry fixes this block pattern and these degeneracies, but not the four numerical values.*

*Proof.* The residual vanishes precisely when $F_E \operatorname{im} C \subseteq \operatorname{im} C$. Since $C$ is isometric, restriction to this invariant subspace gives the unique compression $\Omega_{24} = C^* F_E C$. $A_5$-equivariance and Schur's lemma give one Hermitian $2 \times 2$ orientation matrix $M_\lambda$ on each of the multiplicity-free summands $1, 3, 3', 5$. Reversal covariance gives $R_{24}\Omega_{24}R_{24} = -\Omega_{24}$, so each orientation block is traceless and has eigenvalues $\pm\omega_\lambda$. Tensoring with $V_\lambda$ gives the displayed degeneracies. □

**Corollary 6.10** (Raw labels are modular eigenchannels only at equal affinity)**.** *If each raw register slot is realized as a normalized sum of transitions with that label and distinct labels have disjoint edge support, exact channel closure requires*

$$a_e = a_{e'} \quad \textit{whenever } e \textit{ and } e' \textit{ carry the same raw label.}$$

*Otherwise the raw slot basis is not a modular eigenbasis; any invariant 24-channel space must instead use source-derived linear combinations.*

**Proposition 6.11** (Operator-frame identification boundary)**.** *Let $V_a \in \mathcal{A} \subseteq B(\mathcal{H})$ be realized transition operators and let $\omega_a$ be source-derived frequencies. If self-adjoint $K$ and $K'$ both solve*

$$[K, V_a] = \omega_a V_a, \qquad [K', V_a] = \omega_a V_a \quad \textit{for every } a,$$

*then $K - K'$ lies in the commutant of the algebra generated by the $V_a$. Hence the channel frame identifies a noncommutative modular Hamiltonian only modulo that represented commutant. If, in addition, $K, K' \in \mathcal{A}$ and the $V_a$ generate $\mathcal{A}$, the ambiguity lies in $\mathcal{A} \cap \mathcal{A}' = Z(\mathcal{A})$.*

*Proof.* Subtracting the two commutator equations gives $[K - K', V_a] = 0$ for every $a$; the generated-algebra statement follows. □

*Remark* 6.12 (Rate scale, clock scale, and geometric modular identification). Multiplying every equilibrium rate by one constant leaves every affinity, the reconstructed stationary law, $K$, and $\Omega_{24}$ unchanged, while rescaling the relaxation generator. Rate ratios therefore determine a dimensionless potential; rate magnitudes determine kinetics. Neither fixes a frequency in hertz. Operational time requires the independent clock instrument and affine calibration used later in this paper. A physical Bisognano–Wichmann (BW) identification further requires a refinement-natural comparison showing that the gear-derived state is the same support-visible cap state used by the geometric modular theorem, the independently complete mixed Gelfand–Naimark–Segal common-comparison package on that same tower, its regularized cutoff schedule, and the independent $2\pi$-normalization receipt.

These theorems classify finite register states and operators; the symmetry fixes block patterns and degeneracies, never numerical gaps or clock units. A physical gauge-current algebra additionally requires the full-rank, compact skew-adjoint, commutator-closed current lift, its inner $A_5$ action, and refinement-natural physical descent used by the gauge and particle layers [14, 16].

On the literal transitive twelve-port set $\mathsf{P}_{12} = A_5/C_5$, an irreducible $A_5$-equivariant rate law has a unique invariant equilibrium distribution, hence $\pi_p = 1/12$ and constant $K_p$. On $\mathsf{P}_{12} \times \{+,-\}$, with $A_5$ acting on the port coordinate, an invariant potential has at most two values $K_+$ and $K_-$. The geometry can therefore carry one orientation gap $\Delta K = K_- - K_+$. It cannot supply twenty-four independent energy levels. The state set $X$, oriented transition set $E^{\mathrm{or}}$, and channel label map $\lambda : E^{\mathrm{or}} \to \mathsf{R}_{24}$ are separate objects; $\lambda$ need not be a bijection. In particular, twenty-four oriented edge instances contain only twelve unoriented edges and cannot connect a graph on twenty-four states.

# 7 Patch federation and overlap synchronization

Given a federation of echosahedral patches, overlaps are formed by routing ports into declared interface pairs or interface hyperedges. For an edge $e = \{(i,a),(j,b)\}$, the exposed packets are

$$x_{i,a} = \rho_{i,a}(s_i), \qquad x_{j,b} = \rho_{j,b}(s_j),$$

and the edge mismatch is a nonnegative function

$$\Phi_e(x_{i,a}, x_{j,b}) \geq 0.$$

The total visible mismatch is

$$\Phi(s) = \sum_{e \in E} \Phi_e\big(\rho_{i,a}(s_i), \rho_{j,b}(s_j)\big).$$

Accepted repairs do not increase the declared touched-overlap mismatch, and exact fixed-cutoff branches lower the relevant mismatch unless the local visible datum is repaired. The implementation is a bounded patch operation:

1. read the exposed overlap packets;
2. compare them through a declared commensurability map;
3. choose an allowed local update or rerank move;
4. write a record of the move;
5. expose the resulting port packet;
6. let the exact verifier decide any high-level task claim.

**Proposition 7.1** (Federated synchronization contract)**.** *Suppose a finite patch federation has a declared mismatch functional $\Phi$ with values in a finite ordered set and a finite repair menu. Treat a repair of a locally repaired datum as the identity, not as a step. Assume every accepted nonidentity repair strictly lowers $\Phi$, and every state outside the declared visible local normal forms admits such a repair. Then every maximal accepted-repair sequence is finite and ends at a visible local normal form. If the union-collar gluing and repair-completeness hypotheses of the OPH consensus theorem hold together with protected-support-complete transactions, protected-conflict-complete dependencies, and an independently checked quotient local diamond, the terminal physical observable state is schedule-independent on that carrier. The carrier is thereby a finite constraint-code implementation surface; no QECC distance, min-cut formula, spectral mixing rate, or wall-clock liveness bound follows unless the corresponding extra certificate is supplied.*

*Proof.* The first conclusion is the finite Lyapunov argument: the declared score strictly decreases on every allowed step, so an infinite accepted-repair sequence is impossible; completeness makes a terminal state a declared local normal form. The second conclusion consumes the quotient-local confluence package of the OPH consensus theorem. Atomic commits alone do not supply its protected-support, coherent-payload, or local-diamond receipts. The result applies to the federated carrier and visible interface defined above. □

The schedule-independent conclusion above is a same-source statement. Equality of endpoints reached from different interiors exposing the same protected boundary requires injectivity of the boundary map on the consistent quotient, as characterized in Ref. [10]; weak normalization and all-schedule liveness are separate hypotheses. The same reference shows that a total exact collar-preserving local repair exists only when every admissible collar value has a locally consistent extension. Neither condition follows from finite descent alone.

# 8 Native repair histories and finite scale transfer

## 8.1 Native stationary repair and absolute record power

Consider a connected finite undirected port graph with positive edge attempt rates $w_e$, weighted Laplacian $L$, and $p$ ports. In a balanced integer sector, subtract the common load floor and write

$$n_i \in \{0,1\}, \quad \sum_i n_i = r, \quad 0 < r < p, \qquad z = n - \frac{r}{p}\mathbf{1}, \quad \Pi = I - \frac{1}{p}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \quad \kappa = \frac{r(p-r)}{p(p-1)}.$$

Independent Poisson attempts on edge $e$ exchange its endpoint loads with probability $1/2$ and otherwise leave them unchanged. This is the fair nearest-integer agreement rule continued after first settlement. It is a specified observer-like system: bounded local port state, seam interaction, readback, and records of local repair attempts. Its stationary history is a different observable from a configuration retained at the end of its first settling sweep.

**Theorem 8.1** (Stationary density and occupation records)**.** *The uniform law $\pi$ on the $r$-subsets is the unique stationary law. The process is reversible, its configuration generator $G$ obeys $Gz = -Lz/2$, and in stationarity*

$$\mathbb{E}_\pi[zz^{\mathsf{T}}] = \kappa\Pi, \qquad \mathbb{E}_\pi[z(t)z(0)^{\mathsf{T}}] = \kappa e^{-|t|L/2}\Pi.$$

*For each fixed matrix $B$, at fixed graph and from any initial law,*

$$T^{-1/2}\int_0^T Bz(t)\,dt \implies \mathcal{N}(0, 4\kappa BL^{+}B^{\mathsf{T}}).$$

*For the literal time average $f_T = T^{-1}\int_0^T z(t)\,dt$, its stationary variance along a Euclidean-unit eigenvector of positive eigenvalue $\lambda$ is instead*

$$\mathrm{Var}(v^{\mathsf{T}}f_T) = \kappa h(\lambda T/2), \qquad h(u) = \frac{2(u-1+e^{-u})}{u^2}, \quad h(0) = 1.$$

*Proof.* Each configuration swap has the same reverse rate. Connectedness makes the fixed-occupancy chain irreducible, giving uniform detailed balance and uniqueness. Sampling without replacement gives the displayed equal-time covariance. The mean swap increment on an oriented edge with incidence vector $b_e$ is $-b_e b_e^{\mathsf{T}} z/2$; summing rates proves the generator identity, and its

semigroup gives the unequal-time covariance. On the mean-zero subspace the Poisson solution is $H = 2L^+z$, with $GH = -z$. The time integral is its Dynkin martingale plus a bounded endpoint term. Ergodicity identifies the martingale bracket and bounded jumps imply its central limit condition; the bracket is equivalently $2\int_0^\infty \kappa e^{-tL/2}\Pi\,dt = 4\kappa L^+$. Finally, integrating $\kappa e^{-\lambda|t-s|/2}$ over the observation square and dividing by $T^2$ gives $h$. □

This specializes the finite reversible additive-functional limit [21] to the actual repair rule. It does not assert a joint refinement and long-time limit. On the refining source screens of the executable observer construction, actual density snapshots have finite-dimensional joint Gaussian limits with covariance $\langle f, e^{-|\tau-u|A}g\rangle$, where $A$ is the native continuum spatial generator and $\tau$ is the attempt count divided by $1260 \cdot 16^l$ at level $l$; uniform fixed-number preparation, independent uniform seam scheduling and full occupation readback are declared there, and no path convergence is claimed [24]. The sector, edge rates, elapsed-time convention, memory duration and record gain are declared. In particular, the two normalizations above differ by $\sqrt{T}$; their powers differ by $T$. A change of time units rescales rates inversely and leaves $\lambda T$ unchanged.

A specified leaky local record gives a bounded-amplitude alternative:

$$\dot{y}_i = z_i - \gamma y_i, \qquad R_\gamma = \sqrt{2\gamma}\,y, \qquad \mathrm{Cov}(R_\gamma) = 4\kappa(L + 2\gamma I)^{-1}\Pi, \quad \gamma > 0.$$

In joint stationarity, $y_i(t) = \int_0^\infty e^{-\gamma s} z_i(t-s)\,ds$. Integrating the same unequal-time covariance proves the formula, and $|y_i| \le \max(r/p, 1-r/p)/\gamma$. This is a covariance identity; the finite-leak field need not be Gaussian. The leak, output gain and stationary preparation are specified, and finite-precision storage or saturation requires a separate implementation bound.

**Theorem 8.2** (Nonlinear readout projection at every finite window)**.** *For a centered vector-valued current-state readout $q$, set*

$$B = \kappa^{-1}\mathbb{E}_\pi[qz^{\mathsf{T}}], \qquad q = Bz + g.$$

*Then $\mathbb{E}_\pi[gz^{\mathsf{T}}] = 0$. Writing $C_T(q)$ for the stationary covariance of its time average and $S = -G$,*

$$C_T(q) = \langle q, h(TS)q^{\mathsf{T}}\rangle_\pi = BC_T(z)B^{\mathsf{T}} + C_T(g), \qquad C_T(g) \succeq 0.$$

*The asymptotic fluctuation covariance is $4\kappa BL^+B^{\mathsf{T}} + 2\langle g, S^+g^{\mathsf{T}}\rangle_\pi$.*

*Proof.* The components of $z$ span a $G$-invariant subspace. Self-adjointness in $L^2(\pi)$ makes its orthogonal complement invariant too, so the cross correlations vanish at every time lag. Spectral integration of the observation square gives $h(TS)$; it is positive. The inverse-generator formula follows by integrating the decaying semigroup on centered functions. □

On all $\binom{12}{6} = 924$ states of the half-filled production carrier, $\kappa = 3/11$ and the positive graph eigenvalues are $5 - \sqrt{5}, 6, 5 + \sqrt{5}$, with multiplicities $3, 5, 3$. These are graph bands. At unit attempt-rate normalization the port-average root mean square of $f_T$ is respectively $0.497739, 0.478409, 0.352867, 0.141920, 0.0459402$ at $T = 0.01, 0.1, 1, 10, 100$. Independent moment evolution checks the absolute finite-window formula for load, local Laplacian drive and a centered two-port product; the product has a nonzero orthogonal remainder. No geometric volume law is assigned to these test readouts.

## 8.2 Locally sampled attempt records

Let $D = \mathrm{diag}(d_i)$, where $d_i = \sum_{e\ni i} w_e$, and let $D_e$ have diagonal entries $1/d_i$ at the endpoints of $e$, zero elsewhere. At *every* attempt, including null attempts, append the pre-attempt vector $D_ez$. Let $A_T$ be its accumulated value on a common elapsed-time slice. Each port uses its local load, calibrated local attempt rate and the prepared filling $r/p$.

**Theorem 8.3** (Attempt-record covariance)**.** *At fixed graph, from any initial law,*

$$T^{-1/2}A_T \Longrightarrow \mathcal{N}(0,\Sigma_{\mathrm{ev}}), \qquad \Sigma_{\mathrm{ev}} = 4\kappa L^{+} - \kappa(\Pi D^{-1} + D^{-1}\Pi) + J, \quad J = \kappa \sum_e w_e D_e \Pi D_e.$$

*The correction satisfies*

$$\|\Sigma_{\mathrm{ev}} - 4\kappa L^{+}\| \le \frac{3\kappa}{d_{\min}}, \qquad \left| \frac{v^{\mathsf{T}}\Sigma_{\mathrm{ev}} v}{4\kappa/\lambda} - 1 \right| \le \frac{3\lambda}{4d_{\min}}$$

*for any mean-zero unit eigenvector $Lv = \lambda v$.*

*Proof.* The conditional reward rate is $\sum_e w_e D_e z = z$. With $H = 2L^{+}z$, reward plus $\Delta H$ is a martingale increment. The bounded endpoint correction has no effect on the limit. Its stationary bracket contains the occupation contribution $4\kappa L^{+}$, reward square $J$, and cross bracket $-\kappa L^{+} L D^{-1} = -\kappa \Pi D^{-1}$ plus its transpose. Here $b_e b_e^{\mathsf{T}} D_e = b_e b_e^{\mathsf{T}} D^{-1}$. Since $0 \preceq \Pi \preceq I$, one has $0 \preceq J \preceq \kappa D^{-1}$. The triangle inequality gives the operator bound and its Rayleigh restriction gives the relative bound. Finite state space and bounded event marks justify the martingale limit. □

The counting correction is correlated with subsequent repair. Treating it as independent noise is incorrect: the two-port control gives one half of the occupation-record variance. Recording successful moves only, subtracting a deterministic mean from raw uncentered marks, or stopping each patch after its own fixed event count defines a different observable. The asymptotic accumulator is unbounded; finite memory requires its own storage, reset or saturation law. These qualifications matter when attaching the record to a physical observer clock.

## 8.3 Temporal filtering and fixed-area normalization

For a common deterministic real kernel $w \in L^2([0,\infty))$, extended by zero, let $\widehat{w}$ be its Fourier transform. The stationary filtered density mode of decay rate $a = \lambda/2$ has variance

$$V(\lambda) = \frac{\kappa}{\pi} \int_{\mathbb{R}} |\widehat{w}(\omega)|^2 \frac{a}{a^2 + \omega^2}\, d\omega.$$

Consequently $\lambda V(\lambda)$ is nondecreasing and, for a nonzero kernel, $-1 \le d\log V / d\log \lambda \le 1$. Indeed, $a^2/(a^2+\omega^2)$ is nondecreasing in $a > 0$; the logarithmic derivative of the Lorentzian lies between $-1$ and 1, and the integrated derivative is its positive weighted average. Domination on compact positive $a$-intervals justifies differentiation. Under a separately established three-dimensional metric identification $\lambda \propto k^2$, this common temporal filter cannot generate power redder than $k^{-2}$. It therefore does not supply $k^{-3}$ bulk curvature power. This statement excludes nonlinear, spatially varying, adaptive and nonstationary readouts from its scope.

There is also a normalization bound independent of spherical quadrature. For arbitrary fixed port directions $\hat{n}_i$, define empirical harmonic coefficients with weights $4\pi/p$. If the same deterministic, trajectory-independent temporal kernel has $L^1$ gain at most $W$, then the ensemble mean harmonic pseudo-power obeys

$$\widetilde{C}_\ell := \frac{1}{2\ell+1} \sum_m \mathbb{E}|a_{\ell m}|^2 \le \frac{4\pi\kappa W^2}{p} \le \frac{\pi W^2}{p-1}.$$

To see this, stationary covariance and semigroup contraction bound every coefficient variance by $\kappa W^2 \|\Pi \alpha_{\ell m}\|^2$. The spherical-harmonic addition theorem gives $(2\ell+1)^{-1}\sum_m \|\alpha_{\ell m}\|^2 = 4\pi/p$, even for irregular or repeated directions. Ordinary averages have $W = 1$, so their fixed-area expected pseudo-power vanishes on refinement. This is not a pointwise bound on every realized record. The amplified $T^{-1/2}$ history has growing gain and does not satisfy a uniform $W$ premise. This is a restriction on the specified load record, not a theorem that geometric curvature vanishes.

**Proof and evidence scope.** The canonical native-repair evidence package contains the stationary, finite-window, event-mark and filter controls, including all 924 carrier states and exact rational four-port witnesses. The finite Lean witnesses (`NativeRepairFourCycle` and `TemporalFilterMonotonicity`) certify generator conservation and positivity, uniform stationarity, density drift, static and Green covariances, a Poisson solution, distinct load/drive spectra and the pointwise filter inequality. The probability limits and Fourier-integral arguments above are analytical proofs; they are not claimed as formalized general stochastic theorems. Production source pins, specifications, receipts and independent controls are archived with a reproduction entry in the public repository. Neither the native density nor its nonlinear test product has been identified with the geometric scalar $\frac{1}{3}\log(J_X/\bar{J}_X)$. No primordial amplitude, tilt, physical time unit, or photon transfer follows from the graph covariance alone.

### 8.4 Conditional scale transfer in the large settled configurations

The retained level-eight through level-ten configurations comprise sixteen repair schedules per level, conditional on one initial load field at each level. For equal-count ancestral groups, let $x$ and $y_s$ be the initial and terminal group means, centered with the same group weights. The response is $b_s = \langle x, y_s\rangle/\langle x, x\rangle$. For $S$ schedules define

$$P_{\mathrm{tot}} = \frac{1}{S}\sum_s \|y_s\|^2, \quad P_{\mathrm{noise}} = \frac{1}{S-1}\sum_s \|y_s - \bar{y}\|^2, \quad P_{\mathrm{common}} = \frac{1}{S(S-1)}\sum_{s\neq t}\langle y_s, y_t\rangle.$$

Then $P_{\mathrm{tot}} = P_{\mathrm{noise}} + P_{\mathrm{common}}$ exactly. Conditional independent schedule sampling makes these unbiased estimates of schedule variance and squared conditional mean. The finite common-power estimate may be negative and is not clipped. For disjoint groups with $m_g$ ports and normalized weights $a_g$, the uniform balanced fixed-total null gives the expected centered group variance $\kappa\sum_g(a_g - a_g^2)/m_g$.

| Carriers per group | Mean level-ten response | Schedule-noise fraction |
|---:|---:|---:|
| 1 | 0.05643 | — |
| 16 | 0.42876 | — |
| 64 | 0.66796 | — |
| 256 | 0.82285 | — |
| 1,024 | 0.90758 | 1.241% |
| 4,096 | 0.95232 | 0.594% |
| 65,536 | 0.98983 | 0.143% |

At the largest group the level-ten terminal variance is 11.048 times the stationary balanced reference. At 1,024 carriers the noise fraction across the three levels remains between 1.12% and 1.26%. These measurements show retained coarse input information and conditional schedule variability. Equal ancestral counts are not equal physical areas; group size is not a measured wavelength. The stopping artifact is the end of the first settling sweep, during which balanced swaps can continue. The spatial configuration retains this texture; its public component multiset does not encode the spatial placement. These are not independent cosmological initial-condition samples or a measured causal horizon.

The estimator, weighting controls, small statistics and 48 terminal-state hash cross-pins are preserved with their producer in the native-repair evidence package. Receipt checking without the remote arrays verifies provenance and retained statistics; recomputing their spatial readback requires those arrays.

**Binary port-pattern census.** The sixty proper rotations of the twelve-port carrier act on its binary occupancy patterns. Burnside's lemma gives the orbit polynomial

$$\frac{(1+x)^{12}+15(1+x^2)^6+20(1+x^3)^4+24(1+x)^2(1+x^5)^2}{60}.$$

The coefficients for zero through twelve raised ports are $1,1,3,5,12,14,24,14,12,5,3,1,1$, totaling 96 rotation orbits. Each of four retained schedules at each of levels six, eight, nine and ten contains all 96 classes. The census receipts and an independent polynomial recalculation accompany the same verifier. This counts local binary readout classes, without identifying them as particle species or replacing the spatial configuration by its orbit counts.

### 8.5 Port transport, internal response and metric sensitivity

The same seam law has mean propagator $e^{-tL/2}$. For $N$ twelve-port carriers, resolved low eigenpairs give $N\lambda_2$ equal to $0.882851, 0.825266, 0.817473, 0.816553, 0.815798$ at levels zero through four. Separate level-five through level-seven solves give $0.815612814, 0.815530307, 0.815499887$; their largest relative eigenpair residuals are $2.92\times10^{-10}, 2.24\times10^{-9}, 1.18\times10^{-9}$. These are finite floating-point solves, not interval bounds on the gap or a proof of a refinement limit.

In orthonormal carrier-mean and internal-port coordinates, write

$$L=\begin{pmatrix} A & B \\ B^{\mathsf{T}} & C \end{pmatrix}.$$

Internal connectivity makes $C$ positive definite. Completing the square proves the exact static identity

$$\min_v \begin{pmatrix} u \\ v \end{pmatrix}^{\mathsf{T}} L \begin{pmatrix} u \\ v \end{pmatrix} = u^{\mathsf{T}}(A-BC^{-1}B^{\mathsf{T}})u, \qquad v=-C^{-1}B^{\mathsf{T}}u.$$

At level zero this Schur complement reduces dipole-rate error from 42.9–44.2% to 0.348–0.363%, with internal stationarity residual $1.20\times10^{-15}$. It is a zero-frequency reduction; exact dynamic elimination replaces $C^{-1}$ by the frequency-dependent resolvent. At level four the first mode has only $1.46\times10^{-5}$ of its squared norm in internal variation, yet the naive coordinate-trial rate is about 50% too large.

Band averages do not establish rotational invariance. At level four the relative spread of the seven-mode band is 14.642% for the actual port graph, 13.115% for unweighted cell adjacency and 0.01561% for a separately supplied round-metric cotangent operator. This exploratory metric control is a different operator. Inverting a precision does not remove its anisotropy: on the mean-zero space, $[L^{-1},U]=-L^{-1}[L,U]L^{-1}$. A heat-trace calculation with bounded omitted modes admits a level-two window of screen spectral dimension 1.938–2.016; the fixed higher-level eigenpair budgets do not resolve the prescribed heat-trace window. This is a boundary-screen diffusion diagnostic, not a measurement of bulk spacetime dimension. For the native Green field on the refining spherical screens, with carrier interiors eliminated, the executable observer construction proves operator-norm convergence along a common spatial subsequence, and the limiting generator $A$ obeys $\lambda_1(A)\geq 3/4096$ and $n/49920\leq\lambda_n(A)\leq 3n/2$, giving heat-trace spectral dimension two on the two-dimensional source readout domain [22, 23]. The source-pinned solves and metric controls reside in the same evidence package.

## 8.6 Absolute refinement covariance and nested records

The retained refinement sample contains 61 chains: 18 level-seven to level-eight chains, 32 level-eight to level-nine chains, and 11 shuffled controls. Four child carriers inherit their parent's twelve-port record; independent fresh integer loads are drawn uniformly from $0, \dots, 4$. All 61 retained chains settle. Eight ancestral group sizes give 488 covariance rows. Let $x, f, y$ denote centered inherited, fresh and settled group means. Seeded reconstruction of $\mathrm{Var}(f)$, the archived variance ratio and the simple and joint normal equations recover their absolute covariance matrix. All rows satisfy the retained positive-semidefinite and regression checks; 256 independently retained parent variances agree to maximum relative error $4.15 \times 10^{-13}$. Fresh arrays were not independently hashed at production, so that part of the reconstruction remains conditional on the pinned producer and random number generator semantics, including the shuffled-control permutation.

For unshuffled level-eight to level-nine chains, the mean inherited-only residual variance falls from 0.0108181853 at four carriers per group to $2.55666955 \times 10^{-6}$ at 65,536. At the latter scale, removing the fitted fresh-load response too leaves $4.00020779 \times 10^{-8}$. Multiplying the inherited-only variance by group size gives 0.16755, close to the supplied fresh port-average variance $2/12 = 1/6$. The joint inherited response at group sizes $256, 1024, 4096, 16384, 65536$ is respectively $0.905427, 0.956213, 0.978740, 0.991953, 0.994902$. The corresponding $-\log_2 b^2$ values vary from 0.286815 to 0.014788; they are not one common spectral tilt.

For equal nested bins, lift all residuals to the same uniformly selected finest bin. If $e_M$ is the least-squares residual after averaging to groups of size $M$, then

$$\mathrm{Cov}(e_M, e_N) = \mathrm{Var}(e_{\max(M,N)}).$$

Indeed, averaging a fine residual into a coarse bin differs from the coarse residual by a linear combination of the coarse predictors. The coarse normal equations make this difference orthogonal to the coarse residual. Thus successive scale differences are orthogonal and their variances are differences of the residual variances. Orthogonality is not independence. This identity concerns the same nested location; it does not determine covariance at separated sky positions. Parent schedules share initial data, and linear spatial residuals are not innovations conditioned on a complete observer history. The chain receipts, absolute extractor and nested transformation are preserved in the native-repair package.

## 8.7 What a fixed-geometry repair history identifies

The repair carrier has locally readable port loads, seam attempts and records. Its supplied cell chart and incidence are not updated by a load repair. This distinguishes two possible source constructions: reading the geometry present in that implementation, and attaching a proposed volume law to its load history.

**Proposition 8.4** (Frozen geometric readout)**.** *Let a state be $(\Gamma, x)$, and suppose every allowed repair has the form $(\Gamma, x) \mapsto (\Gamma, F_e(x, \xi))$. For any positive cell-volume readout $V_i(\Gamma)$, prepared reference $\bar{V}_i = V_i(\Gamma)$, and finite repair history,*

$$q_i = \frac{1}{3} \log \frac{V_i(\Gamma)}{\bar{V}_i} = 0.$$

*Every unequal-position covariance and every linear coarse/fine covariance of this field is zero, under every load preparation, schedule and recording clock. The conclusion applies separately at every refinement level.*

*Proof.* Induction over the history preserves the first component. Its readout equals the reference pointwise, so the log contrast vanishes before any expectation or coarse graining is taken. □

This proposition concerns perturbations of the supplied geometry. It says neither that the background geometry is flat nor that all possible observer dynamics have zero curvature. It also does not prohibit an operational metric reconstructed by a new constitutive rule from load-dependent probes. Such a rule would add the missing metric-volume interface.

On the native geodesic carrier, a concrete chart diagnostic is the positive cone volume $V_i = |\det(v_{i0}, v_{i1}, v_{i2})|/6$ of a triangular cell. It is an imported Euclidean chart volume, not a measured physical collar volume. The finite controls apply the production integer repair to both a uniform half-filled preparation and a clustered half-filled preparation at 20 and 80 cells, with 240 and 960 ports. All 64 attempts in each control are retained and independently replayed. Rotating the chart preserves these volumes; scaling all coordinates by two multiplies them by eight. Both operations leave the log contrast with the matched reference zero. Comparing absolute cell sizes at different refinements does not turn this background change into a stochastic perturbation.

**Proposition 8.5** (Equal local variance does not identify source shape)**.** *Consider the isolated twelve-port carrier's internal icosahedral Laplacian $L$, $\Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/12$, and native stationary fixed-occupancy load $z$. Both $z$ and the local seam drive $Lz$ are locally computable, mean-zero, and equivariant under carrier rotations. After separately normalizing their covariance traces to one, they give*

$$C_{\mathrm{load}} = \frac{\Pi}{11}, \qquad C_{\mathrm{drive}} = \frac{L^2}{360}.$$

*Both diagonals are $1/12$. At antipodal ports their covariances are $-1/132$ and $0$, respectively; at adjacent ports they are $-1/132$ and $-1/45$. Thus locality, carrier symmetry, mean removal and equal one-point variance leave separated-position power unselected.*

*Proof.* Native stationarity gives $\mathrm{Cov}(z) = \kappa\Pi$. Since $L\mathbf{1} = 0$, the drive covariance is $\kappa L^2$. The graph is five-regular, so $(L^2)_{ii} = 30$ and $\mathrm{tr}(L^2) = 360$. Adjacent vertices have two common neighbors, yielding $(L^2)_{ij} = -10 + 2 = -8$; antipodal vertices have neither an edge nor a common neighbor, yielding zero. Division by the respective traces proves the statement. Equivariance follows because every carrier automorphism commutes with $L$. □

This is the exact internal seam graph used by native repair and by its isolated-carrier control; inter-carrier seams are excluded from this twelve-port calculation. Its covariance matrices are not asserted to be the full coupled federation covariances. The filling, stationary reference and gains used to normalize the traces are declared inputs. The normalization is a diagnostic comparison, with no observed spectrum used to choose either gain. Exact enumeration retains both the one-raised and six-raised stationary preparations, with 12 and 924 configurations and $\kappa = 1/12$ and $3/11$. It checks all covariance entries, including two disjoint antipodal two-port blocks and their fine/coarse cross covariance within that carrier. This block comparison is not a cosmological separation or a continuum refinement limit. For a centered current-state readout, $q = Bz + g$ with $B = \kappa^{-1}\mathbb{E}[qz^{\mathsf{T}}]$, the same calculation retains the absolute density component and residual. The local mismatch readout $\sum_{j\sim i}(n_i - n_j)^2$, centered under its preparation, has $B = 0$ and a nonzero residual at half filling: complementing every load leaves this readout invariant while reversing $z$. Nonlinearity therefore supplies an additional load observable, but by itself no geometric meaning.

The exact record controls use $P = I + G/30$, with the continuous generator $G$ at unit attempt rate per seam: one tick is a uniformly selected seam attempt with a fair endpoint coin. For

$m = 1, 2, 4$ retained readings and $c = 1, 2$ attempts between readings, they check

$$\operatorname{Cov}\left(\frac{1}{m}\sum_{a=0}^{m-1} q(X_{ca})\right) = \frac{mC_0 + \sum_{a=1}^{m-1}(m-a)(C_{ca} + C_{ca}^{\mathsf{T}})}{m^2}, \qquad C_t = \mathbb{E}[q(X_0)q(X_t)^{\mathsf{T}}].$$

These altered clocks change record covariances and leave the supplied geometry invariant. They are finite exact checks, not a physical time calibration or an asymptotic refinement experiment.

For any of these load observables $u_i$, one may attach positive labels $J_i/\bar{J}_i = \exp(3u_i)$, giving a formal log-volume contrast $u_i$ without changing a single native load transition. These are unselected extensions, not certified physical models. Even gain calibration cannot choose between the distinct covariance shapes above. A nontrivial geometric source needs an actual metric or collar-volume update, its relation to native records, an independently specified reference slice, and a physical calibration. The fixed-geometry construction supplies none of those by relabelling a load fluctuation. Nonstationary preparations, spatial memory or new couplings are possible mechanisms, but each must expose that interface.

Reconstructing a chart solely from the exact native linear response does not introduce a stochastic geometry update: the fair repair law has $Gx = -Lx/2$ with the same prescribed $L$. Any fixed reconstruction from this operator is independent of the realized load preparation. A state-dependent metric extracted from other response or fluctuation data would require its own observable and constitutive identification.

The general finite-history and observability statements, together with a four-cycle equal-diagonal covariance witness, are kernel-checked in the Lean geometry library.[10] The twelve-port rational enumeration and native trace checks are reproduced by an independent verifier with retained inputs and outputs.[11] No primordial amplitude, tilt, physical expansion or photon transfer is inferred from these controls.

## 9 Records, observers and checkpoints

An observer is not added as a metaphysical extra. In this paper an observer is an operational pattern in a patch or patch subfederation with persistent access to:

1. an observer-facing local algebra;
2. a record algebra;
3. a stable readout/update interface;
4. enough checkpoint data to define future observer-accessible probabilities.

**Maintained records in a noisy coherent source.** A declared coherent-code source realizes bounded self-reading patches with local state, ports, readback, noisy feedback and actual public record banks. Complete transfer/reset services reduce carrier leakage, and a protected quantum computation exports into a maintained classical archive. With a fixed positive local service error below a derived library bound, $O(\log\log q)$ quantum protection levels, staged decoding and expander refresh of $O(\log q)$-bit words make the joint decoded quantum and live record error tend to zero.

[10] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/Lean/Geometry/NativeGeometricSourceIdentification.lean

[11] https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/code/native_geometric_source

The archive remains readable throughout each refresh; no perfect bare decision bit or fault-free runtime controller is assumed. The analytical argument imports computation-code fault tolerance and an explicit expander bound; finite tests check its interfaces and fault controls, without executing the enormous all-size apparatus. Coherent code gluing, the local noise law, external schedule and cofinal control/inventory resources are hypotheses. They are not consequences of classical record agreement. Logical accounting, physical energy, calibrated time and source-selected matter dynamics are distinct.

For a separate deterministically prepared filled vacuum, the same massive walk has a positive normal-ordered excitation Hamiltonian with exactly its integer-time Fock channels. Both carrier bands enter the local covariance limit. Complete resolved-mode number reads retain the occupied background and have bounded mean excitation cost in the stated smooth carrier class; a finite analytical clock witness has probability error below 0.038 and swing above 0.923. Its enormous three-dimensional apparatus is not simulated. The longer sea preparation has its own charged protection and storage bounds, so the empty-vacuum resource estimates do not apply to it. High-band Floquet shifts preserve these integer-time reads and infrared limits while changing ultraviolet excitation costs. This positive stroboscopic generator identifies neither laboratory energy nor the work of the supplied controls.

**Definition 9.1** (Federated observer checkpoint)**.** *For an observer-supporting patch subfederation $O$, a checkpoint at semantic cut $D$ is*

$$\mathrm{Chk}_O(D) = (\mathcal{R}_O(D), \rho_O^{\mathrm{acc}}(D), \mathfrak{I}_O^{\mathrm{ext}}(D), \mathcal{L}_O^{\mathrm{sem}}(D), \mathfrak{B}_O(D)),$$

*where $\mathcal{R}_O(D)$ is the observer-accessible record algebra, $\rho_O^{\mathrm{acc}}(D)$ is the state restricted to observer-accessible records and visible interfaces, $\mathfrak{I}_O^{\mathrm{ext}}(D)$ is the external port-interface tuple, $\mathcal{L}_O^{\mathrm{sem}}(D)$ is the semantic future-law class for the same physical instruments and event algebra, and $\mathfrak{B}_O(D)$ is the public or internal provenance bundle needed to replay the checkpoint at the declared accuracy. Worker IDs, queue positions, retry counters, repair-cycle indices, timestamps, and packet latencies live in $\mathfrak{B}_O(D)$ unless a branch declares them physical inputs; they are not observer identity or observer time.*

**Theorem 9.2** (Checkpoint continuation at fixed cutoff)**.** *If two federated observer checkpoints agree exactly on $\mathcal{R}_O(D)$, $\rho_O^{\mathrm{acc}}(D)$, $\mathfrak{I}_O^{\mathrm{ext}}(D)$, and $\mathcal{L}_O^{\mathrm{sem}}(D)$, then they induce the same future probability law on the observer-accessible event algebra. If their accessible states differ by trace distance at most $\varepsilon$, then the induced future history laws differ in total variation by at most $\varepsilon$.*

*Proof.* All future observer-accessible probabilities are computed by applying the same completely positive update maps and the same event-readout maps from the same semantic future-law class to the same accessible algebra and external interface data. Exact equality gives equality of all future event probabilities. In the approximate case, contractivity of trace distance under completely positive trace-preserving maps gives the stated total-variation bound. □

**Definition 9.3** (Public checkpoint capacity interface)**.** *For a finite federated checkpoint family, the capacity-facing export is*

$$\mathsf{PubChk}_{r,D} = \big(\{X_O\}, \{X_e\}, \{r_{Oe}\}, \mathsf{Reach}, \mathfrak{P}, \mathfrak{K}, \mathcal{H}_{\mathrm{cap},r,D}, \{P_x\}, \mathsf{Ext}, \mathsf{Ref}\big),$$

*with local central-record atoms $X_O = \mathrm{At}\,\mathcal{R}_O$, interface-record atoms $X_e$, completed-instrument atom readouts $r_{Oe} : X_O \to X_e$, a semantic event DAG* Reach *witnessing endogenous public-record reachability, a frozen publicness policy $\mathfrak{P}$, source-derived joint checkpoint kernels $\mathfrak{K}$ with the required marginals, a capacity carrier of dimension $D$ with nonzero record projections $P_x$ for reachable public classes, and separate capacity-extension and fixed-capacity refinement packets.*

These fields are an interface contract, not automatic outputs of a local checkpoint: the visible algebra maps $\pi_{i,e}$ do not supply the record-atom maps $r_{Oe}$, and observer-by-observer marginals do not determine the joint kernels $\mathfrak{K}$. In the observers layer [12] the packet defines the multiplicative readback $M_0(q) = \alpha(G_q)$, where $G_q$ is the compound confusability graph of reachable public records. When the complete terminal fibre scalarizes, $M_0(\mathfrak{U}_N) = \widehat{F}_{r,0}(e^N)$, and the universe-level equation is $N = \log M_0(\mathfrak{U}_N)$: $M_0$ counts records and $N$ is the logarithmic capacity. The correctable-code capacity and finite-size closure theorems are carried by [15].

## 10 Central records and measurement

The measurement package is a finite central-record package. Records are exposed by observer-facing patch subfederations.

Let $\mathcal{Z}_{\mathrm{rec}}(t)$ be the commutative algebra generated by the completed, observer-accessible record projectors at cycle $t$. An event $E$ is a projector in $\mathcal{Z}_{\mathrm{rec}}(t)$. Declare a finite-dimensional $*$-algebra $\mathcal{A}_t$, a normalized state $\rho$, and an embedding of $\mathcal{Z}_{\mathrm{rec}}(t)$ into its center. These quantum representation data are inputs rather than outputs of the repair process. The declared operational measurement rule is:

$$\Pr(E) = \mathrm{Tr}(\rho E), \qquad \rho \mapsto \frac{E\rho E}{\mathrm{Tr}(\rho E)} \quad \text{when } \Pr(E) > 0.$$

**Theorem 10.1** (Conditional federated central-record measurement)**.** *On a fixed-cutoff federated patch carrier, suppose a completed write/verify slice exposes a finite commutative central record algebra $\mathcal{Z}_{\mathrm{rec}}(t)$ with the declared algebra-state representation above. The Born probability and declared Lüders conditioning rule above define the operational measurement package on the observer-accessible event surface. Re-reading the same completed record event has probability one, conditional on that record event.*

*Proof.* The theorem is the standard finite-dimensional algebra-state argument. Because all declared record events commute and belong to the observer-accessible center for the completed slice, they define an ordinary finite classical event algebra. Probabilities are the supplied Born traces on that event algebra, and conditioning on an event uses the declared Lüders update. The effect projectors alone do not select this update; same-effect non-Lüders instruments exist. After conditioning on $E$, the same central projector $E$ is true with probability one. Each finite-matrix step of this argument (commutativity and centrality of the record span, Born traces, the Lüders update with its fixed-point law, and the collapse of record conditioning to the normalized projector) is machine-checked in the companion Lean 4 event-algebra development [11]. □

### 10.1 Record-conditioned modular cap responses

For a quotient-visible record token $i$ seen by observer $O$, let $P_{i,O}$ be its central, or declared approximately central, record projector with $\omega_O(P_{i,O}) > 0$. The conditioned record state is the Lüders state $\omega_{i,O}(A) = \omega_O(P_{i,O}AP_{i,O})/\omega_O(P_{i,O})$. For a cap $C$ with declared probe $M_{C,0,O}$, set $M_{C,t,O} := \sigma_t^{C,O}(M_{C,0,O})$ and $R_i(C,t,O) := \omega_{i,O}(M_{C,t,O})$, with the $2\pi$-normalized cap flow $\sigma_t^{C,O} = \alpha_{\lambda_C(2\pi t)}$ of the spacetime layer [20]. A point-source localization branch is a hypothesis or finite receipt, not a consequence of a record projector. It must exhibit

$$R_i(C,t,O) = b(C,t,O) + a(i,C,t,O)\,\psi_{C,t,O}\left(\frac{\eta(X_i(t), n(C,t,O))}{R_H}\right) + \xi(i,C,t,O)$$

with declared gains, baselines, kernel shape, cap normal and modular normalization; centrality defect, calibration, modular transport, kernel mismatch, finite-record noise and point-model error all enter the single localization error norm $\sigma_{\mathrm{record}}$. Carrier presentations inducing the same record token, conditioned state, cap probe, cap geometry, normalization and calibration produce the same response vector and the same $H^3$ localization ball. A fitted point that fails held-out cap residuals, mixture controls or the error bound is a failed or approximate branch.

### 10.2 A finite Bell candidate

A federated carrier may contain two observer-facing wings with commuting setting and outcome records on one compare slice. The commuting record indicators alone fix neither a two-wing representation nor a joint quantum law. The binary-icosahedral spinor adapter gives one exact finite candidate: its unique invariant singlet and an incidence-defined 120-row setting family give

$$|S_{\mathrm{CHSH}}| = 1 + \frac{3}{\sqrt{5}} > 2.$$

All 720 Pauli covariance identities and the joint law are checked by an independent exact verifier. The complete twelve-port census has 960 maximizing quadruples, so the 120-row family is not selected uniquely. The source supplies no completed two-wing record instrument or source-selected setting mechanism, so this is a finite projective candidate and no physical Bell prediction.

## 11 Edge sectors and the compact-gauge layer

At fixed cutoff, let $\alpha$ label a finite set of exposed edge sectors on one declared overlap collar, with stationary weights

$$\pi_\beta(\alpha) = \frac{d_\alpha e^{-\beta C_2(\alpha)}}{Z(\beta)}.$$

For finite groups $C_2(\alpha)$ is the declared sector penalty or finite-group Casimir surrogate; for compact groups the Peter–Weyl lift belongs to the compact-gauge construction.

**Theorem 11.1** (Fixed-cutoff edge-sector law)**.** *If a declared finite overlap collar has sector labels $\alpha$, degeneracies $d_\alpha$, and a local repair/thermalization generator whose detailed-balance stationary law is $\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)}$, then the collar gives the fixed-cutoff Casimir edge law used by the compact-gauge construction. The compact-group heat-kernel lift requires the separate companion theorem.*

*Proof.* This is a finite stationary-measure statement on the declared sector algebra; the compact-group lift uses the Peter–Weyl construction and normalization of the compact-gauge construction. □

The carrier supplies the compact-gauge layer with the fixed-cutoff patch, overlap, record and repair interface, the finite edge-sector law, and local repair semantics read in controlled quotient form. Complete reversible port response and endogenous overlap transport force the abstract Lie-algebra type $\mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3)$ under the premises of the gauge layer [14]. The finite cylinder system and its projective GNS extraction are proved under the compact-gauge premises. Identification with the four-dimensional Yang–Mills state, reflection-positive reconstruction, noncollapse, and equality between the Yang–Mills gap and the repair gap require separate certificates [19], and an OPH-native vacuum needs the quotient-ensemble selector, source Euclidean slab data and transfer reconstruction of Section 12. A positive repair relaxation gap is not a physical mass gap.

## 12 Finite quotient ensemble compatibility

**Finite quotient ensemble theorem.** Fix a finite regulator $r$. The physical presentation space is $\Sigma_r$, the presentation redundancy groupoid is $\Gamma_r$, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map $n_r$ is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to $N_r$, or set $S_r = +\infty$ outside $N_r$. A separately specified prior $\nu_r$ on $Q_r$ instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_{\#}\nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x)=q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and $\mu_r$ has no mass outside $N_r$. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims, and a physical branch declares which one it uses.

**Normal-form projector non-selection.** For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to $Q_{\mathrm{nf}}$ is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on $Q_{\mathrm{nf}}$ is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine $X$ and its quotient-visible invariants, but they do not choose a member of $X$. If two laws $\mu, \nu$ are supported on $X$ and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite $Q$, positive $m$, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q)\exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state $\sigma_r$, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log\sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr}\exp(\log\sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection $P$, first restrict to their joint support. The same multiplier argument uses $P(\log\sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q')=q} m_s(q')e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of $q$. Then

$$(c_{sr})_{\#}\mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0}\left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r\right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta)\cdot F_r(q_r)$. If the residual is uniformly bounded by $\varepsilon$, the induced total-variation defect is bounded by $\tanh\varepsilon$.

**Implementation invariance and representative lifting.** If implementations $A, B$ have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel $\widetilde{P}(x, y)$ descends to $Q_r$ only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations $E_C$ are defined on $L^2(X_r, \pi_r)$, so the reference law $\pi_r$ is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where $P_0$ projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^{\perp} \neq \{0\}$, since zero Dirichlet energy would put $f$ in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \operatorname{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by $s$, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

**Finite evidence accuracy.** For bounded coarse observables $O$, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to $\epsilon_{\mathrm{ref}}$, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Physical-vacuum requirement.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q,q') = J_r(q',q) \ge 0$, local potential $V_r$, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q,q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state $\Omega_r$, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with $\mu_r^{\mathrm{vac}}$. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family. On the native source screens the actual density law has such a reflection-positive reconstruction along one common refining subsequence; its one-particle generator is the native continuum operator $A$ rather than $\sqrt{A}$, and it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry.[12]

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance. On the native source screens the continuum energy is not boundedly congruent to the static boundary

[12] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/observer_time_fields/REFINEMENT.md

action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like $n$, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply.[13]

# 13 Conclusion

The carrier layer proposes that observers execute on federations of twelve-port icosahedral carriers repaired by the native seam law. On one carrier, oriented incidence and the declared counting grammar select the twelve unit port lines, inverse pairing, proper $A_5$ action and rank-three Gram frame; four stated requirements characterize the classical seam rule; its record laws form a five-multiplier convex family; and the uniform repair mean selects the normalized rank-three Gram band as its slowest response. The twelve-port orbit fixes the directional content of every invariant cosine kinetics through angular rank five and confines its first residue to the unique rank-six line. The seam complex carries exact finite counterparts of current, continuity, Gauss law, local face action and stable step-scaled evolution, and native repair histories carry exact stationary covariance and record-power laws. Central records, checkpoints and the edge-sector law give the measurement and continuation interface consumed by the observer layer.

Three joins leave this layer. Cross-carrier routing must become record-derived and refinement natural, since the existing port-ancestry map fails seam adjacency. The rank-three response band must acquire physical position, clock and energy maps before it describes space. And a protected matter defect must arise from overlap-transport data that the repair law supplies, because the carrier's face complex has vanishing first cohomology and the tested native double-load candidate is transient.

# A The edge-center scalar register

The dark-sector and coherent-matter branches consume one fixed-cutoff export of the carrier: a quotient-visible scalar register on a connected collar cut, the protected $\mathbb{Z}_6$ reserve on the same register, and the finite-thickness survival coefficient built from them. Every result in this appendix is conditional on the completeness and reserve premises stated in it; none fixes a physical coefficient by itself.

## A.1 Rank-one scalar-slot completeness

**Definition A.1** (Finite scalar-slot register)**.** *Fix a regulator $r$, physical quotient $Q_r = \Sigma_r/\Gamma_r$, and collar or local support region $C$. Let $E_{\nu,r}(C)$ be the finite set of edge-center scalar slots on the physical quotient. For each $e \in E_{\nu,r}(C)$, let $p_{e,\nu} : Q_r \to \{0,1\}$ be the quotient-visible central indicator that slot $e$ carries an active scalar repair opportunity, and let $a_e > 0$ be the declared scalar-slot weight. The canonical quotient-edge scalar opportunity count and normalized readout are*

$$\mathcal{N}_{\nu,r,C}(q) = \sum_{e \in E_{\nu,r}(C)} a_e\, p_{e,\nu}(q), \qquad S_{\nu,r,C}(q) = \frac{\mathcal{N}_{\nu,r,C}(q)}{\sum_e a_e}.$$

*The indicators live on $Q_r$; gauge representatives, port labels, implementation coordinates and accepted repair schedules are removed before $\mathcal{N}_{\nu,r,C}$ is read.*

[13] https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/physical_space_time/OCCUPATION_FIELD.md

**Theorem A.2** (Rank-one scalar-slot completeness on the quotient-edge branch)**.** *Let $G_C$ be the quotient-preserving relabeling group of scalar slots in collar $C$. Assume that $G_C$ preserves the physical quotient and the weights $a_e$, that the scalar branch keeps only quotient-visible central observables invariant under $G_C$, and that the scalar slots form one scalar orbit after transverse finite-thickness averaging. Then every first-order quotient-visible scalar observable on the register has the form $F_{\mathrm{sc}}(q) = \alpha + \beta\,\mathcal{N}_{\nu,r,C}(q)$; after constants are removed, $\mathcal{S}^{(1)}_{\mathrm{sc}}(C) = \mathrm{span}\{\mathcal{N}_{\nu,r,C}\}$.*

*Proof.* A first-order scalar observable has the form $F(q) = \alpha + \sum_e b_e\, p_{e,\nu}(q)$. Invariance under the transitive action of $G_C$ on the slot orbit forces $b_e/a_e$ to be constant, so $F = \alpha + \beta\mathcal{N}_{\nu,r,C}$. With finite transverse thickness the same argument applies orbitwise before the declared weight integral. □

The theorem concerns only the scalar quotient of the edge-center register. It selects no primitive observer size, hardware carrier, probability law on $Q_r$ or finite covariant stress parent.

## A.2 Edge-center algebra, scalar activation and the protected reserve

Throughout, $\tau_{q,r,C}$ is the normalized quotient expectation on the finite physical collar quotient algebra, so $\tau_{q,r,C}(1) = 1$. For a projection $p$ with $\tau_{q,r,C}(p) > 0$, the reciprocal reserve depth is $d^{\#}_{q,r,C}(p) := \tau_{q,r,C}(p)^{-1}$. Survival coefficients use the normalized mean, never the reciprocal depth.

**Definition A.3** (Physical collar quotient and edge-center map)**.** *Fix a finite regulator $r$ and a connected collar cut $C$. Let $Q_{r,C}$ be the physical collar quotient after removing gauge representatives, hidden carrier coordinates, port labels, repair-schedule identifiers, mesh labels, shard or worker identifiers and inert ancillary labels, and let $\mathcal{A}^{\mathrm{phys}}_{r,C}$ be the finite physical collar algebra. The cut exposes a quotient-visible edge-center sector map $\eta_{r,C} : Q_{r,C} \to \Xi_{r,C}$ onto the finite set of edge-center cut sectors of the repaired collar normal form. For $\xi \in \Xi_{r,C}$, $e_{\xi,r,C}$ is the central sector projection of the fibre $\eta^{-1}_{r,C}(\xi)$.*

**Definition A.4** (Edge-center quotient algebra)**.** $\mathcal{E}_{r,C} := \mathrm{Alg}\{e_{\xi,r,C} : \xi \in \Xi_{r,C}\} \subseteq Z(\mathcal{A}^{\mathrm{phys}}_{r,C})$*, so that* $\mathcal{E}_{r,C} \cong \ell^\infty(\Xi_{r,C})$ *and* $1_{\mathcal{E}_{r,C}} = \sum_\xi e_{\xi,r,C}$.

**Theorem A.5** (Edge-center quotient algebra)**.** *On the fixed-cutoff connected-collar branch, $\mathcal{E}_{r,C}$ is a finite commutative central subalgebra of $\mathcal{A}^{\mathrm{phys}}_{r,C}$.*

*Proof.* The fibres partition the collar quotient, so $e_\xi e_{\xi'} = \delta_{\xi\xi'} e_\xi$ and $\sum_\xi e_\xi = 1$. The edge-center label is quotient-visible superselection data of the cut, so the projectors are central; finitely many orthogonal central projections generate $\ell^\infty(\Xi_{r,C})$. □

**Definition A.6** (Quotient scalar readout algebra)**.** $\mathsf{Scal}_{r,C} \subseteq \mathcal{A}^{\mathrm{phys}}_{r,C}$ *is the subalgebra generated by the quotient-local collar readouts invariant under orientation, frame and bulk-coordinate choice, hidden carrier coordinates, port relabeling, accepted repair schedule, mesh and shard labels and inert ancillary labels.*

**Assumption A.7** (Edge-center scalar completeness)**.** *On the declared fixed-cutoff scalar branch,* $\mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}$*: no independent scalar event algebra is carried by bulk coordinates, screen-area labels, noncentral matrix directions, hidden carrier coordinates, port labels or repair-schedule identifiers after the physical quotient.*

**Definition A.8** (Scalar recoverability defect and activation projector)**.** *Let $R_{r,C} \in \mathsf{Scal}_{r,C,+}$ be the quotient-local scalar recoverability defect on the cut; on the positive scalar source branch $R_{r,C} = I(A : D \mid B)$ or its quotient-local finite-collar representative. The scalar activation projector is* $\Pi^{\mathrm{scal}}_{r,C} := \mathbf{1}_{(0,\infty)}(R_{r,C})$.

**Theorem A.9** (Scalar activation is an edge-center event)**.** *Under edge-center scalar completeness,* $\Pi^{\mathrm{scal}}_{r,C} \in \mathcal{E}_{r,C}$.

*Proof.* $R_{r,C} \in \mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}$, and functional calculus is internal to the finite commutative algebra $\mathcal{E}_{r,C}$. □

**Definition A.10** (Protected $Z_6$ reserve projector)**.** *Let the declared faithful matter image on this finite protected-center branch be* $G_{\mathrm{packet}} = (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$*; this declaration does not select the physical global form. Let* $\zeta^{(j)}_{6,r,C} : \Xi_{r,C} \to \{0,1\}$, $j \in \mathbb{Z}_6$, *be six pairwise disjoint quotient-visible indicators of the protected reserve classes, and set*

$$Z^{(j)}_{6,r,C} := \sum_{\xi:\zeta^{(j)}_{6,r,C}(\xi)=1} e_{\xi,r,C}, \qquad Z^{\mathrm{tot}}_{6,r,C} := \sum_j Z^{(j)}_{6,r,C}, \qquad Z_{6,r,C} := Z^{(j_0)}_{6,r,C}$$

*for one declared class* $j_0$*. All of these lie in* $\mathcal{E}_{r,C}$*. The notation does not select* $j_0$ *as the physical blocked class; any use of one class rather than the total projector carries that attachment separately.*

**Theorem A.11** (Same-register and commutation theorem)**.** *Assume edge-center scalar completeness and protected-center realization. Then* $\Pi^{\mathrm{scal}}_{r,C}, Z_{6,r,C} \in \mathcal{E}_{r,C}$*, and therefore* $[\Pi^{\mathrm{scal}}_{r,C}, Z_{6,r,C}] = 0$ *in the physical quotient collar algebra.*

*Proof.* Theorem A.9 and Definition A.10 place both projectors in the commutative algebra $\mathcal{E}_{r,C}$. □

**Assumption A.12** (Scalar activation channel completeness)**.** *The scalar activation channel on the cut is* $\mathcal{C}^{\mathrm{scal}}_{r,C} = \mathrm{Alg}\{\Pi^{\mathrm{scal}}_{r,C}\} \subseteq \mathcal{E}_{r,C}$*; there is no second independent scalar activation generator after the physical quotient.*

**Theorem A.13** (Scalar-channel exhaustion)**.** *Assume edge-center scalar completeness, scalar activation channel completeness and protected-center realization. Then the scalar activation channel is exhausted by* $\Pi^{\mathrm{scal}}_{r,C}$*, and the declared class* $j_0$ *splits it into commuting orthogonal projections,*

$$\Pi^{\mathrm{scal}}_{r,C} = \Pi^{\mathrm{scal}}_{r,C}(1 - Z_{6,r,C}) + \Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C}.$$

*The first summand survives the declared* $j_0$ *event and the second is blocked by it. Calling* $j_0$ *the physical blocked class is a separate attachment, and the total projector gives a different split.*

*Proof.* Channel completeness leaves no activation support outside $\Pi^{\mathrm{scal}}_{r,C}$, and Theorem A.11 makes the two summands commuting orthogonal projections whose sum is $\Pi^{\mathrm{scal}}_{r,C}$. □

**Theorem A.14** (No separate scalar carrier)**.** *Assume edge-center scalar completeness, and let* $B$ *be any proposed alternative scalar activation projector on a bulk, screen-area, noncentral, implementation-local or hidden-carrier algebra. If* $B$ *changes scalar activation without factoring through* $\mathcal{E}_{r,C}$*, the physical quotient observable algebra has been changed. If* $B$ *changes no quotient observable, its scalar content equals its edge-center central support in* $\mathcal{E}_{r,C}$*, and the extra carrier is inert.*

*Proof.* By completeness every physical scalar activation event factors through $\eta_{r,C}$. If $B$ does not, presentations agreeing on the edge-center sector differ in $B$; if that changes activation, activation is not determined by the declared scalar quotient. Otherwise the extra degrees of freedom are silent and only the central support of $B$, which lies in $\mathcal{E}_{r,C}$, contributes. For noncentral $B$ the same argument runs inside $\mathcal{A}^{\mathrm{phys}}_{r,C}$. □

## A.3 Coherent-matter scalar-source forcing

The exhaustion theorem fixes the available local scalar carrier. It does not turn raw mass, heat, stored energy or an engineering substrate into a scalar source.

**Definition A.15** (OPH-coherent material subfederation)**.** *Fix a screen collar $C$, a finite material region $U \subset C$ and a refinement scale $r$. An OPH-coherent material subfederation is a family $\mathfrak{M} = (\mathfrak{F}_U, \mathsf{R}_U, \mathsf{P}_U, \mathsf{C}_U)$, where $\mathfrak{F}_U$ is a self-reading subfederation over $U$, $\mathsf{R}_U$ is a stable internal record readout, $\mathsf{P}_U$ is a boundary prediction operator against neighbouring screen states, and $\mathsf{C}_U$ is a coherent mismatch-reduction certificate. It is nondegenerate when $\mathbf{1}_{\mathrm{self-read}}(\mathfrak{F}_U)\,\mathsf{R}_U\,\mathsf{P}_U\,\mathsf{C}_U > 0$ survives quotienting and refinement. Presentations with the same quotient record, boundary prediction and coherence certificate define the same class.*

**Definition A.16** (Canonical coherent-matter scalar source)**.**

$$S^{\mathrm{can}}_{\mathrm{coh}}(U,t;h) := \mathbf{1}_{\mathrm{self-read}}(\mathfrak{F}_U)\,\mathsf{R}_U(U,t;h)\,\mathsf{P}_U(U,t;h)\,\mathsf{C}_U(U,t;h).$$

*It vanishes exactly when one of the self-read, record, boundary prediction or coherent mismatch-reduction receipts vanishes. Rest mass, heat and arbitrary coherent energy are outside this definition.*

**Claim A.17** (Quotient descent of coherent-material source)**.** $S^{\mathrm{can}}_{\mathrm{coh}}$ *descends to the screen quotient and is invariant under relabelings of internal coordinates, duplicate micro-presentations and record refinements preserving* $(\mathsf{R}_U, \mathsf{P}_U, \mathsf{C}_U)$ *in the quotient.*

*Proof.* Each factor is a screen-record observable, and the quotient identifies presentations with identical boundary-visible records and predictions. □

**Theorem A.18** (Nonzero coherent scalar source)**.** *Every nondegenerate OPH-coherent material subfederation has $S^{\mathrm{can}}_{\mathrm{coh}}(U,t;h) > 0$; the zero branch is exactly the degenerate branch.*

*Proof.* Nondegeneracy is strict positivity of the defining product. □

**Theorem A.19** (Coherent-matter same-channel forcing)**.** *Assume Scalar Edge-Center Exhaustion, Theorem A.13. For every nondegenerate OPH-coherent material subfederation on a collar $C$, the source $S^{\mathrm{can}}_{\mathrm{coh}}$ is valued in the scalar edge-center register $\mathcal{E}_{r,C}$, the same scalar channel that carries the $Z_6$ reserve.*

*Proof.* By Claim A.17 the source is a quotient-local scalar observable with no boundary direction, spin frame or unscreened vector index, so it factors through $\mathcal{E}_{r,C}$; the reserve lies in the same register by Theorem A.11. □

**Corollary A.20** (Finite scalar channel bridge)**.** *On the same branch the coherent-material record slots and the $Z_6$ collar slices form one finite indexed channel $E$: the slots of $\mathcal{E}_{r,C}$. "The same scalar channel" means that both panels are functions of the same finite family $E$; equality of two separately defined numerical counters is insufficient.*

**Theorem A.21** (Single scalar response form with undetermined effective coefficient)**.** *On the weak-field branch, assume that the frequency response is differentiable at zero source and that its quotient-local scalar dependence factors through the edge-center register. Then*

$$\delta\nu_C = \chi^{\mathrm{eff}}_{\nu,C}\,\langle \eta_C, S^{\mathrm{can}}_{\mathrm{coh}}\rangle_C + O\Big((S^{\mathrm{can}}_{\mathrm{coh}})^2\Big)$$

*for a local edge-center test functional $\eta_C$ and an undetermined effective coefficient $\chi^{\mathrm{eff}}_{\nu,C}$. The result fixes the scalar form only; it neither derives the coefficient nor excludes several mechanisms adding in the same channel.*

*Proof.* Differentiability gives a linear functional on the register, written in the displayed form after normalization; same-channel contributions add into the effective coefficient. □

## A.4 The oriented register and the reserve mean

**Definition A.22** (Oriented 24-slot repair register)**.** *On the twelve-port carrier-sieve branch, let* $\mathsf{P}_{12,r,C}$ *be the twelve exposed central carrier ports of the cut. Reversible write/check orientation gives* $\mathsf{R}_{24,r,C} := \mathsf{P}_{12,r,C} \times \{+,-\}$*, with* $|\mathsf{R}_{24,r,C}| = 24$*, and the slot algebra* $\mathcal{R}^{\mathrm{or}}_{24,r,C} := \ell^\infty(\mathsf{R}_{24,r,C})$*. The oriented register refines the bookkeeping of the same cut surface and creates no independent scalar carrier.*

**Proposition A.23** (The oriented register carries no modular clock)**.** *Every faithful state on* $\mathcal{R}^{\mathrm{or}}_{24,r,C}$ *is tracial, so its modular automorphism group is trivial. A nontrivial modular lift requires a noncommutative transition algebra, such as* $B(\ell^2 X)$ *with operators* $V_{u\to v} = |v\rangle\langle u|$*, and a faithful physical state on it.*

*Proof.* The density of a faithful state on a finite commutative algebra is central, hence $\rho^{it} a \rho^{-it} = a$. □

The shared-edge protected-reserve branch has two declared inputs: the shared cut entropy density $\bar{\ell}_{\mathrm{shared}} = P/4$ and $\mathbb{Z}_6$ class equidistribution. Every value downstream of the $P/24$ class mean inherits both. The denominator 24 of the oriented register is a coincidence that carries no corroborating weight for equidistribution.

**Theorem A.24** (Protected reserve mean and reciprocal depth)**.** *On the shared-edge protected-reserve branch, assume* $P > 0$*,* $\tau_{q,r,C}(Z^{\mathrm{tot}}_{6,r,C}) = \bar{\ell}_{\mathrm{shared}} = P/4$*, and* $\mathbb{Z}_6$ *class equidistribution. Then* $\tau_{q,r,C}(Z^{(j)}_{6,r,C}) = P/24$ *for every* $j$*, in particular for* $Z_{6,r,C}$*, and* $d^{\#}_{q,r,C}(Z_{6,r,C}) = 24/P$*. At the comparison coordinate* $P_C = 1.630968209403959$*,*

$$\frac{P_C}{24} = 0.06795700872516496, \qquad \frac{24}{P_C} = 14.715185655746689,$$

*and, under the presence reading of Lemma A.27,* $1 - P_C/24 = 0.9320428293392126\ldots$.

*Proof.* Additivity over six disjoint classes of equal expectation gives $(P/4)/6 = P/24$; the reciprocal depth is its inverse. □

The total and one-class projectors differ, $\tau_q(Z^{\mathrm{tot}}_6) = P/4$ and $\tau_q(Z_6) = P/24$; the $1 - P/24$ coefficient uses the latter, so a physical application must identify one class as the blocked event.

## A.5 Finite-thickness collar survival

**Definition A.25** (Finite-thickness scalar opportunity profile)**.** *Let* $Y_{r,C}$ *be the finite transverse collar-coordinate set with slice projectors* $p_y$*,* $\sum_y p_y = 1$*, and let* $\mathcal{A}^{\mathrm{thick}}_{r,C} \supseteq \mathcal{E}_{r,C} \vee \mathrm{Alg}\{p_y\}$ *be the finite-thickness collar algebra with lifted projectors* $\Pi^{\mathrm{scal,thick}}_{r,C}$ *and* $Z^{\mathrm{thick}}_{6,r,C}$*. Assume that* $\Pi^{\mathrm{scal,thick}}_{r,C}$*,* $Z^{\mathrm{thick}}_{6,r,C}$ *and every* $p_y$ *belong to one common finite commutative event algebra* $\mathcal{C}_{r,C} \subseteq \mathcal{A}^{\mathrm{thick}}_{r,C}$*; mere containment in the thick algebra does not imply this commutation clause. With* $\tau = \tau^{\mathrm{thick}}_{q,r,C}$*, set*

$$a_r(y) := \tau(\Pi^{\mathrm{scal,thick}}_{r,C} p_y), \quad A_r := \sum_y a_r(y) > 0, \quad w_r(y) := \frac{a_r(y)}{A_r},$$

$$\epsilon_{\mathbb{Z}_6,r}(y) := \frac{\tau(\Pi_{r,C}^{\mathrm{scal,thick}} Z_{6,r,C}^{\mathrm{thick}} p_y)}{\tau(\Pi_{r,C}^{\mathrm{scal,thick}} p_y)}$$

*for scalar-active slices, and the scalar-weighted reserve mean* $\bar{\epsilon}_{\mathbb{Z}_6,r}^{\mathrm{scal}} := \sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y)$. *For continuum notation, embed the finite coordinate sets in one measurable space and form* $\mu_r = \sum_y w_r(y)\delta_y$. *A continuum branch must separately certify that* $\mu_r \to w(y)\,dy$ *and that the weighted reserve means converge to* $\int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y)$, *with* $0 \le \epsilon_{\mathbb{Z}_6} \le 1$ *and* $\int w = 1$. *Pointwise arrows on regulator-dependent finite sets do not by themselves imply this integral convergence; only under that continuum convergence receipt do we write* $\bar{\epsilon}_{\mathbb{Z}_6}^{\mathrm{scal}} = \int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y)$.

**Theorem A.26** (Finite-thickness mean reserve disintegration)**.** *At fixed regulator,*

$$\sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{\tau(\Pi_{r,C}^{\mathrm{scal,thick}} Z_{6,r,C}^{\mathrm{thick}})}{\tau(\Pi_{r,C}^{\mathrm{scal,thick}})}.$$

*If this scalar-weighted quotient mean equals* $P/24$, *so does the left side. If a cofinal regulator tail satisfies that receipt and the continuum convergence receipt, then* $\int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y) = P/24$.

*Proof.* The slice factors cancel, and $\sum_y p_y = 1$ sums the numerators to $\tau(\Pi^{\mathrm{scal}} Z_6)$. The continuum statement uses the assumed convergence of weighted means; no pointwise-limit interchange is inferred. □

The identity $\tau_q(Z_{6,r,C}) = P/24$ does not imply the scalar-weighted receipt

$$\tau_q(\Pi_{r,C}^{\mathrm{scal}} Z_{6,r,C})/\tau_q(\Pi_{r,C}^{\mathrm{scal}}) = P/24,$$

which follows from scalar-reserve unbiasedness; co-registration or commutation alone is insufficient.

**Lemma A.27** (Presence reading of the reserve profile)**.** *On the common commutative event algebra of Definition A.25, the finite profile* $\epsilon_{\mathbb{Z}_6,r}(y) \in [0,1]$ *is the conditional probability of the reserve-presence event in a co-registered scalar slot at* $y$; *the continuum profile has this reading only with the measure-convergence receipt. A scalar opportunity survives exactly when the presence event does not occur, so* $\lambda_{\mathrm{slot}}(y) = 1 - \epsilon_{\mathbb{Z}_6}(y)$. *No occupancy variable with that presence probability attains the Poisson zero-count factor* $e^{-\epsilon}$: $1 - \epsilon < e^{-\epsilon}$ *for* $\epsilon > 0$, *and every finite sub-slot refinement* $(1 - \epsilon/m)^m$ *lies strictly below its unattained supremum* $e^{-\epsilon}$. *A mean-count reading, consuming* $\epsilon(y)$ *as the expectation of an* $\mathbb{N}$*-valued occupancy, severs the link to the projection trace and needs a carrier beyond* $Z_{6,r,C}$ *(Remark A.30).*

**Theorem A.28** (Finite-thickness collar survival)**.** *Under the presence reading,* $\lambda_{\mathrm{collar,r}} = 1 - \sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y)$; *with the continuum convergence receipt,* $\lambda_{\mathrm{collar}} = 1 - \int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y)$.

*Proof.* Average the local survival factor of Lemma A.27 over the normalized profile. □

**Corollary A.29** (Exact finite-thickness coefficient from the scalar-weighted receipt)**.** *If* $\int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y) = P/24$, *then* $\lambda_{\mathrm{collar}} = 1 - P/24 = 0.9320428293392126\ldots$ *at* $P_C$. *The scalar-weighted receipt fixes the coefficient by linearity; no uniformity of the profile is consumed. In general* $\lambda_{\mathrm{collar}} = 1 - \bar{\epsilon}_{\mathbb{Z}_6}^{\mathrm{scal}}$.

*Remark* A.30 (Mean-count reading and its Markov band). Under the mean-count reading, Markov's inequality gives $1 - \epsilon \le \Pr[N = 0] \le 1$, hence with the scalar-weighted $P/24$ receipt $1 - P/24 \le \lambda_{\mathrm{collar}} \le 1$. The coefficient is order one on every reading; its exact value is reading dependent, and the presence reading is the one this branch's receipt semantics supply.

**Definition A.31** (Uniform product-thickening branch)**.** *The branch consists of a product collar algebra* $\mathcal{A}^{\mathrm{thick}}_{r,C} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y)$*, a product trace* $\tau^{\mathcal{E}}_{q,r,C} \otimes \tau^{Y}_{r}$*, the reserve pullback* $Z^{\mathrm{thick}}_{6,r,C} = Z_{6,r,C} \otimes 1_Y$*, the disintegration* $\Pi^{\mathrm{scal,thick}}_{r,C} = \sum_y \Pi^{\mathrm{scal}}_{r,C}(y) \otimes p_y$ *with* $\Pi^{\mathrm{scal}}_{r,C}(y) \in \mathcal{E}_{r,C}$*, and slice-wise scalar-reserve unbiasedness,* $\tau^{\mathcal{E}}_{q,r,C}(\Pi^{\mathrm{scal}}_{r,C}(y) Z_{6,r,C}) / \tau^{\mathcal{E}}_{q,r,C}(\Pi^{\mathrm{scal}}_{r,C}(y)) = \tau^{\mathcal{E}}_{q,r,C}(Z_{6,r,C}) = P/24$*. The last clause is essential: product algebra and trace do not force the scalar-active sampler to be reserve-unbiased.*

**Theorem A.32** (Exact uniform product-thickening coefficient)**.** *On the uniform product-thickening branch, under the presence reading,* $\lambda_{\mathrm{collar}} = 1 - P/24 = 0.9320428293392126\ldots$ *at* $P_C$*.*

*Proof.* Unbiasedness gives $\epsilon_{\mathbb{Z}_6}(y) = P/24$ on every scalar-active slice, and $w$ is normalized. □

*Remark* A.33 (Exact value licensing). The exact value $1 - P/24$ is licensed by the scalar-weighted $P/24$ receipt under the presence reading. Without that receipt the theorem-grade coefficient is $1 - \int dy\, w(y) \epsilon_{\mathbb{Z}_6}(y)$, and under the mean-count reading only the Markov band $1 - P/24 \le \lambda_{\mathrm{collar}} \le 1$ is available. No survival coefficient uses $24/P$.

## Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Classical quaternion group representation theory and McKay correspondence [36, 38, 54].
- Conditional variance decomposition, unbiased cross-replicate power and finite-population sampling covariance [31, 44, 35].
- Courant-Friedrichs-Lewy stability certificates for leapfrog schemes [37, 67].
- Discrete exterior calculus and finite lattice Maxwell actions [40, 68].
- Discrete exterior calculus and staggered leapfrog Maxwell integrators on cell complexes [73, 66].
- Discrete Hodge theory, graph Laplacian Green functions, and the Thomson variational principle [52, 41].
- Finite causal sets, signed graph bundles, discrete differential operators, and reproducible scientific workflows [2, 74, 49, 42, 60].
- Finite graph continuity and Gauss laws, cycle-space/Hodge ambiguity, and Heisenberg fixed-observable conservation criteria [29, 42, 52, 26].
- Finite reversible Markov chains, graph Hodge theory, operator compression, and finite-group representation theory [47, 61, 52, 27, 63].
- Finite symmetric exclusion, reversible Markov semigroups and the additive-functional martingale central limit theorem of Kipnis and Varadhan [21, 51, 48].
- Finite tight frames, lattice cosine symbols, Hilbert-space feature maps, and Lipschitz dispersion bounds [70, 67, 62, 69].
- Fourier multiplier convergence and classical Maxwell groups [65, 58].
- Fourier-space vacuum Maxwell curl pairing, vector triple-product identities, and helicity-mode factorizations [46, 28].
- Gaussian fields on the sphere, conformal fractional Laplacians, primordial power spectra, spherical Bessel projection, and spherical Hankel inversion [53, 30, 55, 72].

- Graph nerves, combinatorial manifolds, and refinement limits [43, 59].
- Operational equivalence, quotient systems, and implementation invariance [64, 34].
- Operational quantum systems, finite channel models, holographic code models [39, 71, 57, 25].
- Preregistration checklists and eligibility criteria in experimental protocols [56, 32].
- Schur complements, spectral approximation and reversible finite-chain spectral decomposition [75, 33, 50].
- Variational (symplectic) integrators for Maxwell-type systems, the leapfrog/Courant stability condition, and spectral bounds of cellular Laplacians [66, 37, 45].

## References


[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[8] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[9] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[10] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026.
Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/extra/observable_normal_forms.pdf.

[11] B. Müller, *Verified Projection-Event Calculus in Lean 4: Bundled Arbitrary-Partition Pinching, Lüders Retractions, and CHSH Interoperability*, 2026.
Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/extra/machine_checked_finite_event_algebras.pdf.

[12] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/observers_are_all_you_need.pdf.

[13] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/reality_as_consensus_protocol.pdf.

[14] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf.

[15] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, D. A. Friedman, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/flagship/from_observer_consensus_to_standard_physics.pdf.

[16] B. Müller, M. Poneder, A. Osika, K. Xue, and D. A. Friedman, *Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[17] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Finite provenance and operator producers, independent verifiers, and receipts: https://github.com/muellerberndt/oph-physics-sim/tree/6080f3a4045b7cbd821476a0a8fa0a6b57eeeeab. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/evidence/closure_loop. Causal-poset archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/evidence/source_net_causal_poset. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/r2056/evidence.

[18] FloatingPragma (2026). Observer Patch Holography: Lean library, exact code receipts, finite certificates, and claim ledgers.
Available at https://github.com/FloatingPragma/observer-patch-holography.

[19] B. Müller and J. Hill, *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/extra/yang_mills_gap_clay_problem.pdf.

[20] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/p

aper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.tex.

[21] C. Kipnis and S. R. S. Varadhan, "Central limit theorem for additive functionals of reversible Markov processes and applications to simple exclusions," *Communications in Mathematical Physics* **104** (1986), 1–19. https://doi.org/10.1007/BF01210789.

[22] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/paper/executable_observer_construction.pdf.

[23] FloatingPragma, *A spatial fluctuation field from one native history*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/physical_space_time/OCCUPATION_FIELD.md.

[24] FloatingPragma, *Joint time and field laws on matched source screens*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2056/model/observer_time_fields/REFINEMENT.md.

[25] Almheiri, A., Dong, X., and Harlow, D. (2015). "Bulk locality and quantum error correction in AdS/CFT." *Journal of High Energy Physics* **2015**, 163. https://doi.org/10.1007/JHEP04(2015)163

[26] Arias, A., Gheondea, A., and Gudder, S. (2002). "Fixed points of quantum operations." *Journal of Mathematical Physics* **43**, 5872–5881. https://doi.org/10.1063/1.1519669

[27] Bhatia, R. (1997). *Matrix Analysis*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0653-8

[28] Bialynicki-Birula, I. (1996). "Photon Wave Function." In *Progress in Optics*, 245–294. Elsevier, Amsterdam. https://doi.org/10.1016/S0079-6638(08)70316-0

[29] Biggs, N. (1993). *Algebraic Graph Theory*. Cambridge University Press. https://doi.org/10.1017/CBO9780511608704

[30] Branson, T. P. (1995). "Sharp inequalities, the functional determinant, and the complementary series." *Transactions of the American Mathematical Society* **347**, 3671–3742. https://doi.org/10.1090/S0002-9947-1995-1316845-2

[31] Casella, G., and Berger, R. L. (2002). *Statistical Inference*. Duxbury (Thomson Learning). ISBN 0-534-24312-6.

[32] Chan, A. W., Tetzlaff, J. M., Altman, D. G., Laupacis, A., Gøtzsche, P. C., Krleža-Jerić, K., et al. (2013). "SPIRIT 2013 Statement: Defining Standard Protocol Items for Clinical Trials." *Annals of Internal Medicine* **158**, 200–207. https://doi.org/10.7326/0003-4819-158-3-201302050-00583

[33] Chatelin, F. (2011). *Spectral Approximation of Linear Operators*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611970678

[34] Chiribella, G., D'Ariano, G. M., and Perinotti, P. (2010). "Probabilistic theories with purification." *Physical Review A* **81**, 062348. https://doi.org/10.1103/PhysRevA.81.062348

[35] Cochran, W. G. (1977). *Sampling Techniques*. Wiley. ISBN 0-471-16240-X.

[36] Conway, J. H., and Smith, D. A. (2003). *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*. A K Peters. https://doi.org/10.1201/9781439864180

[37] Courant, R., Friedrichs, K., and Lewy, H. (1928). "Über die partiellen Differenzengleichungen der mathematischen Physik." *Mathematische Annalen* **100**, 32–74. https://doi.org/10.1007/BF01448839

[38] Coxeter, H. S. M. (1973). *Regular Polytopes*. Dover. ISBN 0-486-61480-8.

[39] Davies, E. B., and Lewis, J. T. (1970). "An operational approach to quantum probability." *Communications in Mathematical Physics* **17**, 239–260. https://doi.org/10.1007/BF01647093

[40] Desbrun, M., Hirani, A. N., Leok, M., and Marsden, J. E. (2005). "Discrete exterior calculus." arXiv:math/0508341. https://arxiv.org/abs/math/0508341

[41] Doyle, P. G., and Snell, J. L. (1984). *Random Walks and Electric Networks*. Mathematical Association of America. arXiv:math/0001057. https://arxiv.org/abs/math/0001057

[42] Grady, L. J., and Polimeni, J. R. (2010). *Discrete Calculus*. Springer. https://doi.org/10.1007/978-1-84996-290-2

[43] Hatcher, A. (2002). *Algebraic Topology*. Cambridge University Press. ISBN 0-521-79540-0.

[44] Hoeffding, W. (1948). "A class of statistics with asymptotically normal distribution." *Annals of Mathematical Statistics* **19**, 293–325. https://doi.org/10.1214/aoms/1177730196

[45] Horak, D., and Jost, J. (2013). "Spectra of combinatorial Laplace operators on simplicial complexes." *Advances in Mathematics* **244**, 303–336. https://doi.org/10.1016/j.aim.2013.05.007

[46] Jackson, J. D. (1999). *Classical Electrodynamics*. Wiley, New York. ISBN 0-471-30932-X.

[47] Kolmogoroff, A. (1936). "Zur Theorie der Markoffschen Ketten." *Mathematische Annalen* **112**, 155–160. https://doi.org/10.1007/BF01565412

[48] Komorowski, T., Landim, C., and Olla, S. (2012). *Fluctuations in Markov Processes: Time Symmetry and Martingale Approximation*. Springer, Berlin (Grundlehren der mathematischen Wissenschaften). https://doi.org/10.1007/978-3-642-29880-6

[49] Kwak, J. H., and Lee, J. (1990). "Isomorphism classes of graph bundles." *Canadian Journal of Mathematics* **42**, 747–761. https://doi.org/10.4153/CJM-1990-039-3

[50] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[51] Liggett, T. M. (1985). *Interacting Particle Systems*. Springer, New York (Grundlehren der mathematischen Wissenschaften). https://doi.org/10.1007/978-1-4613-8542-4

[52] Lim, L. H. (2020). "Hodge Laplacians on graphs." *SIAM Review* **62**, 685–715. https://doi.org/10.1137/18M1223101

[53] Marinucci, D., and Peccati, G. (2011). *Random Fields on the Sphere: Representation, Limit Theorems and Cosmological Applications*. Cambridge University Press. https://doi.org/10.1017/CBO9780511751677

[54] McKay, J. (1980). "Graphs, singularities, and finite groups." In *The Santa Cruz Conference on Finite Groups*, 183–186. American Mathematical Society, Providence. https://doi.org/10.1090/pspum/037/604577

[55] Mukhanov, V. (2005). *Physical Foundations of Cosmology*. Cambridge University Press. https://doi.org/10.1017/CBO9780511790553

[56] Nosek, B. A., Ebersole, C. R., DeHaven, A. C., and Mellor, D. T. (2018). "The preregistration revolution." *Proceedings of the National Academy of Sciences* **115**, 2600–2606. https://doi.org/10.1073/pnas.1708274114

[57] Pastawski, F., Yoshida, B., Harlow, D., and Preskill, J. (2015). "Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence." *Journal of High Energy Physics* **2015**, 149. https://doi.org/10.1007/JHEP06(2015)149

[58] Pazy, A. (1983). *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer. https://doi.org/10.1007/978-1-4612-5561-1

[59] Rourke, C. P., and Sanderson, B. J. (1972). *Introduction to Piecewise-Linear Topology*. Springer, Berlin. https://doi.org/10.1007/978-3-642-81735-9

[60] Sandve, G. K., Nekrutenko, A., Taylor, J., and Hovig, E. (2013). "Ten simple rules for reproducible computational research." *PLoS Computational Biology* **9**, e1003285. https://doi.org/10.1371/journal.pcbi.1003285

[61] Schnakenberg, J. (1976). "Network theory of microscopic and macroscopic behavior of master equation systems." *Reviews of Modern Physics* **48**, 571–585. https://doi.org/10.1103/RevModPhys.48.571

[62] Schoenberg, I. J. (1938). "Metric spaces and positive definite functions." *Transactions of the American Mathematical Society* **44**, 522–536. https://doi.org/10.1090/S0002-9947-1938-1501980-0

[63] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[64] Spekkens, R. W. (2005). "Contextuality for preparations, transformations, and unsharp measurements." *Physical Review A* **71**, 052108. https://doi.org/10.1103/PhysRevA.71.052108

[65] Stein, E. M., and Weiss, G. (1971). *Introduction to Fourier Analysis on Euclidean Spaces*. Princeton University Press (Princeton Mathematical Series). https://doi.org/10.1515/9781400883899

[66] Stern, A., Tong, Y., Desbrun, M., and Marsden, J. E. (2015). "Geometric computational electrodynamics with variational integrators and discrete differential forms." In *Geometry, Mechanics, and Dynamics (Fields Institute Communications 73)*, 437–475. Springer. https://doi.org/10.1007/978-1-4939-2441-7_19

[67] Strikwerda, J. C. (2004). *Finite Difference Schemes and Partial Differential Equations*. SIAM. https://doi.org/10.1137/1.9780898717938

[68] Teixeira, F. L., and Chew, W. C. (1999). "Lattice electromagnetic theory from a topological viewpoint." *Journal of Mathematical Physics* **40**, 169–187. https://doi.org/10.1063/1.532767

[69] Trefethen, L. N. (1982). "Group Velocity in Finite Difference Schemes." *SIAM Review* **24**, 113–136. https://doi.org/10.1137/1024038

[70] Waldron, S. F. D. (2018). *An Introduction to Finite Tight Frames*. Birkhäuser/Springer, New York (Applied and Numerical Harmonic Analysis). https://doi.org/10.1007/978-0-8176-4815-2

[71] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[72] Watson, G. N. (1995). *A Treatise on the Theory of Bessel Functions*. Cambridge University Press (Cambridge Mathematical Library). ISBN 0-521-48391-3.

[73] Yee, K. S. (1966). "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media." *IEEE Transactions on Antennas and Propagation* **14**, 302–307. https://doi.org/10.1109/TAP.1966.1138693

[74] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

[75] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

## AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.